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Shelton, IV et al.

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(54) **SURGICAL INSTRUMENT HAVING FORCE
FEEDBACK CAPABILITIES**

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(2013.01); A61B 2017/00199 (2013.01); A61B
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(2013.01);

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(Continued)

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(58) **Field of Classification Search**

CPC A61B 5/180816; A61B 17/068; A61B
17/072; A61B 17/07207; A61B 31/40;
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See application file for complete search history.

(73) Assignee: **Cilag GmbH International**, Zug (CH)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(63) Continuation of application No. 18/117,130, filed on
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(51) **Int. Cl.**

A61B 17/064 (2006.01)

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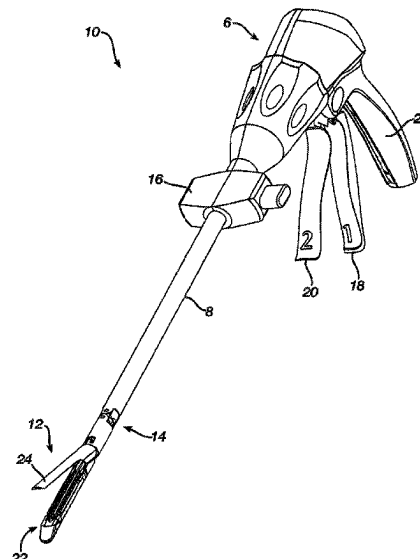
ABSTRACT

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A surgical instrument. The surgical instrument has a force
delivery system which communicates a changing force to a
user of the surgical instrument when a limit of a surgical
function is reached. A sensor senses position of a moving
element to communicate a signal to the force delivery
system. In various embodiments, the system generates a
vibratory force to signal the user.

30 Claims, 53 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/334,878, filed on May 31, 2021, now Pat. No. 11,648,008, which is a continuation of application No. 16/160,460, filed on Oct. 15, 2018, now Pat. No. 11,020,113, which is a continuation of application No. 15/223,791, filed on Jul. 29, 2016, now Pat. No. 10,098,636, which is a continuation of application No. 13/712,090, filed on Dec. 12, 2012, now Pat. No. 9,439,649.

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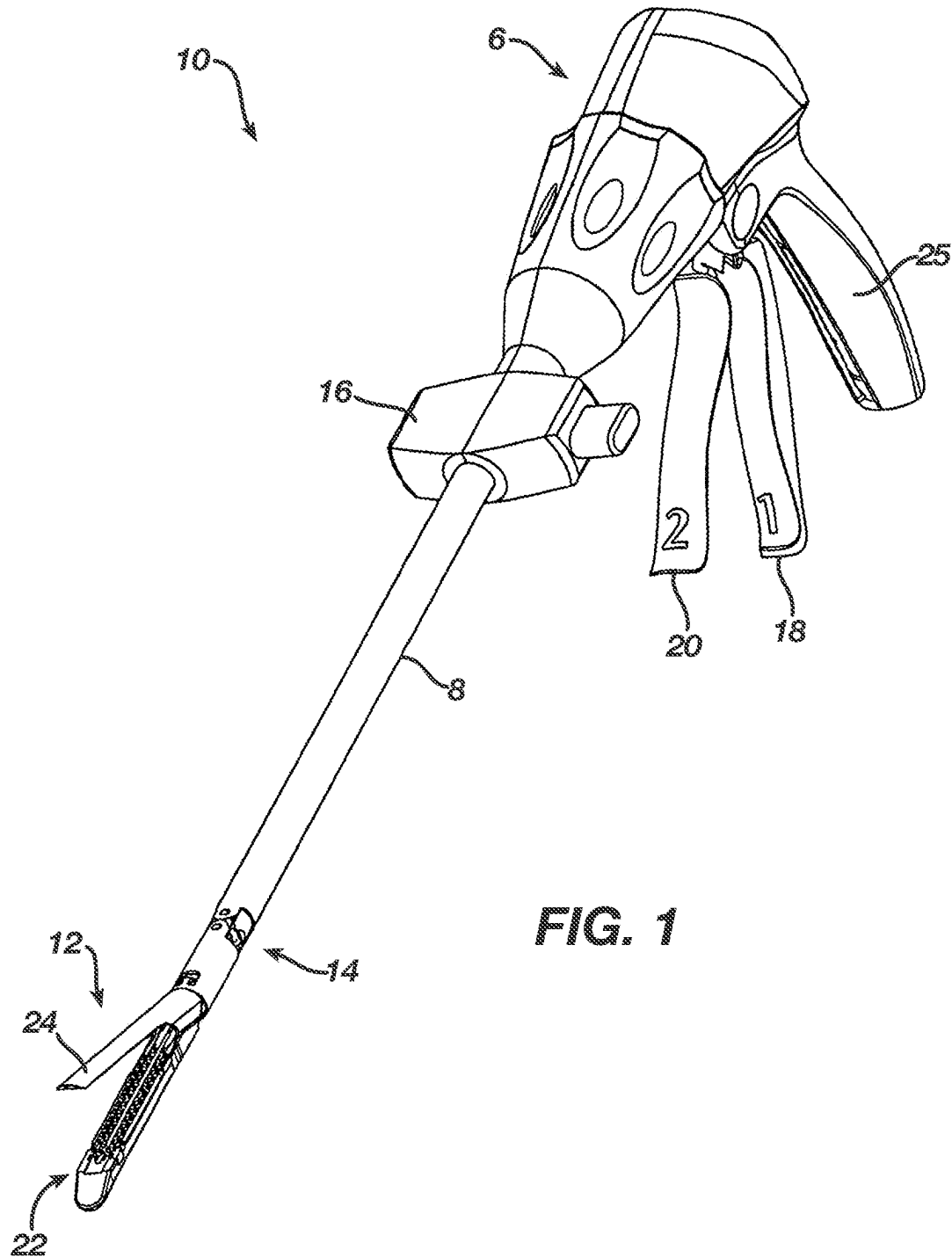
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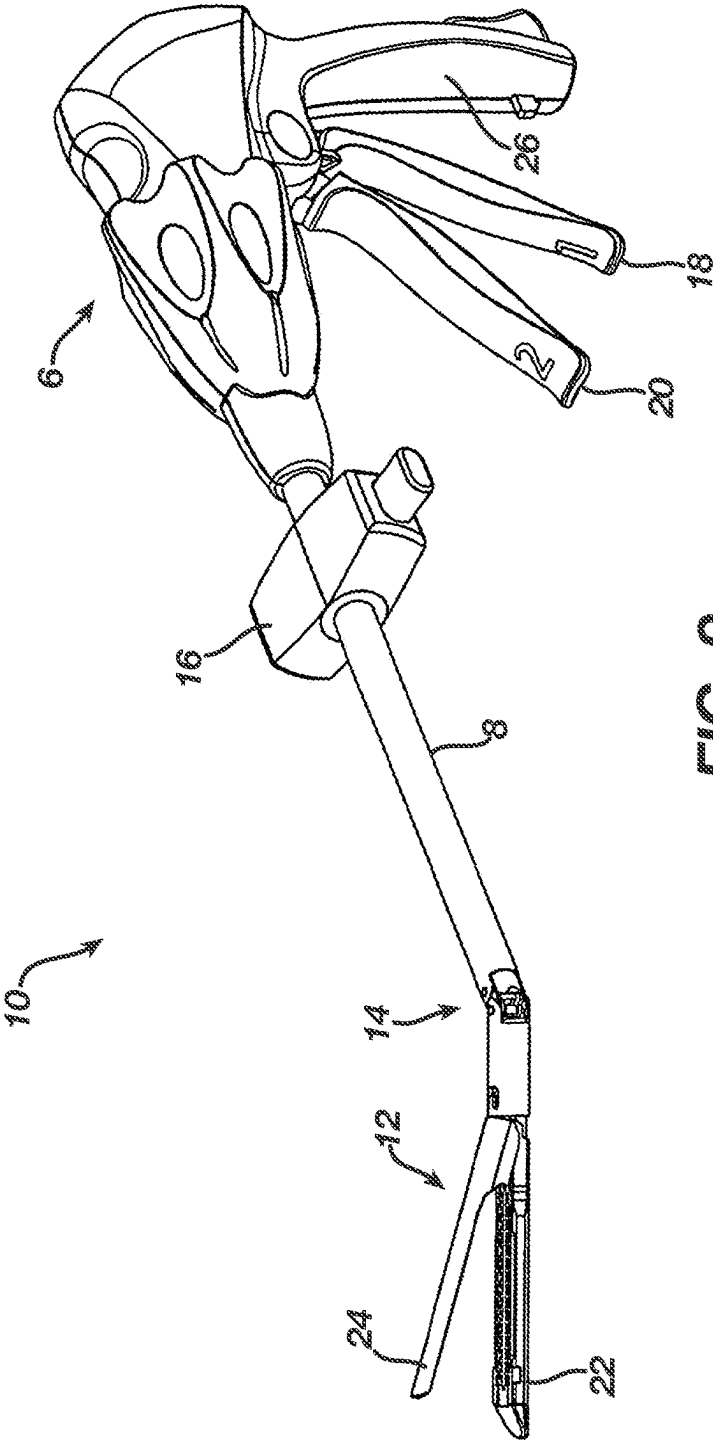
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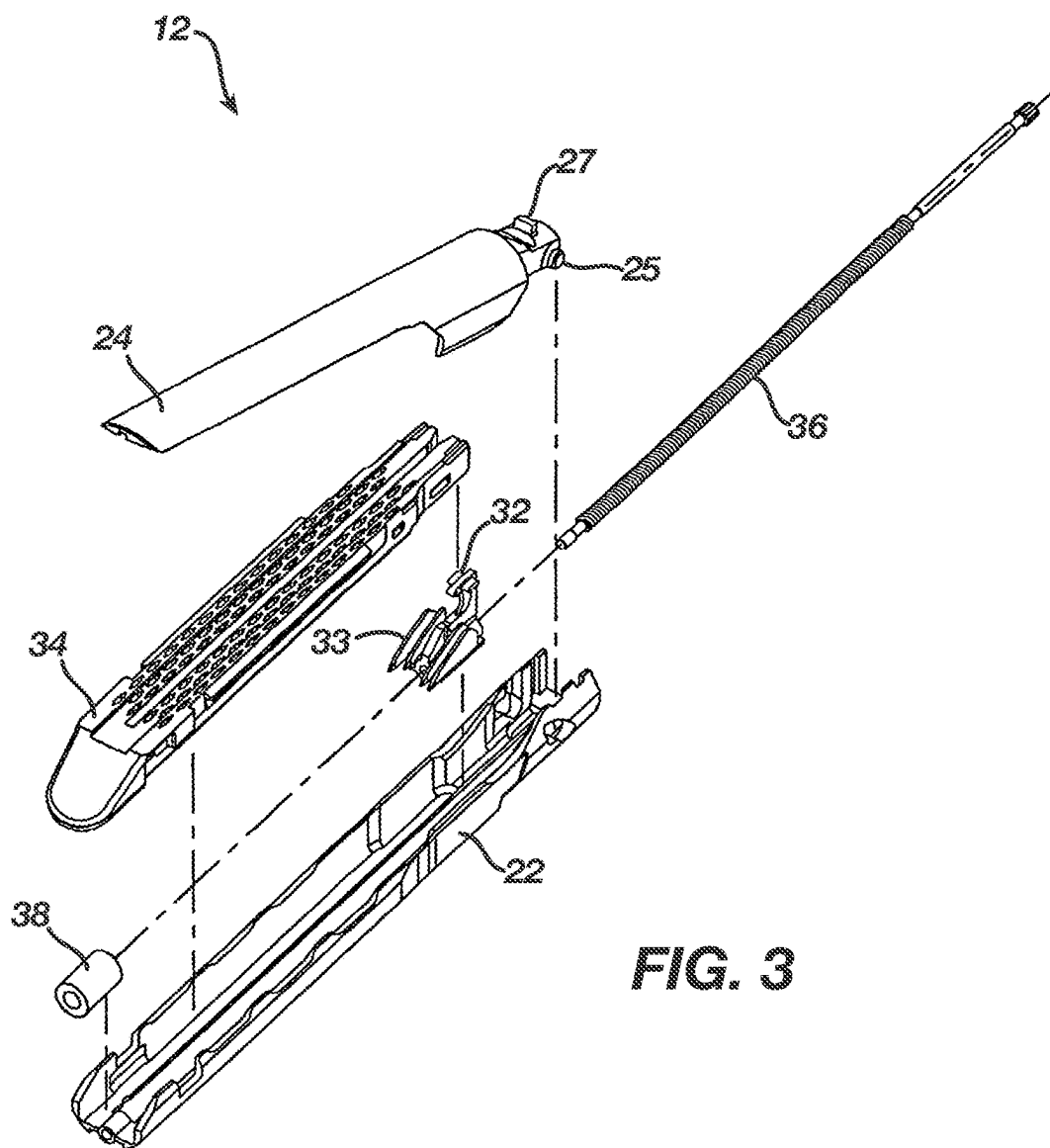
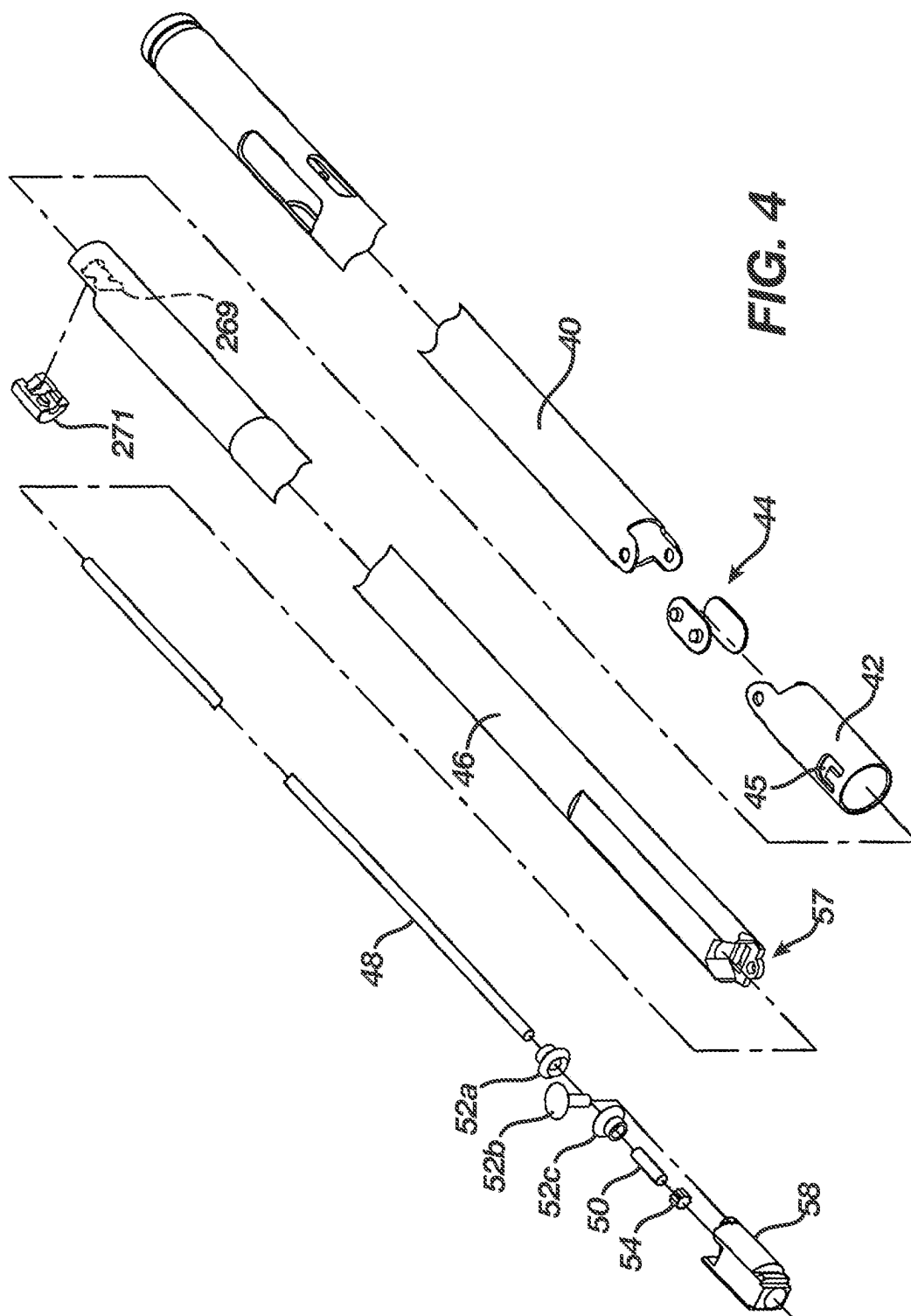
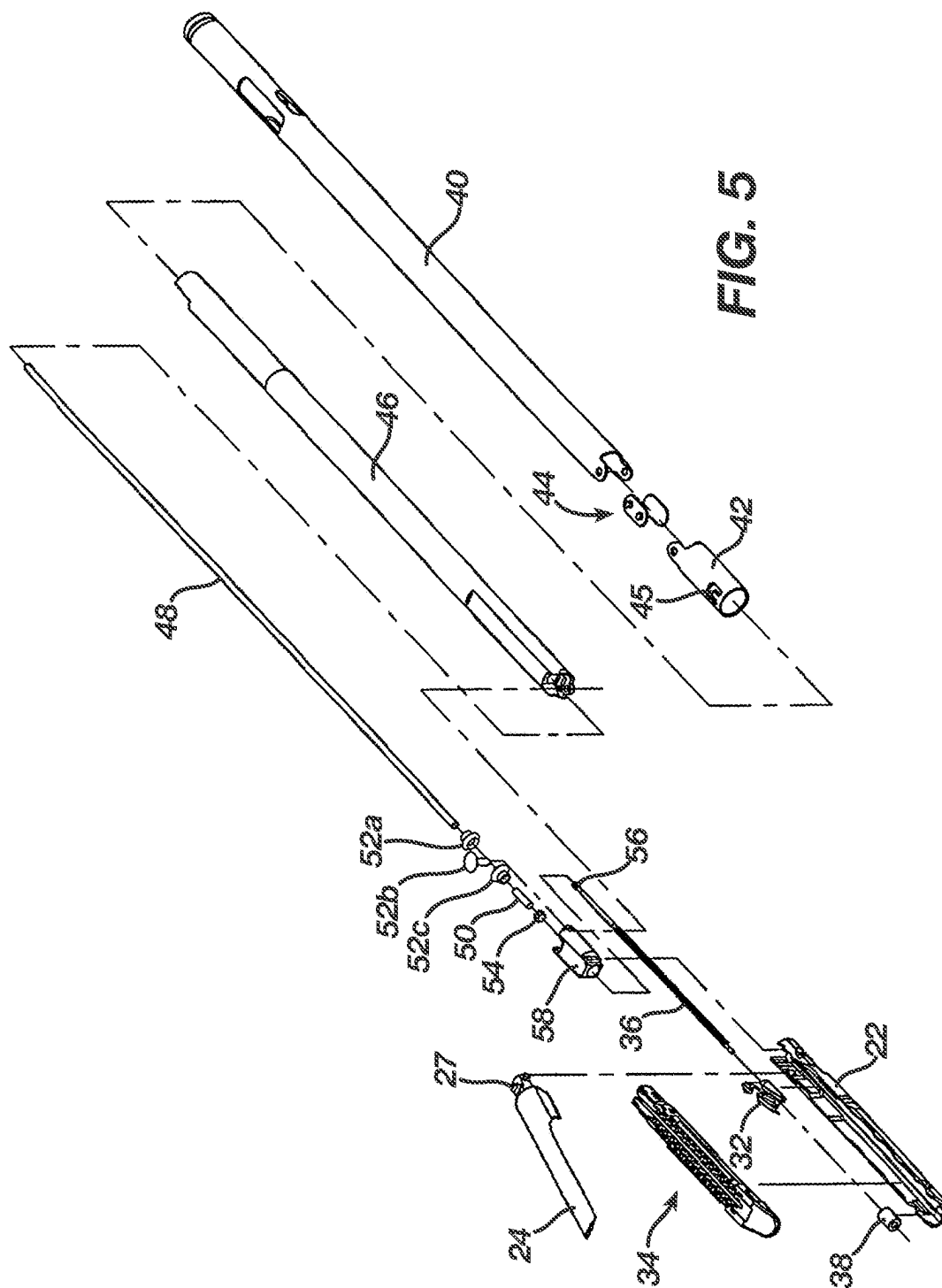


FIG. 3





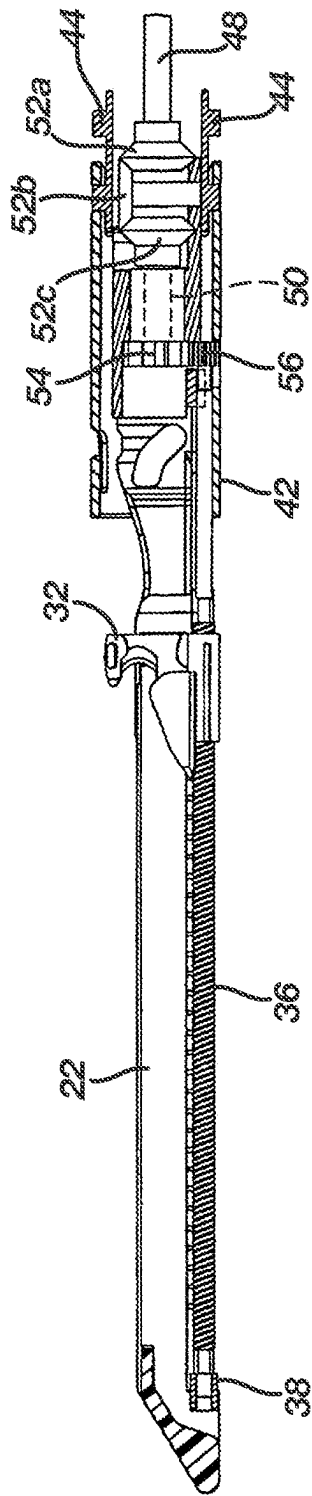


FIG. 6

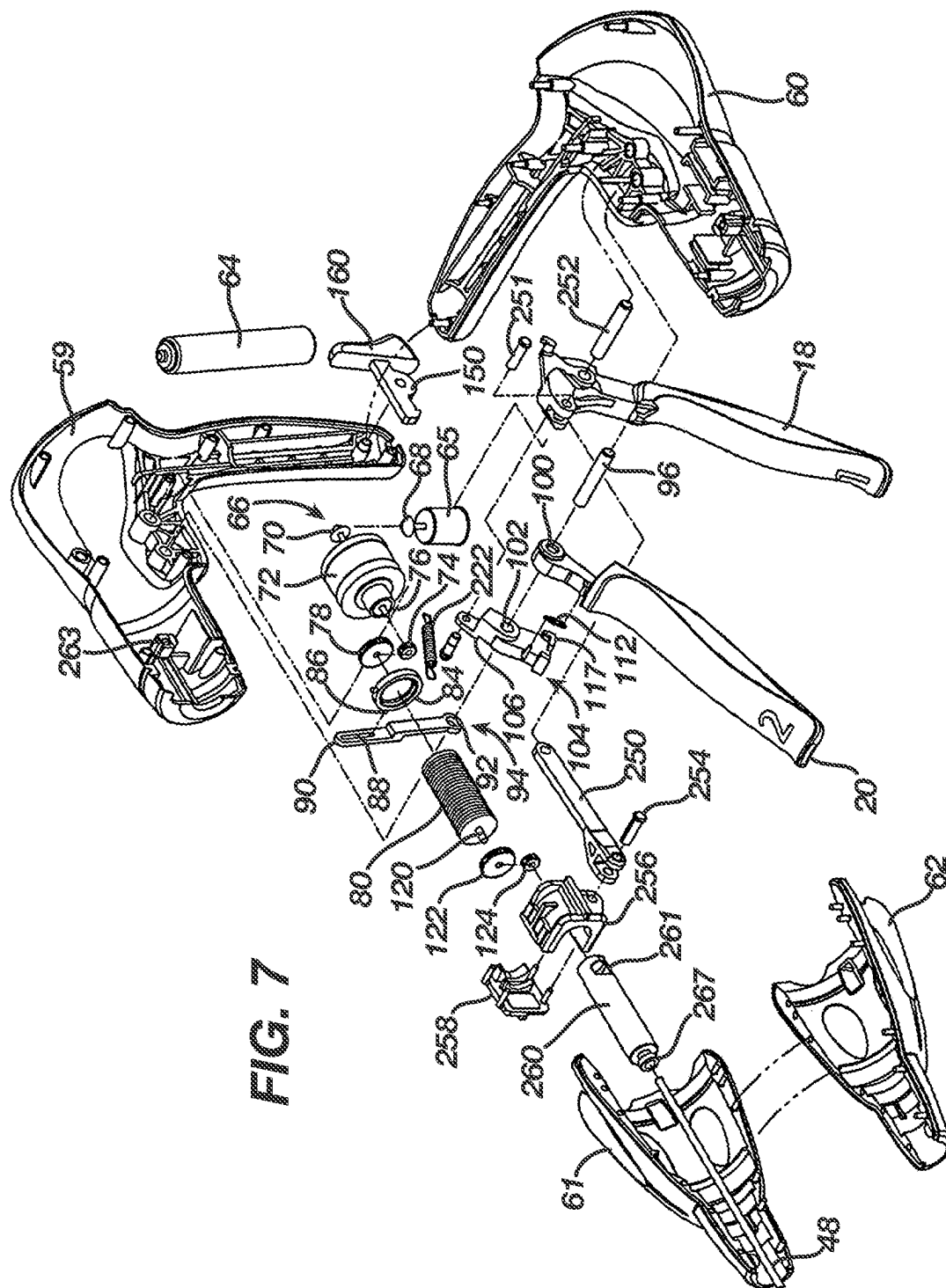
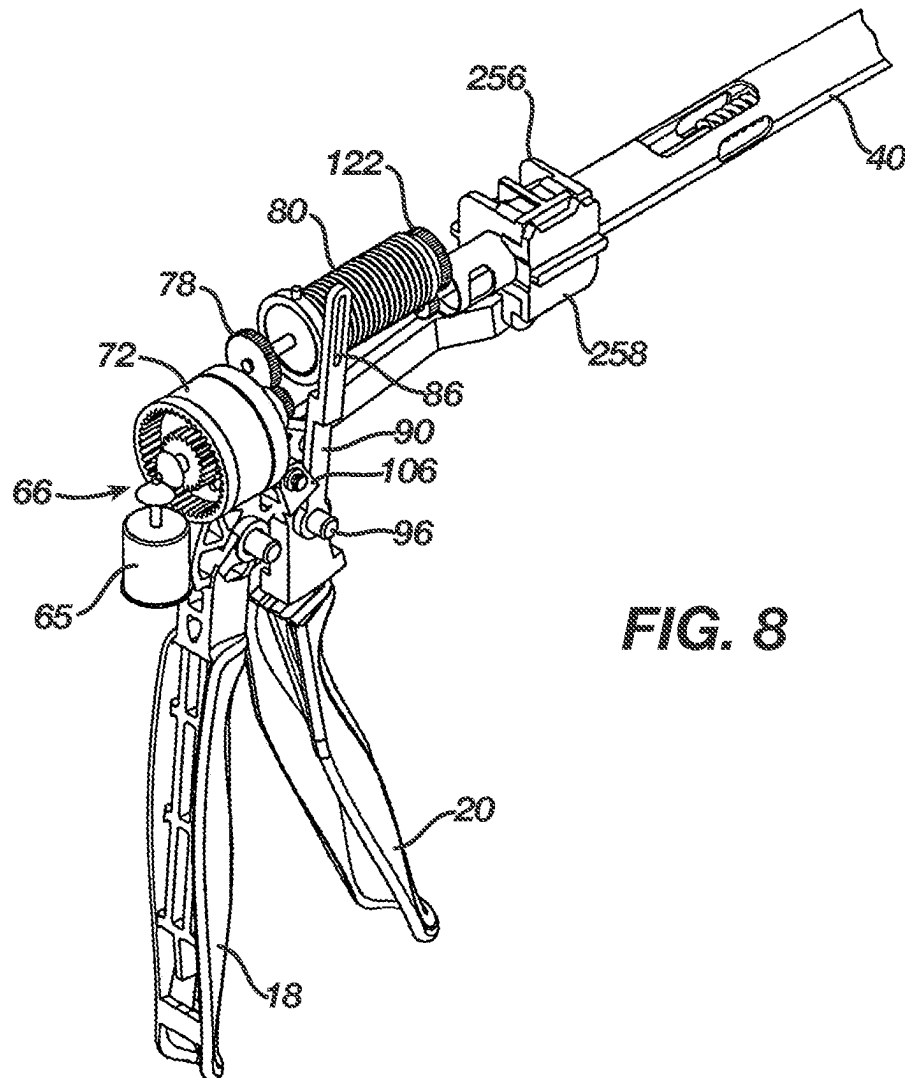
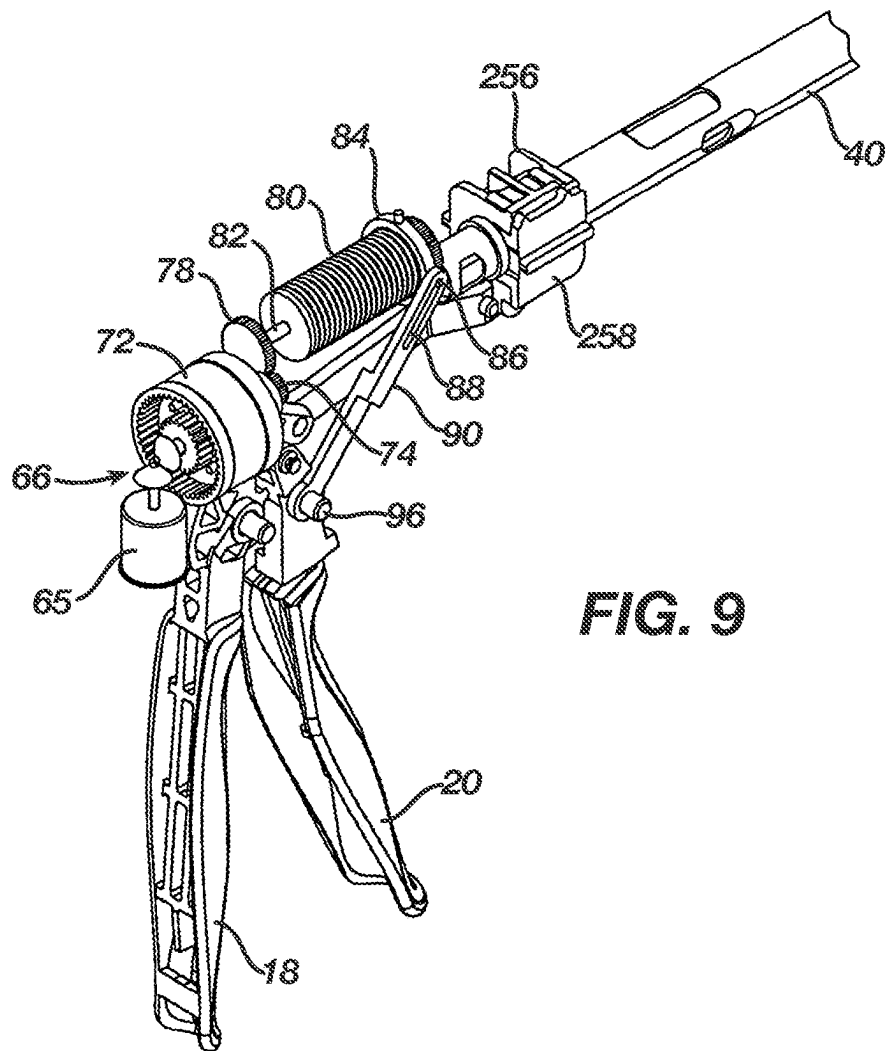
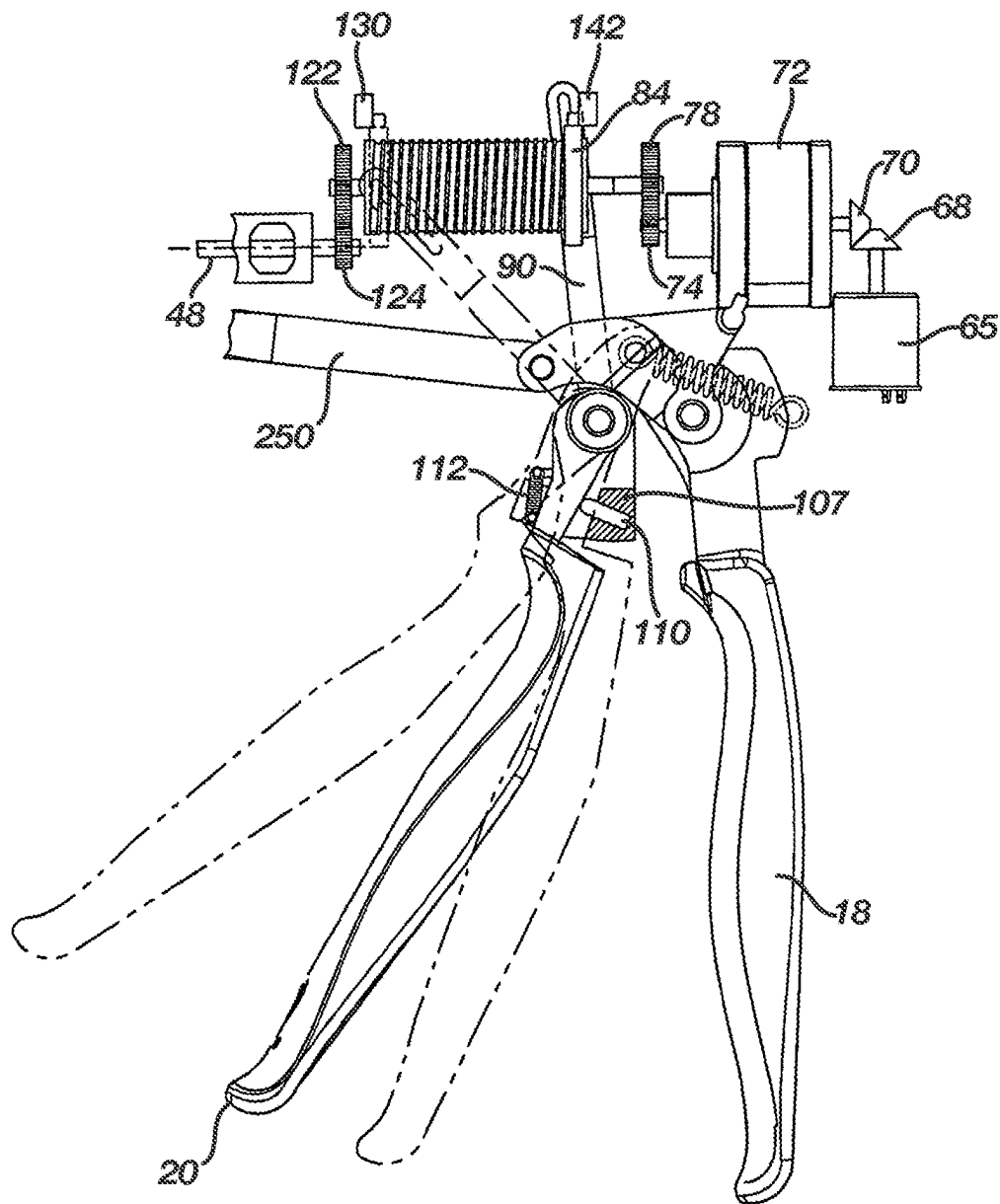
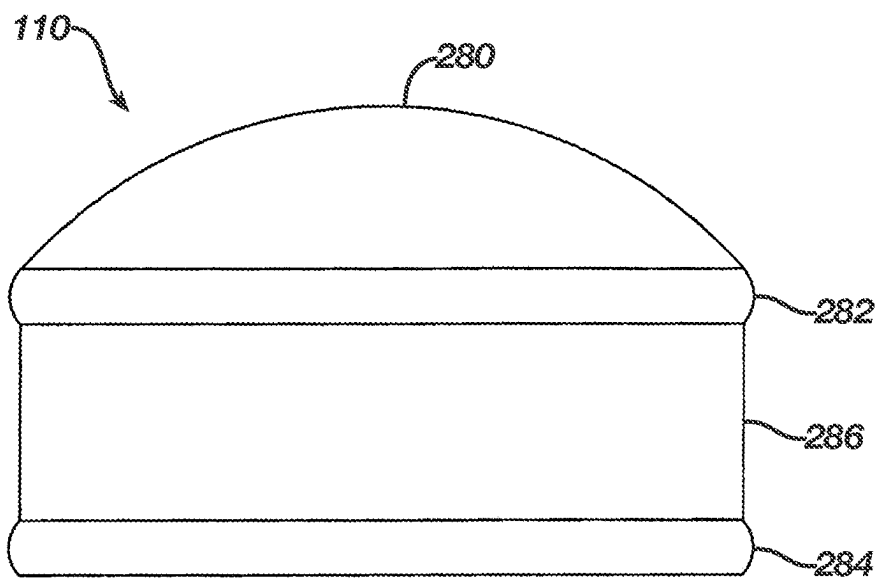
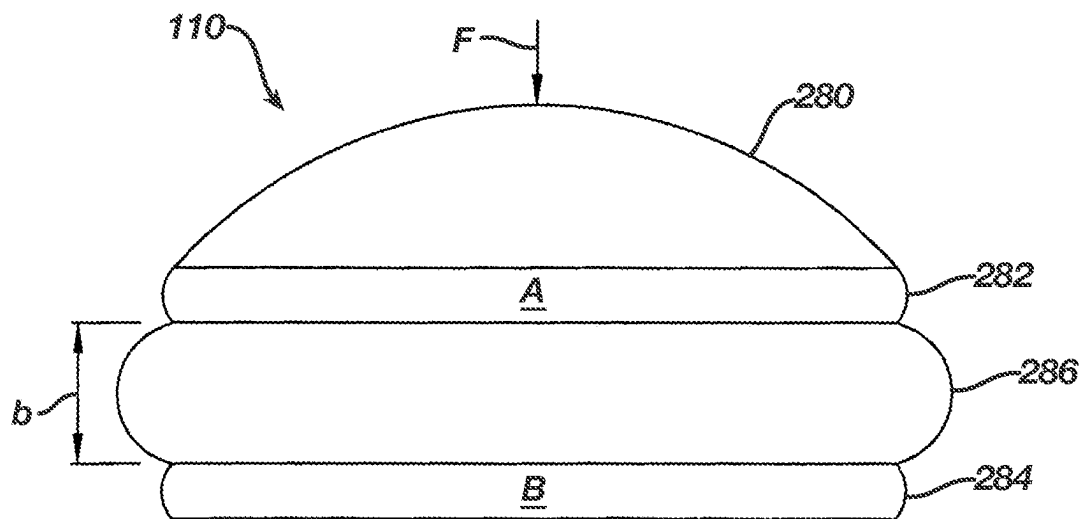


FIG. 7





**FIG. 10**

**FIG. 10A****FIG. 10B**

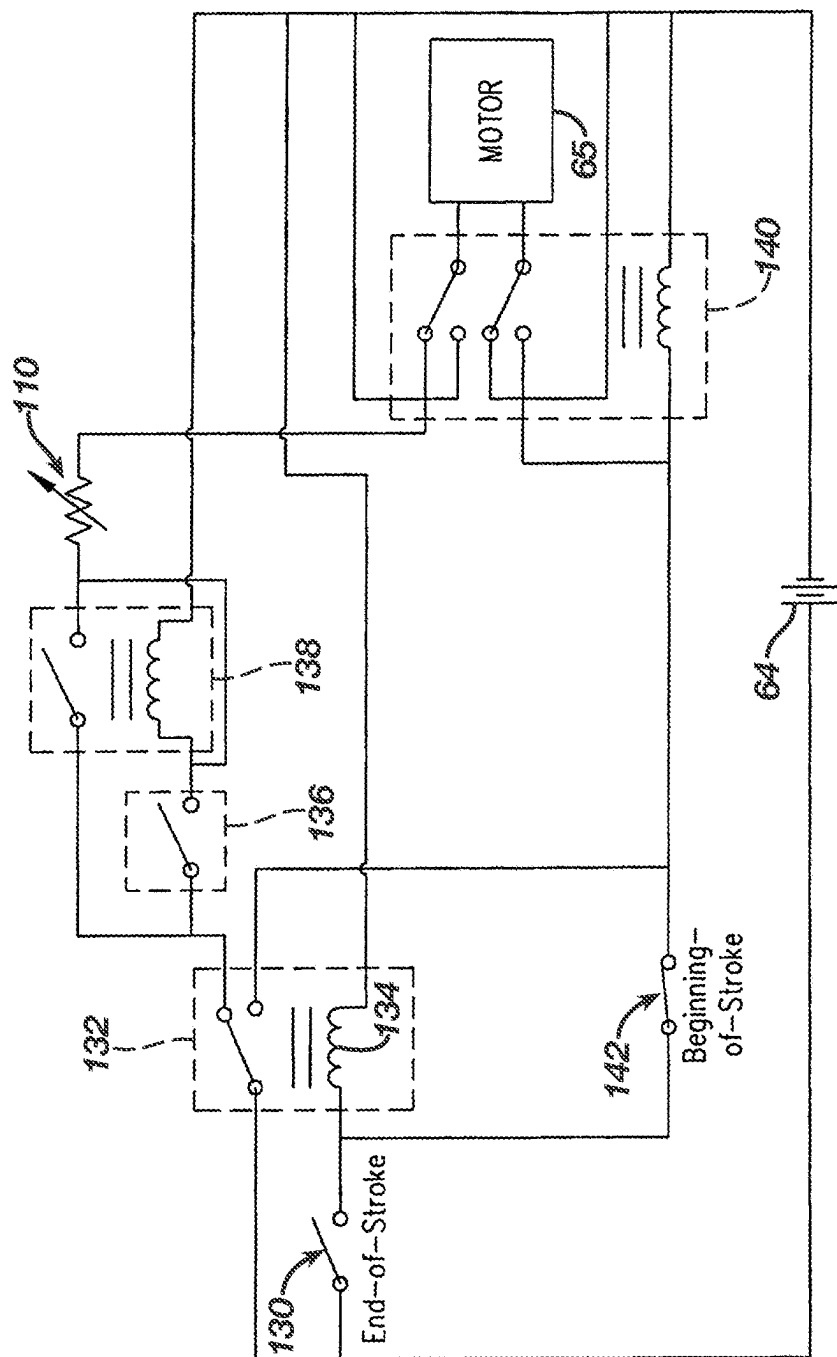
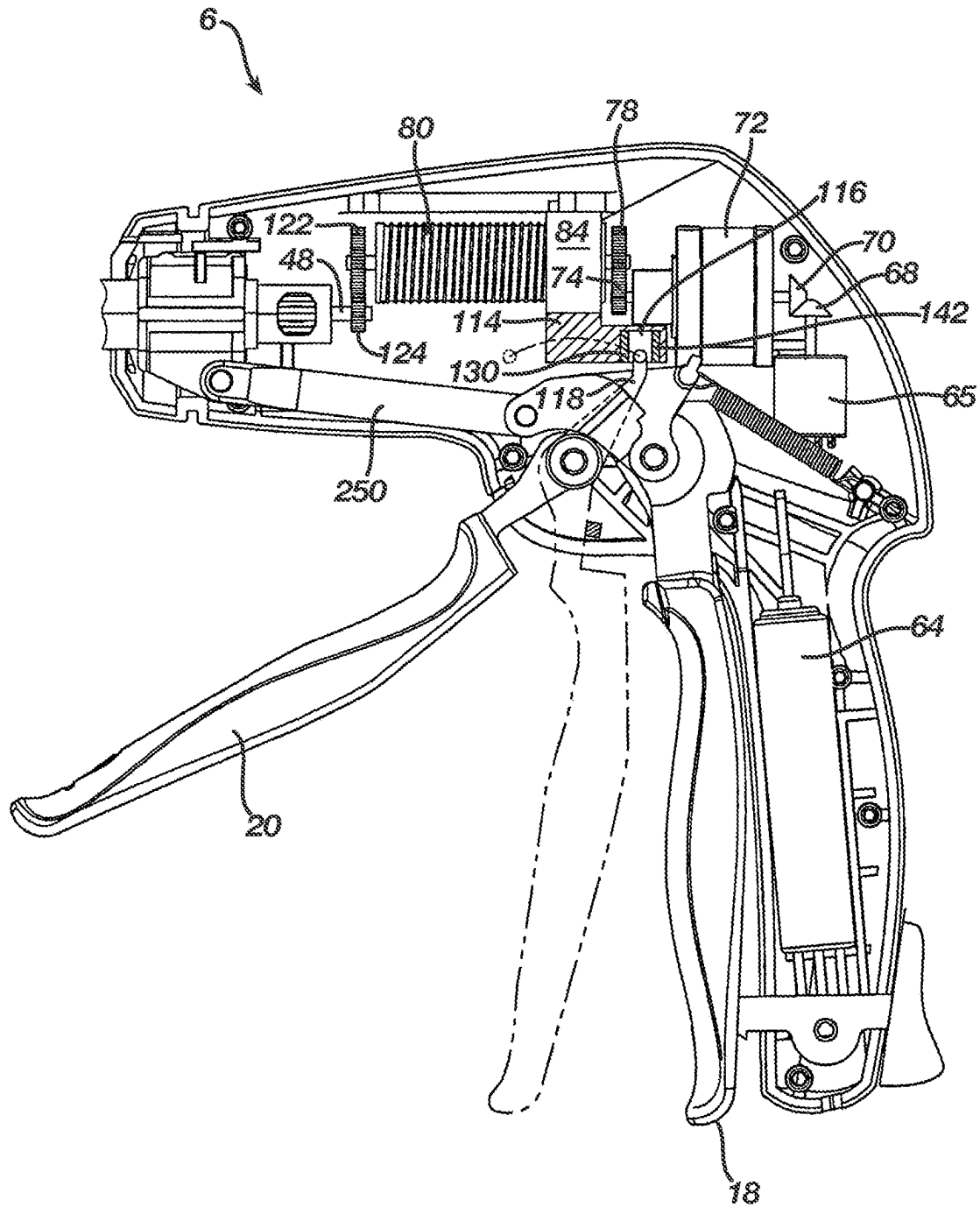


FIG. 11

**FIG. 12**

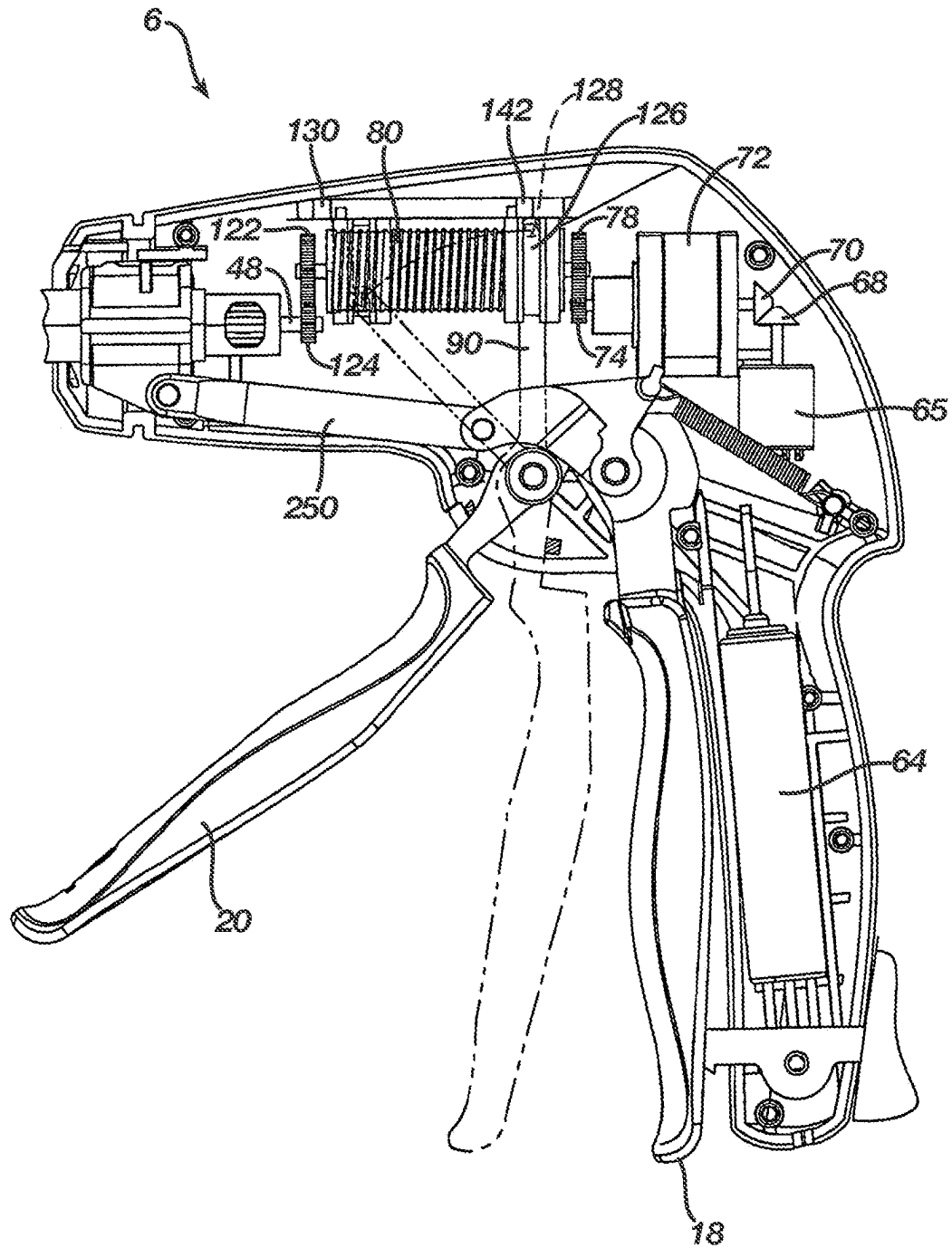
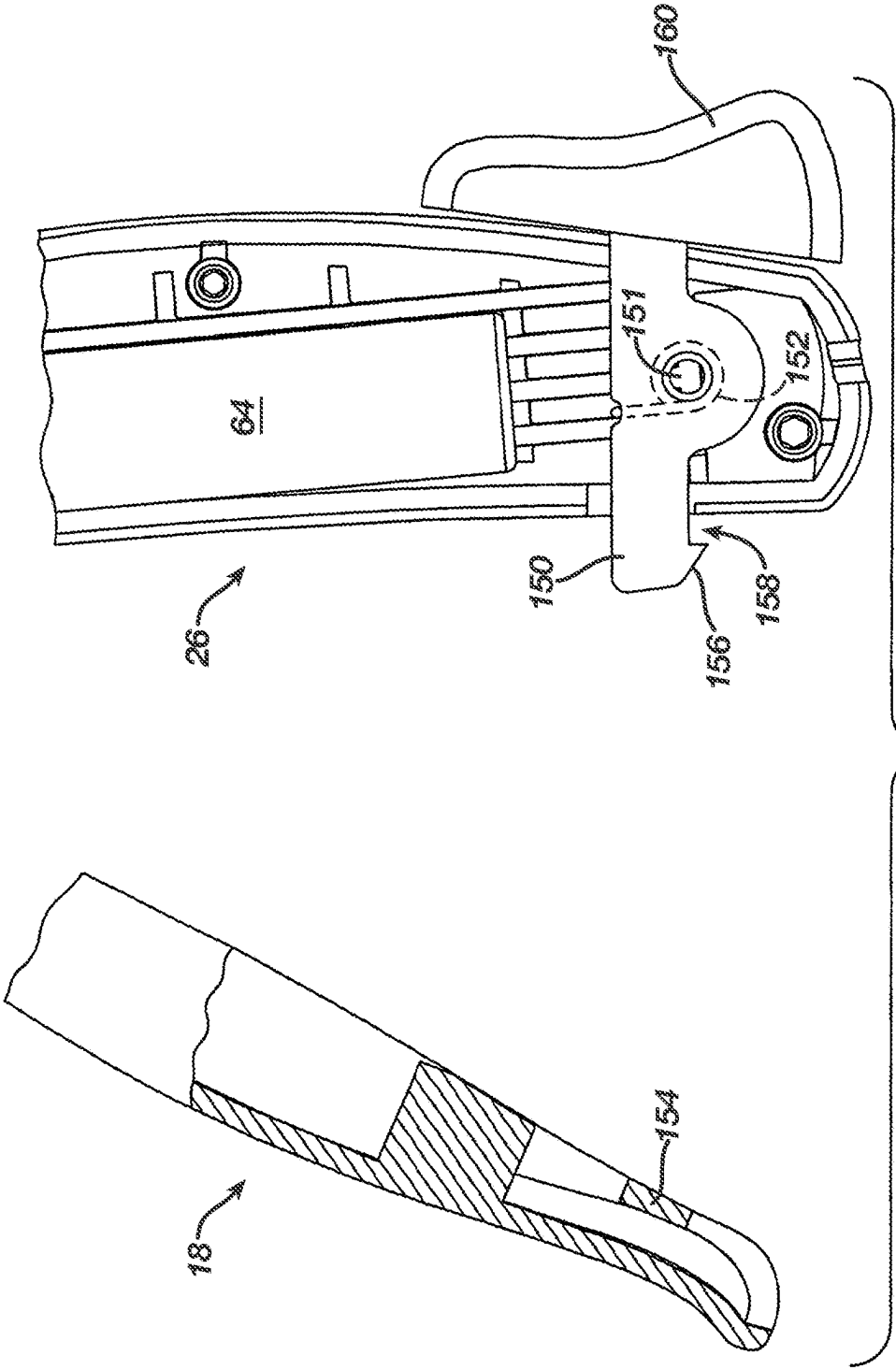
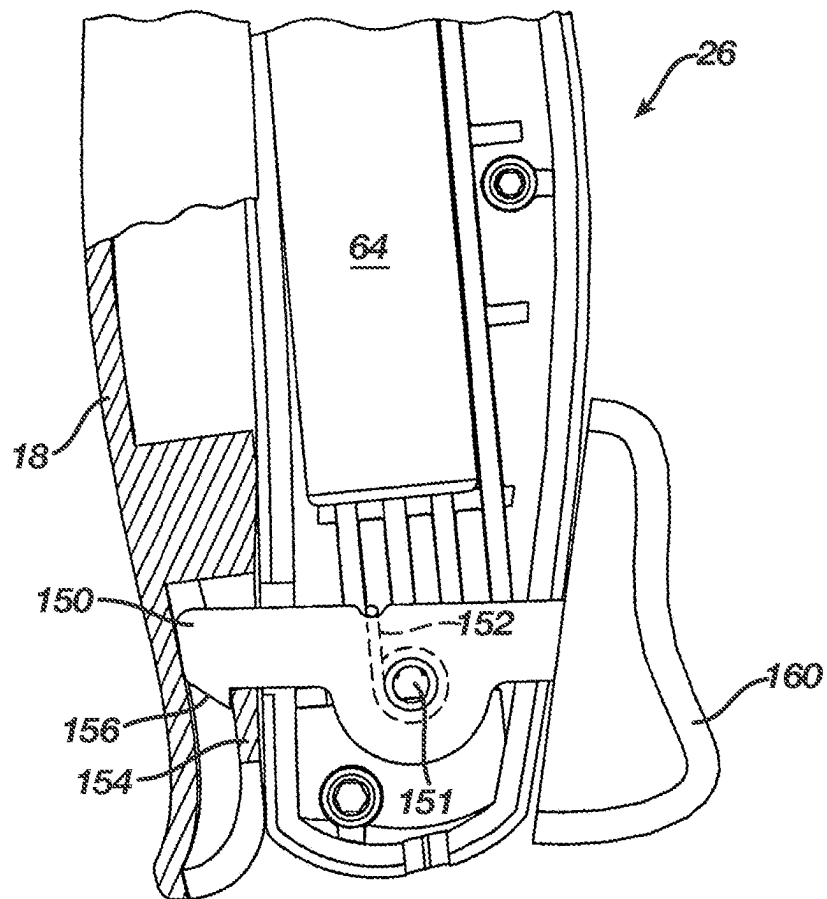
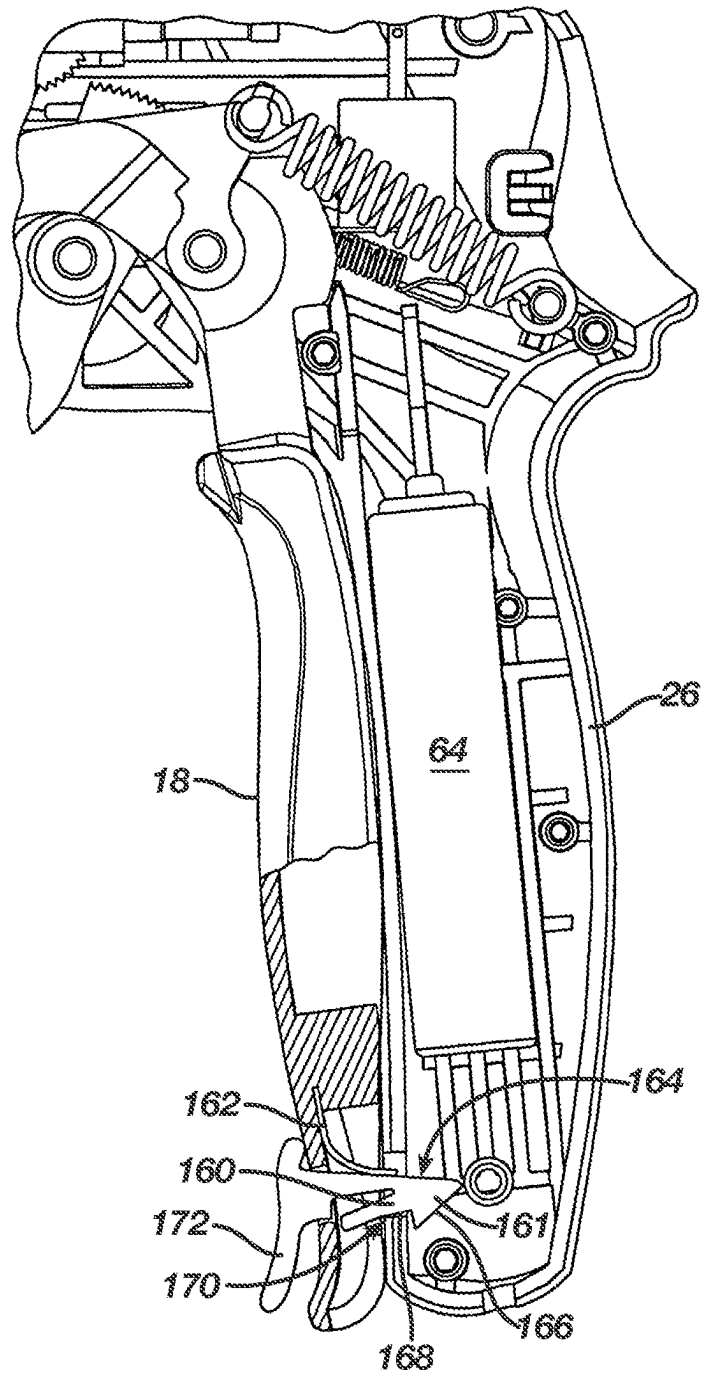


FIG. 13



**FIG. 15**

**FIG. 16**

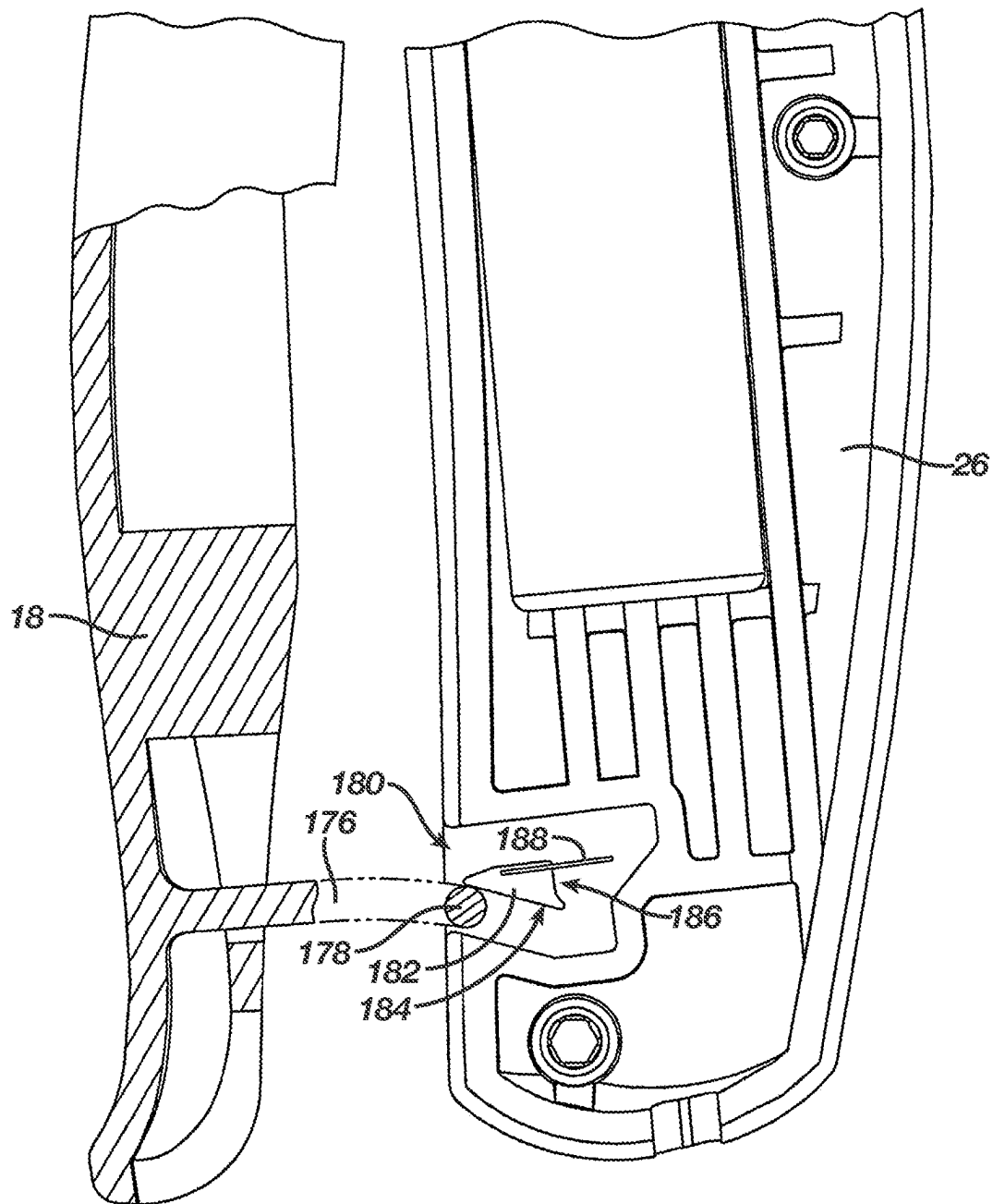


FIG. 17

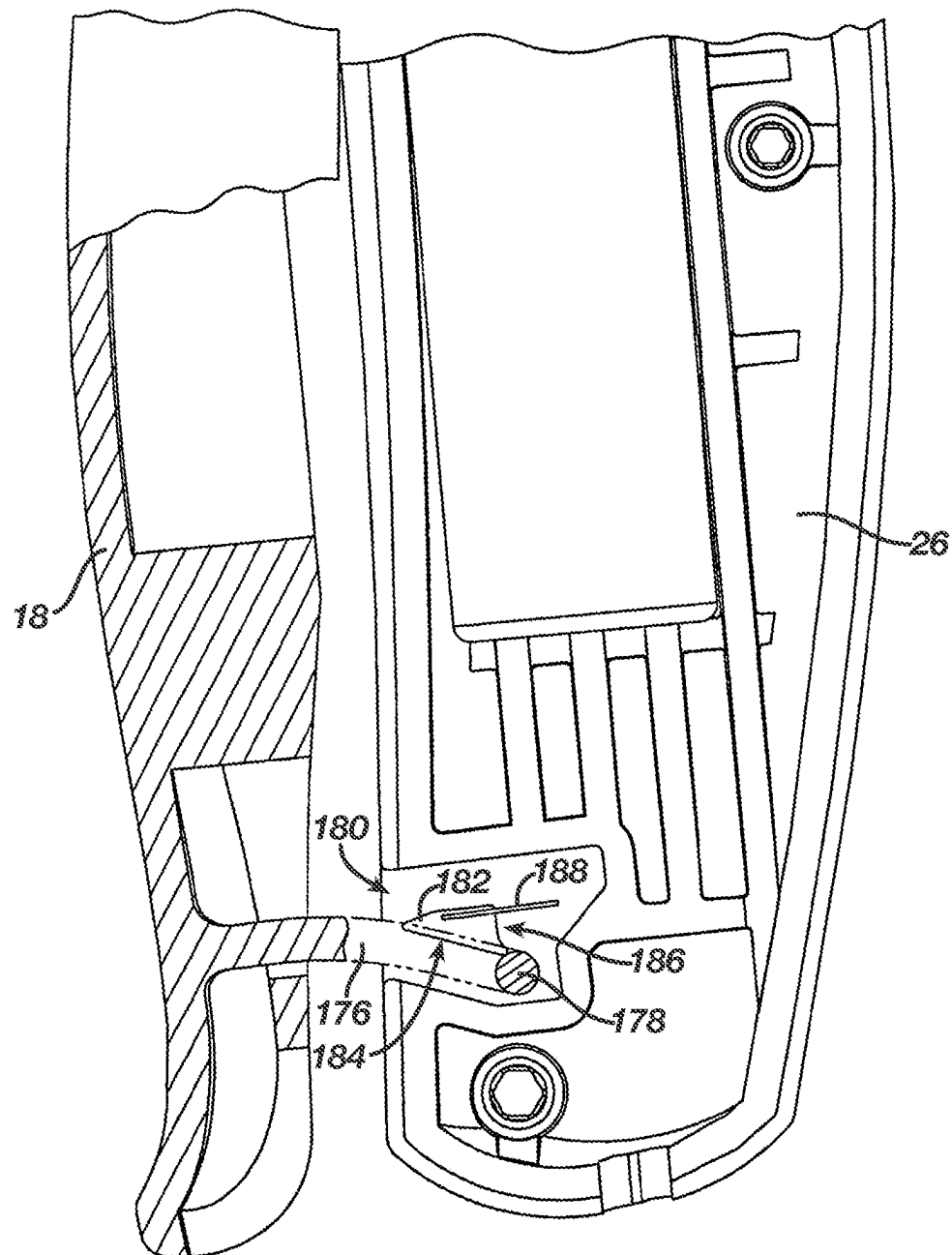


FIG. 18

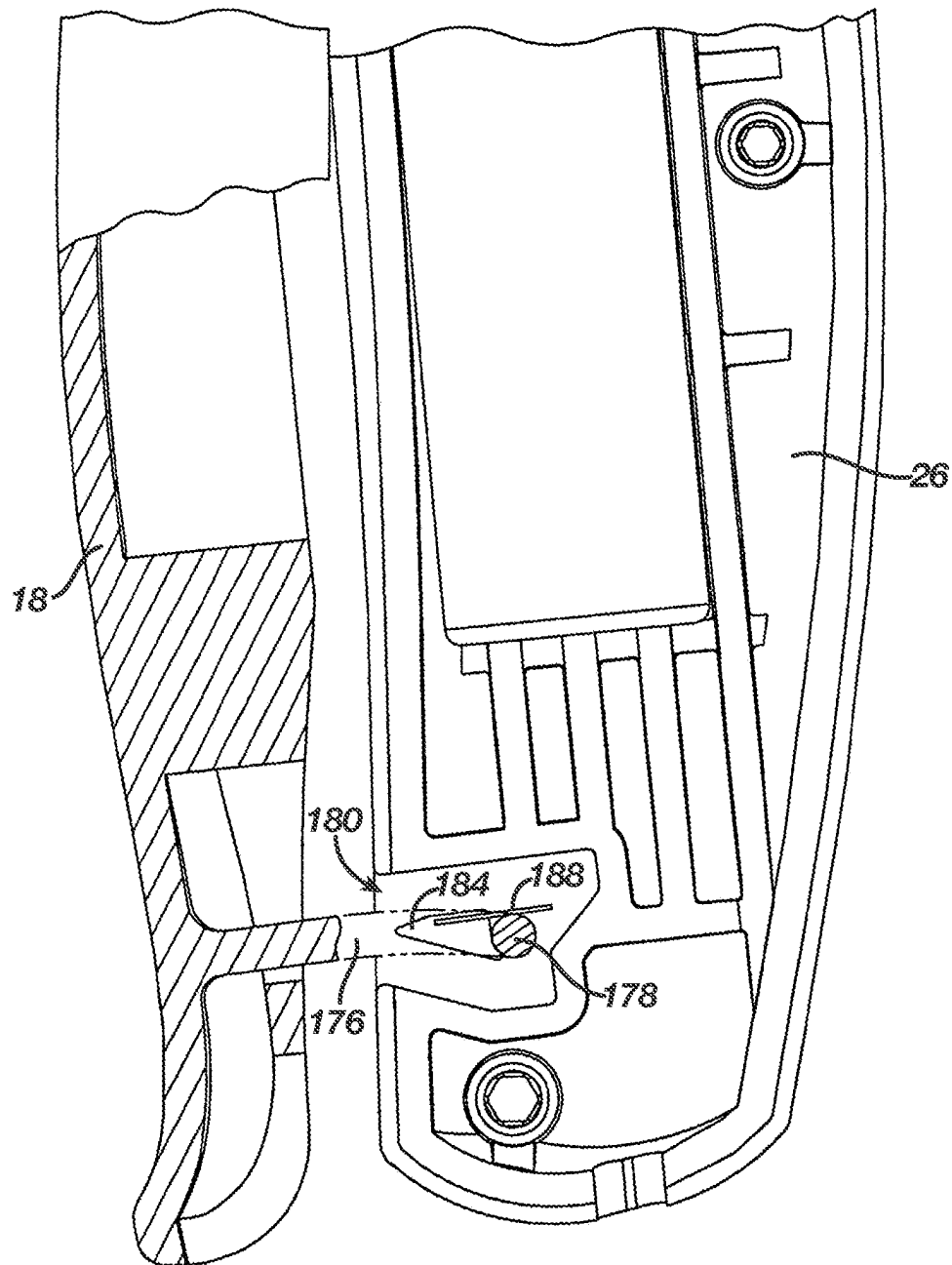
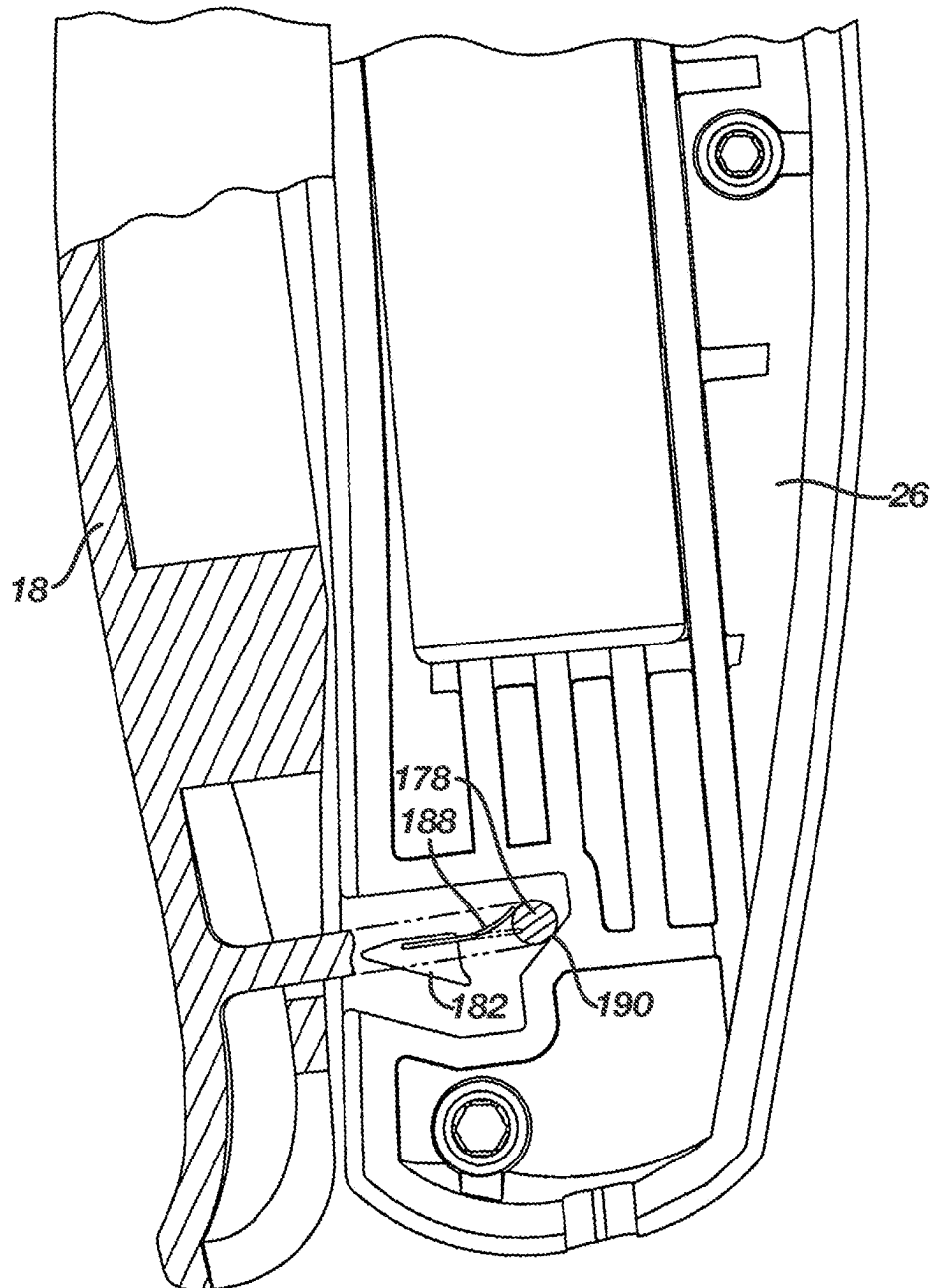
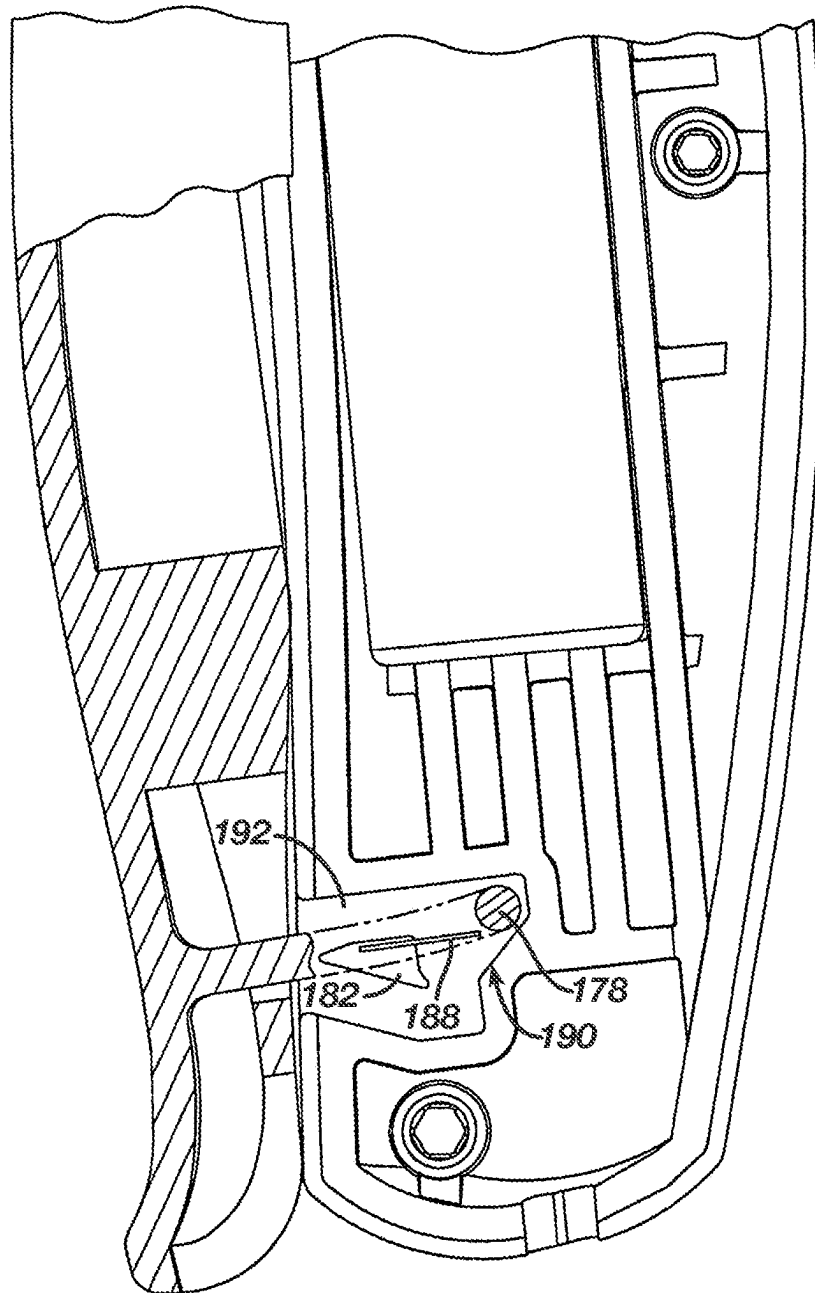


FIG. 19

**FIG. 20**

**FIG. 21**

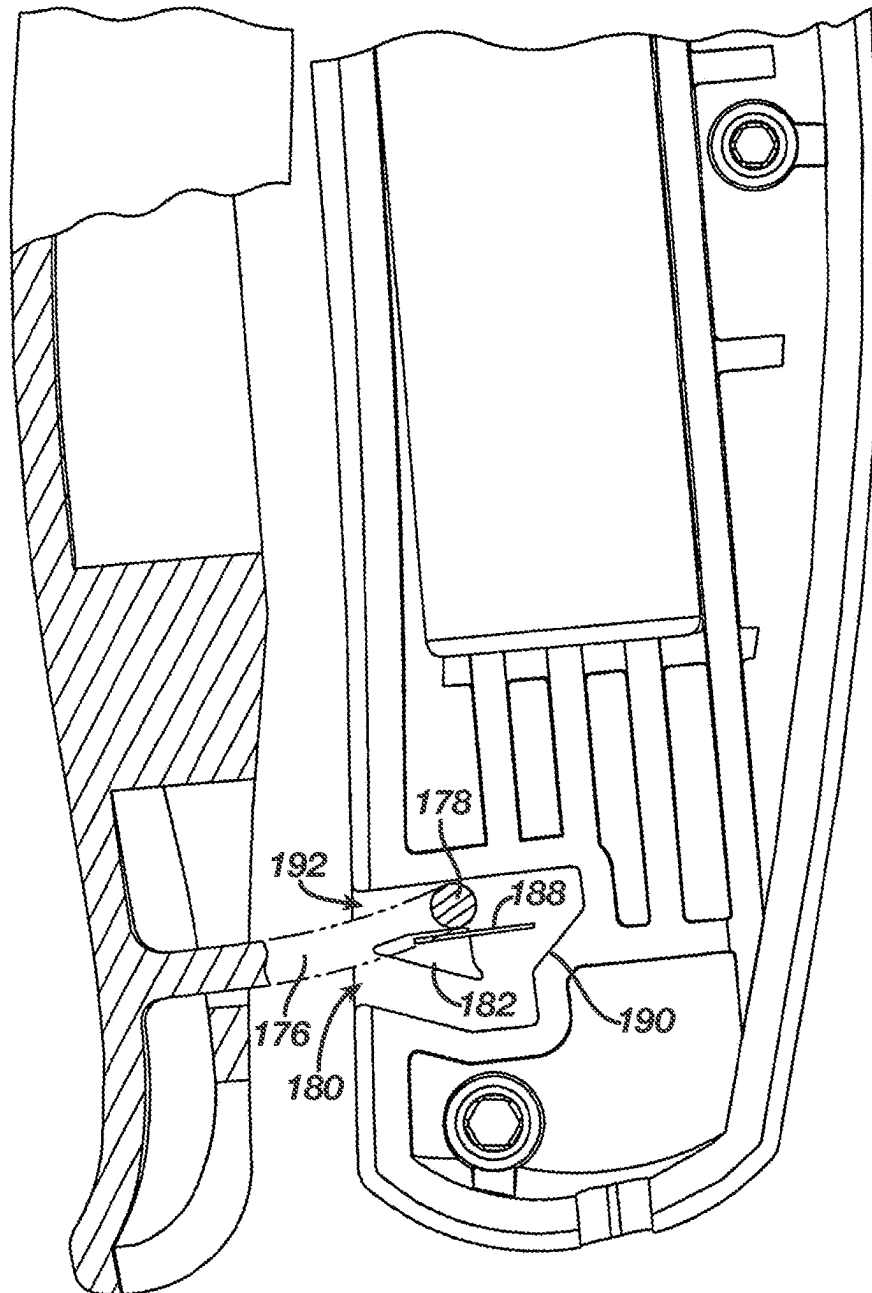
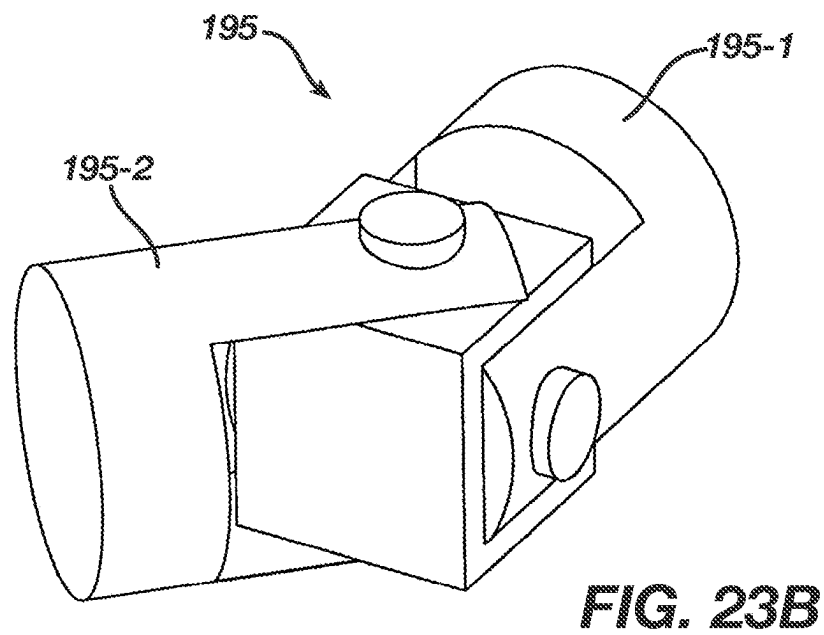
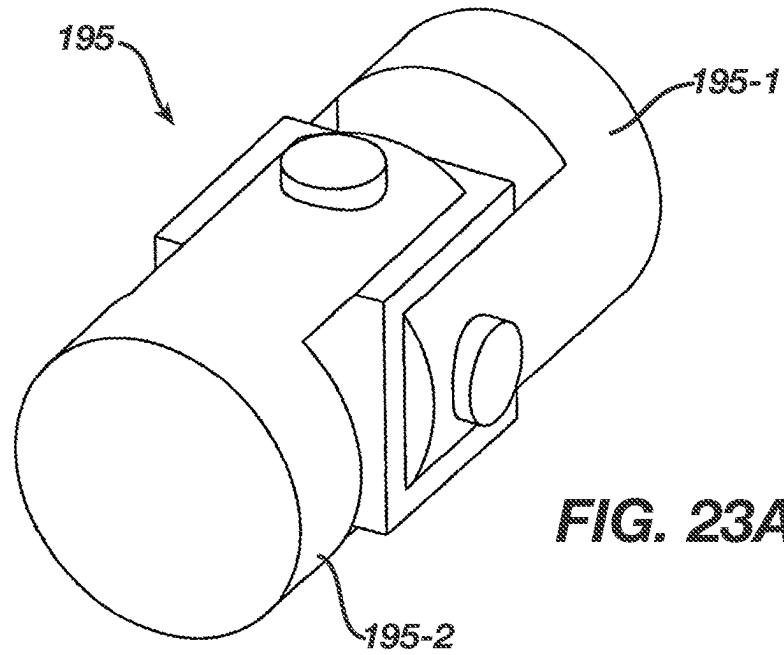


FIG. 22



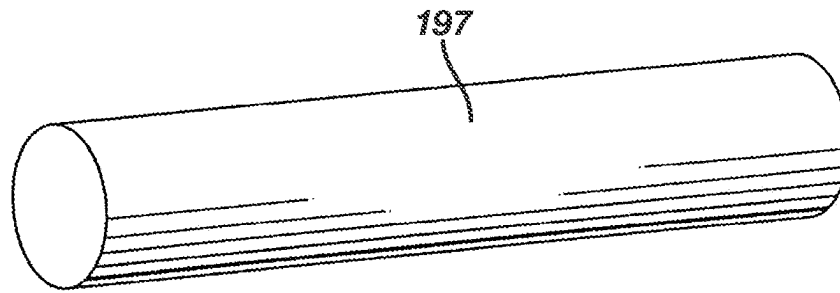


FIG. 24A

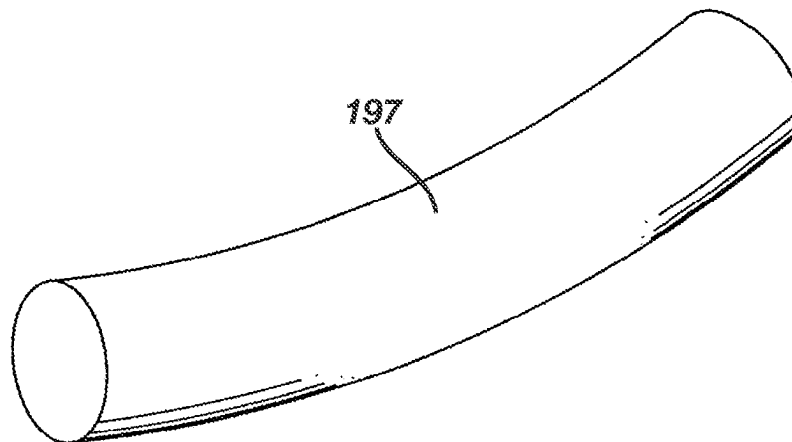
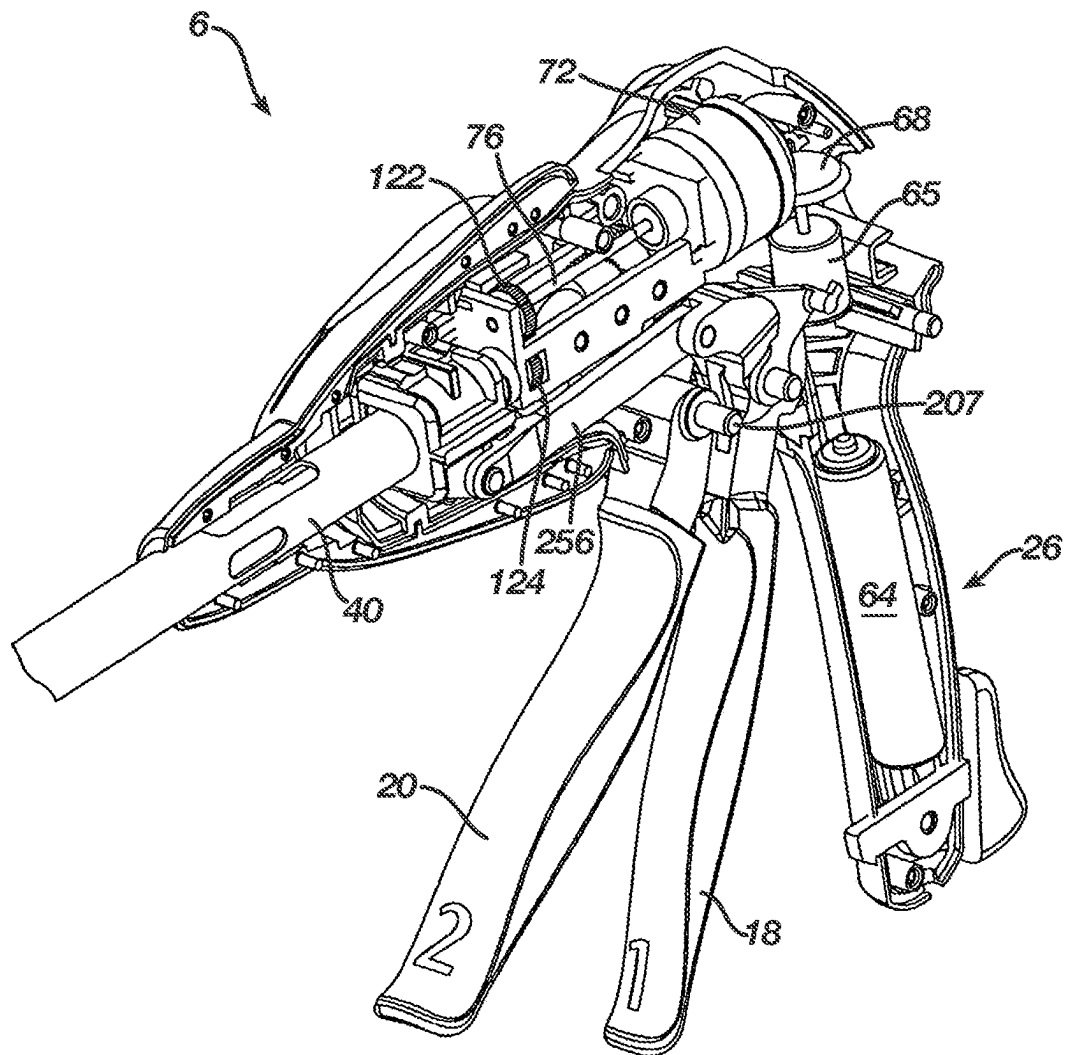


FIG. 24B

**FIG. 25**

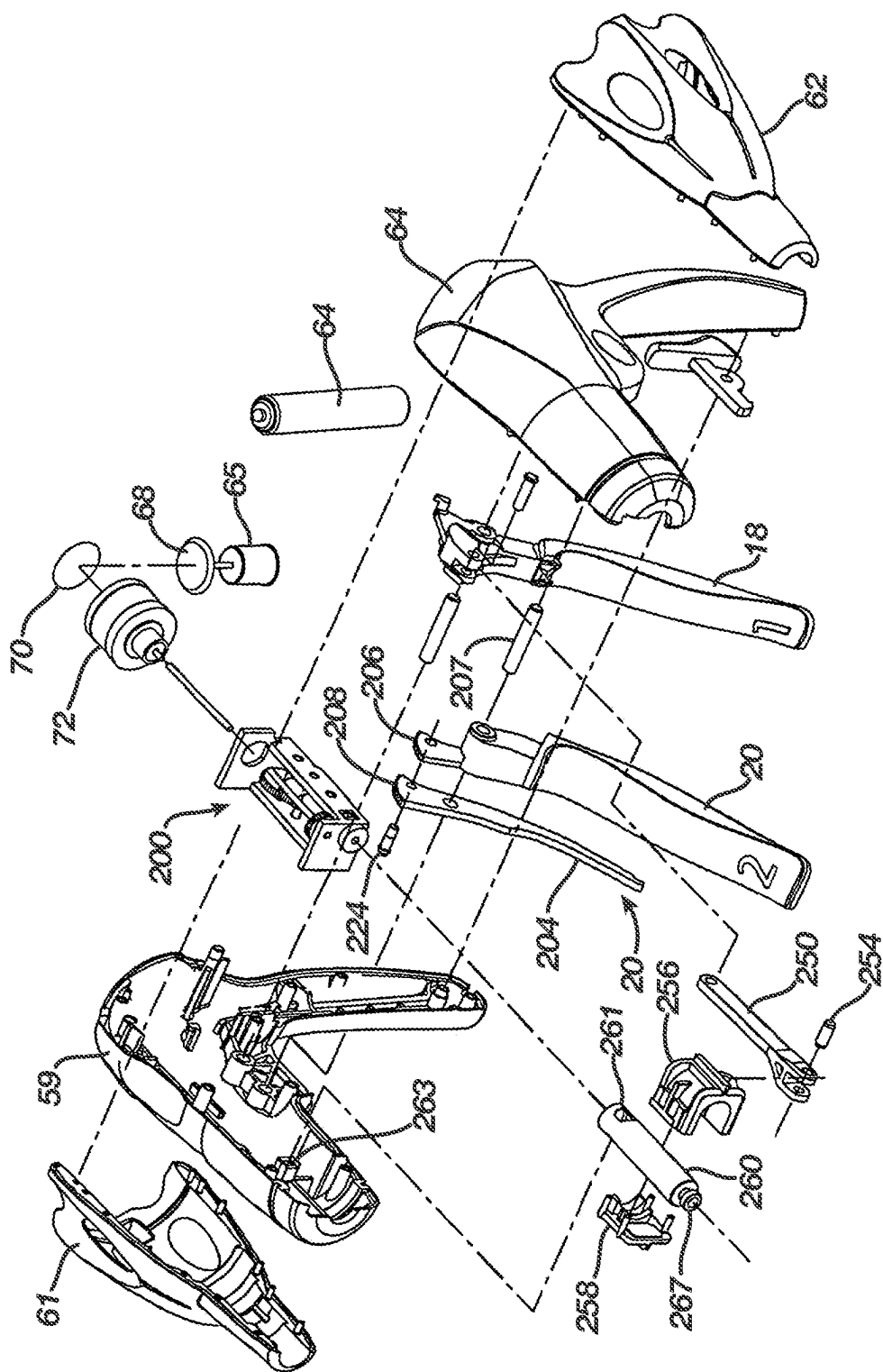
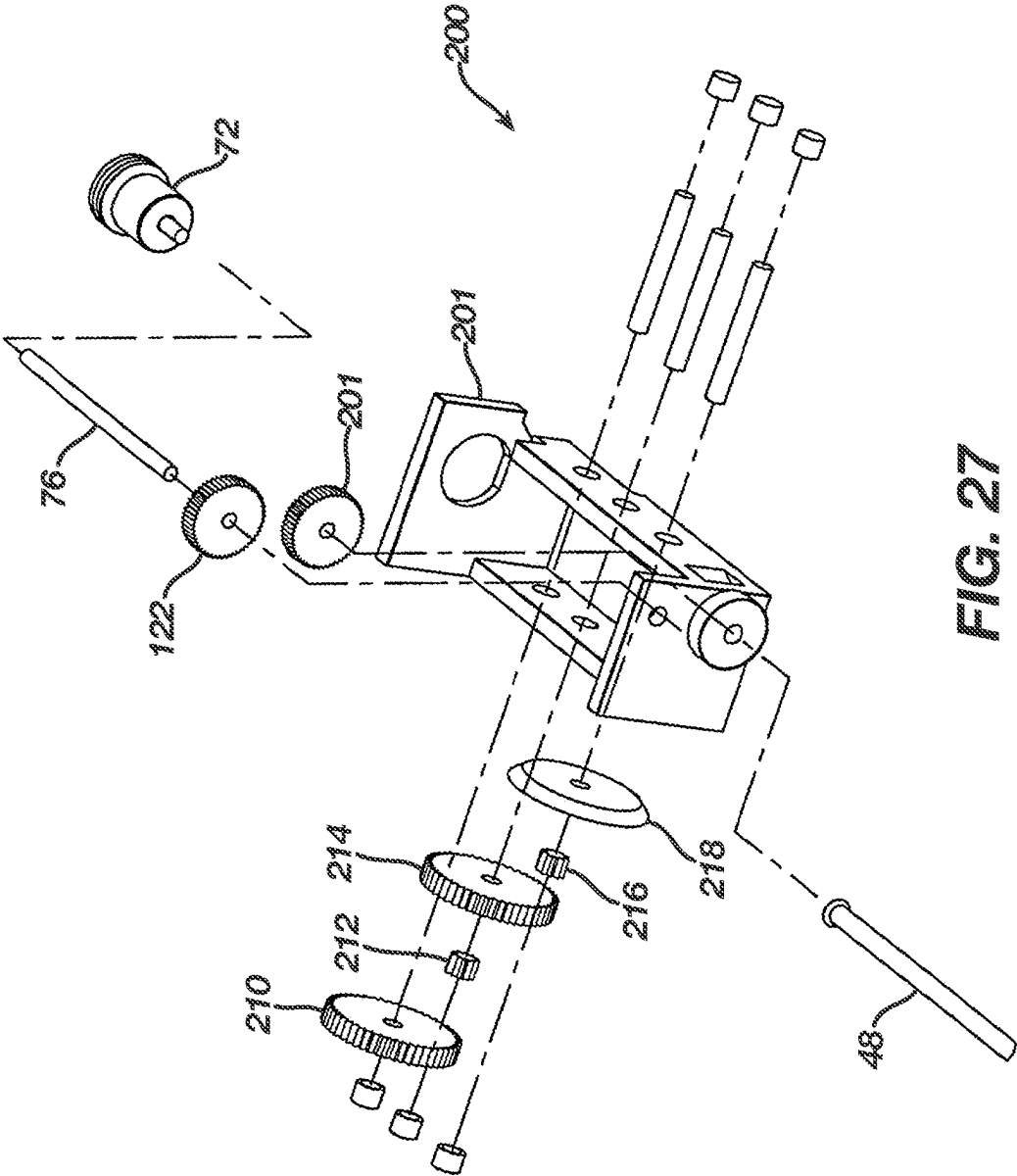


FIG. 26



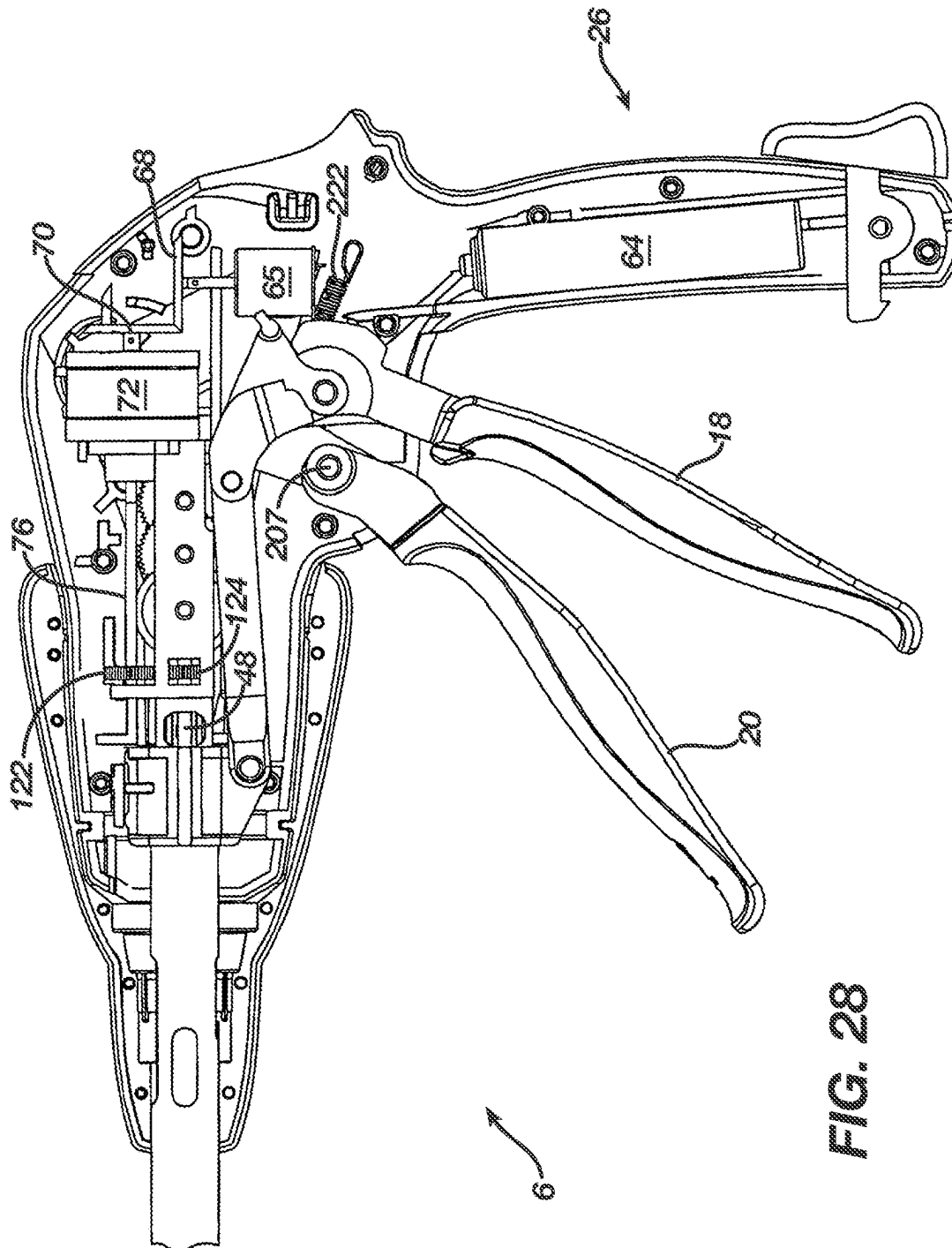
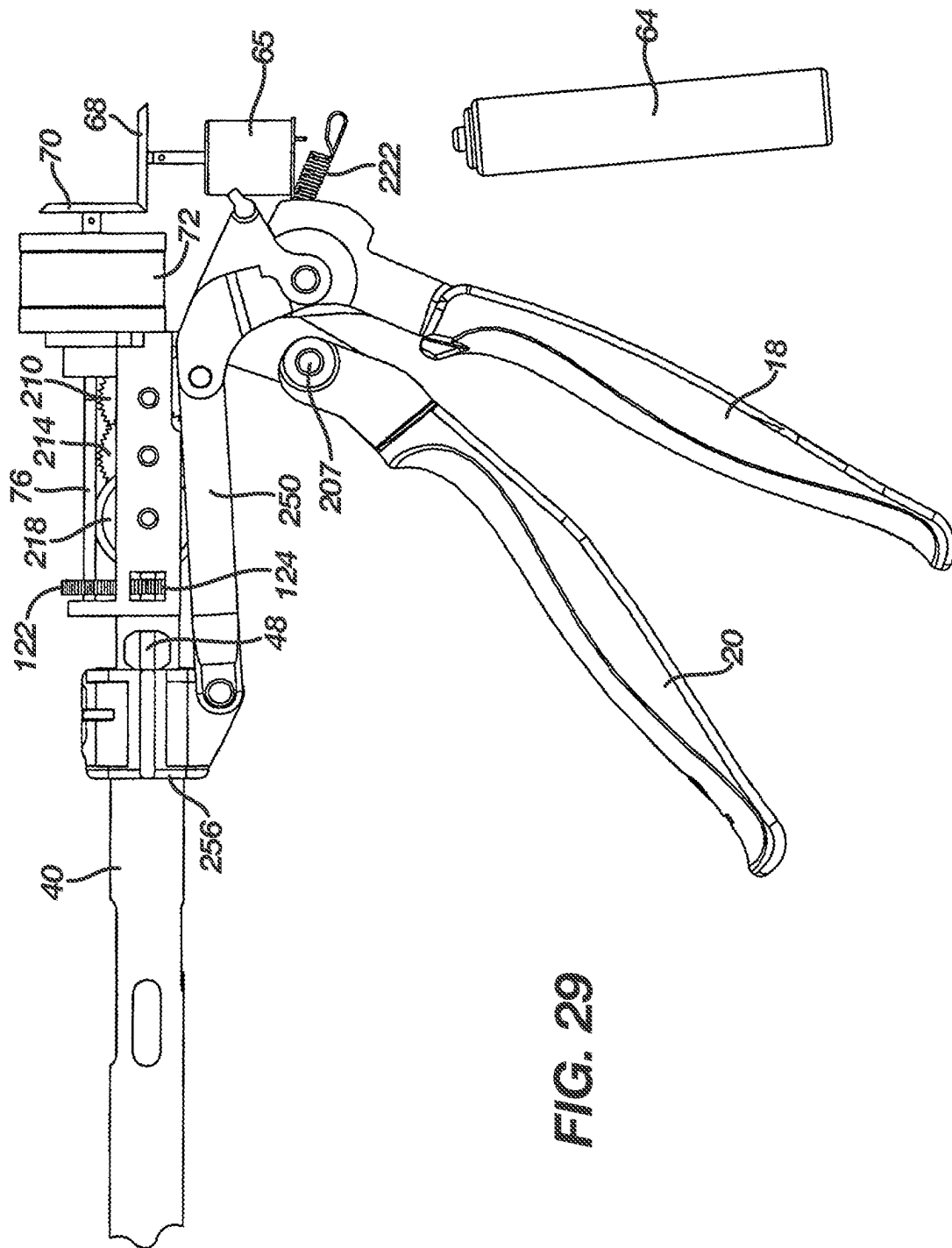


FIG. 28



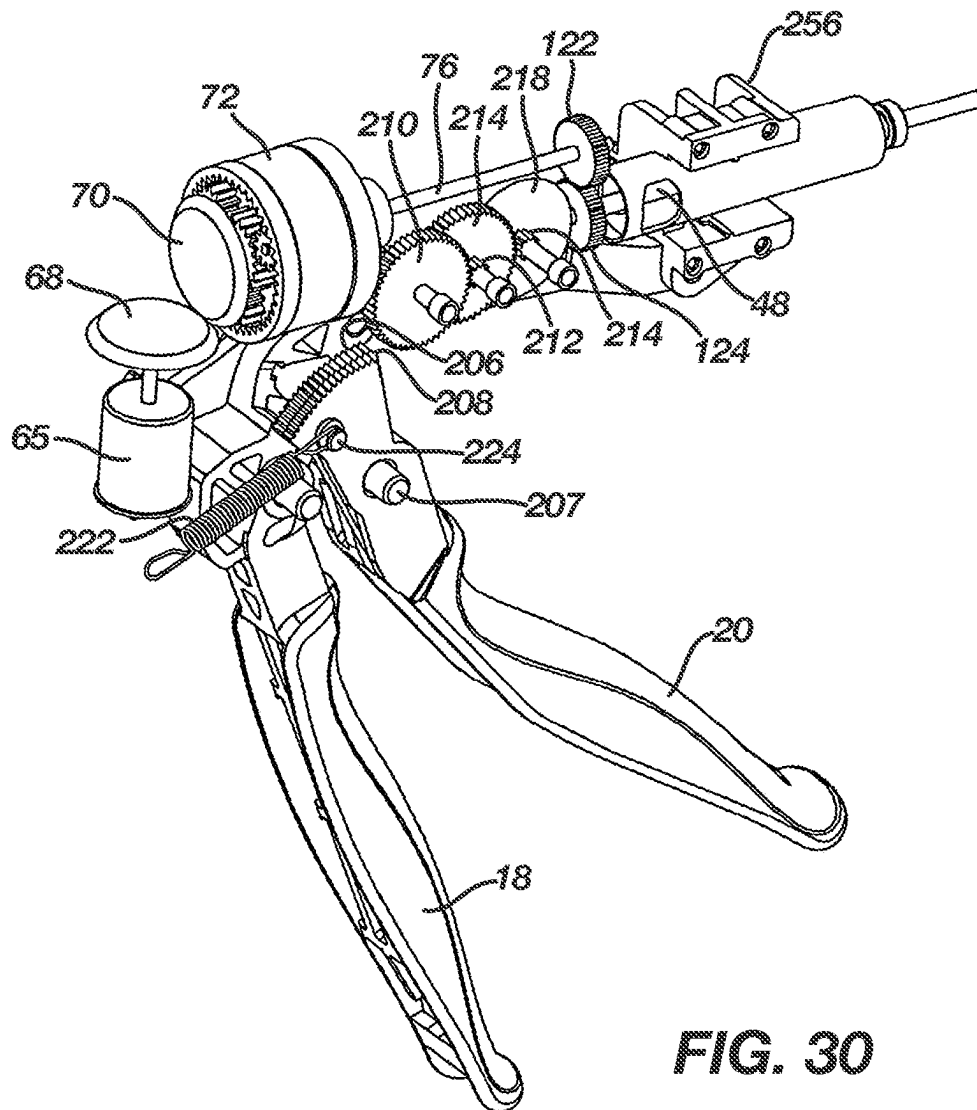
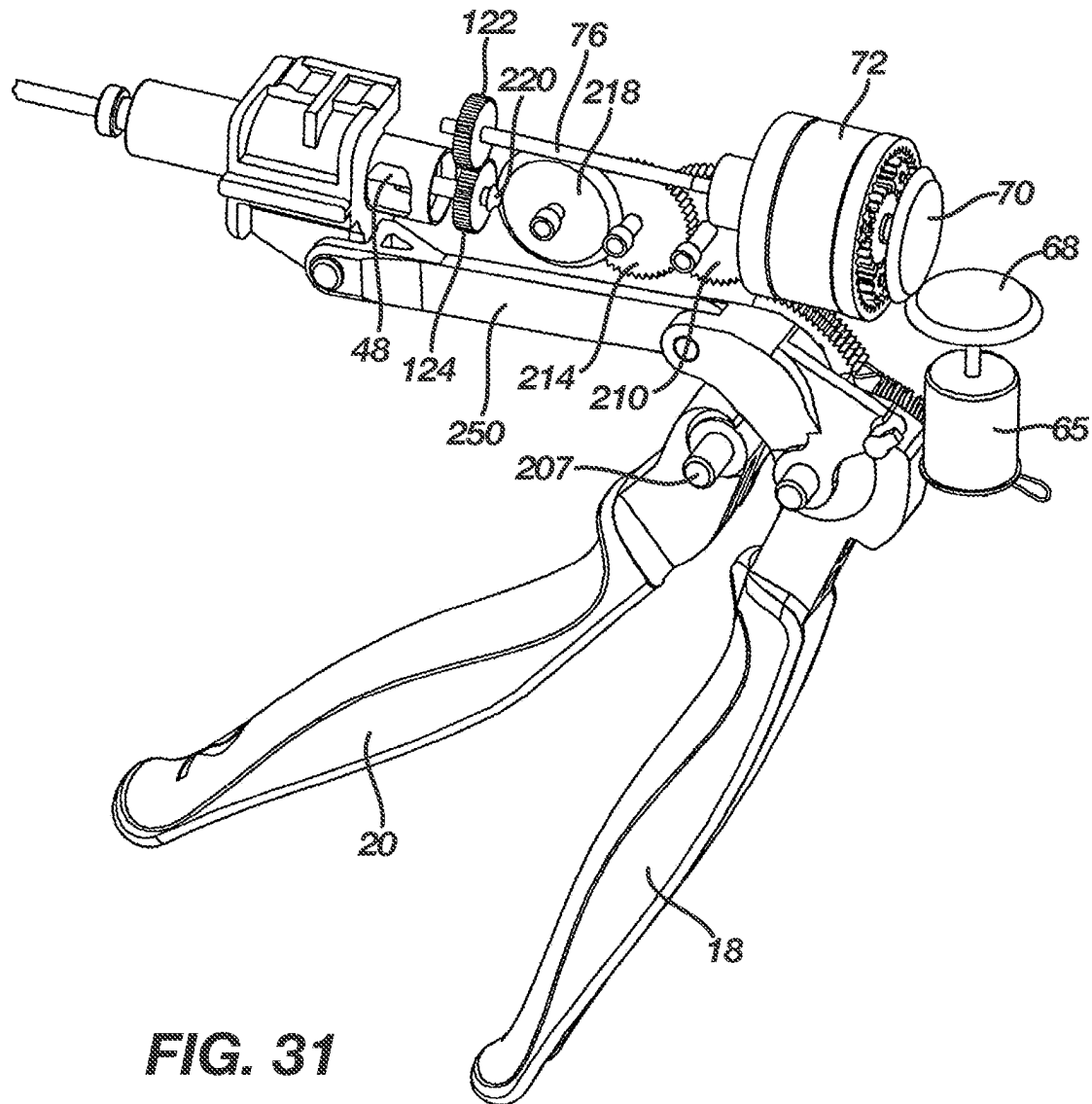


FIG. 30



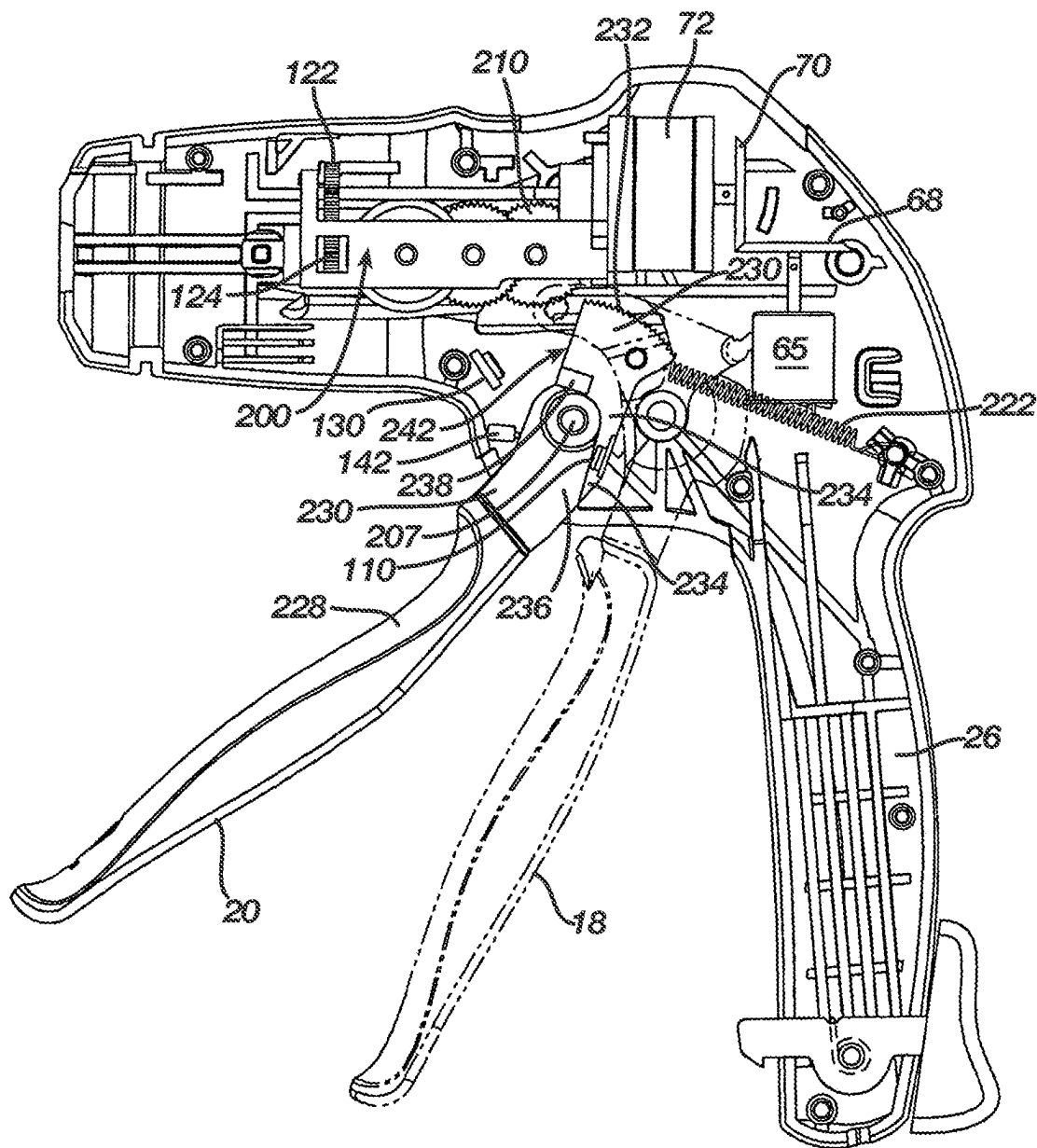


FIG. 32

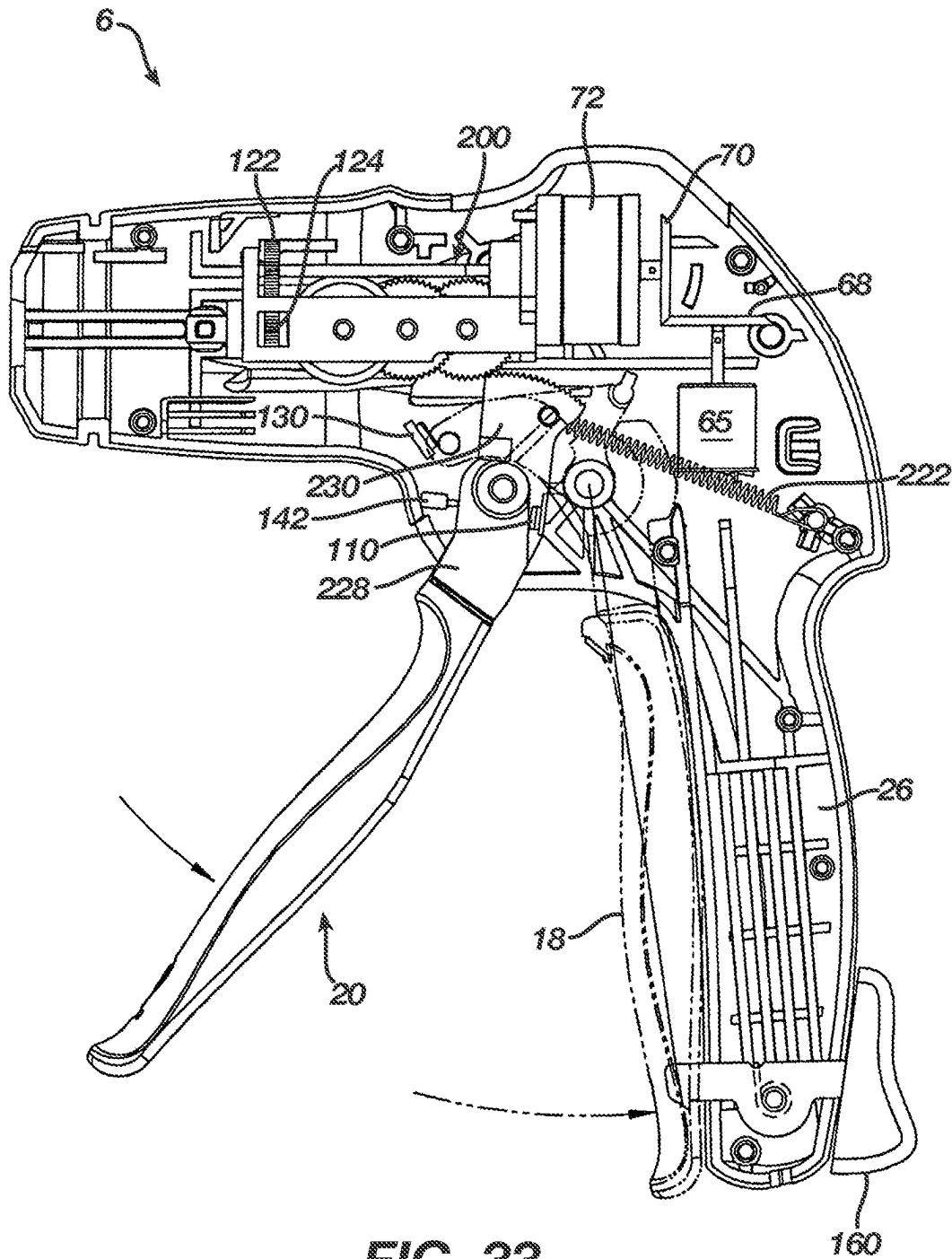


FIG. 33

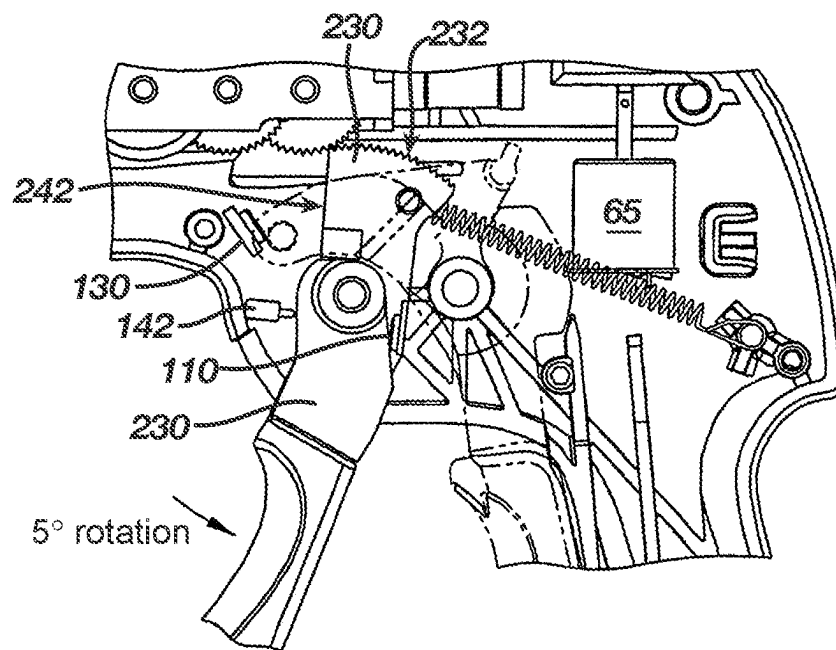


FIG. 34

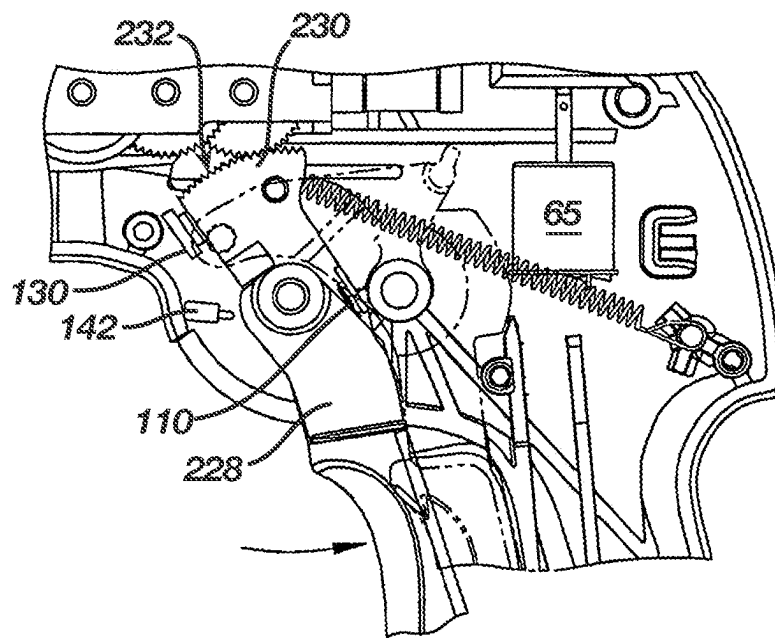
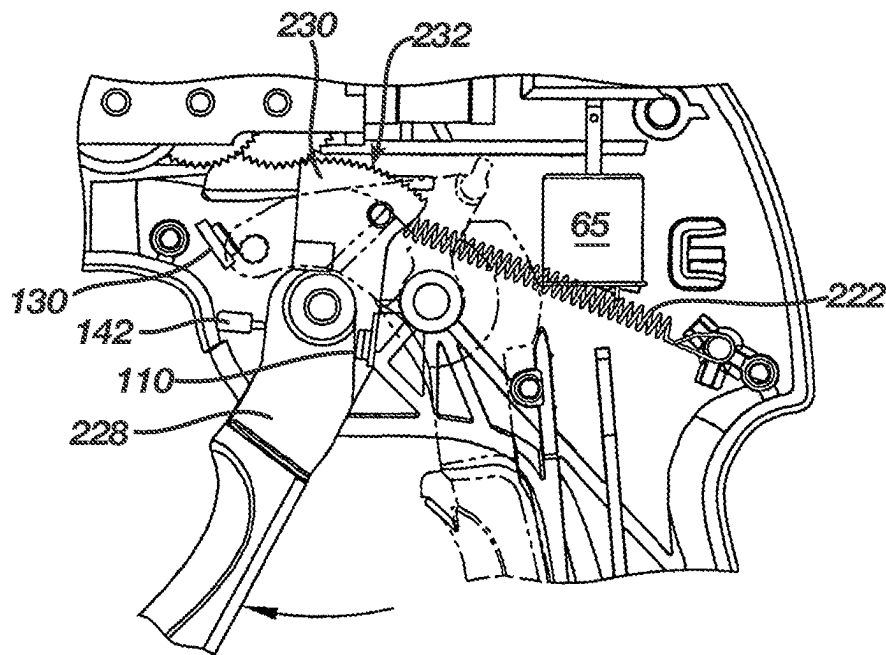
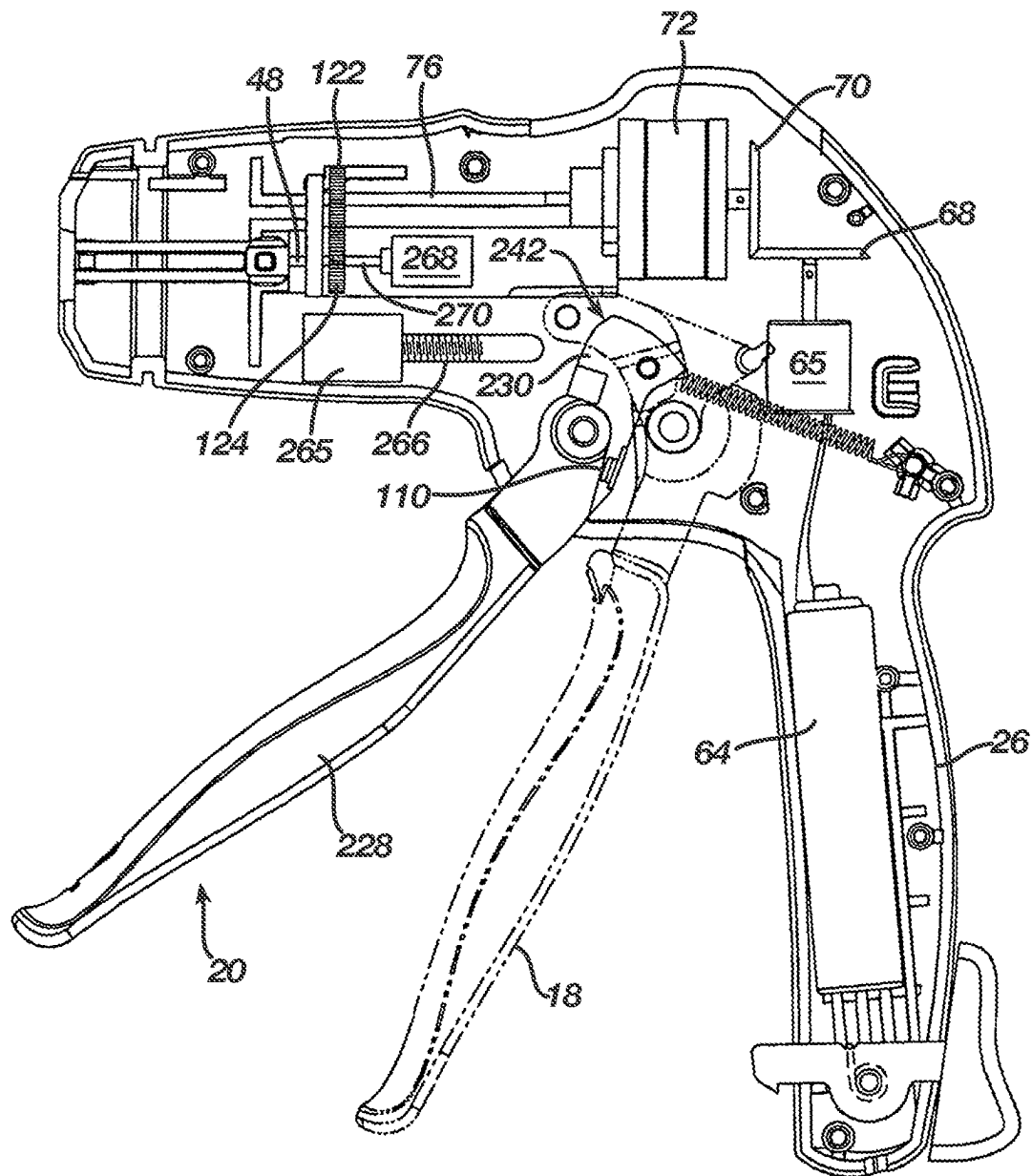
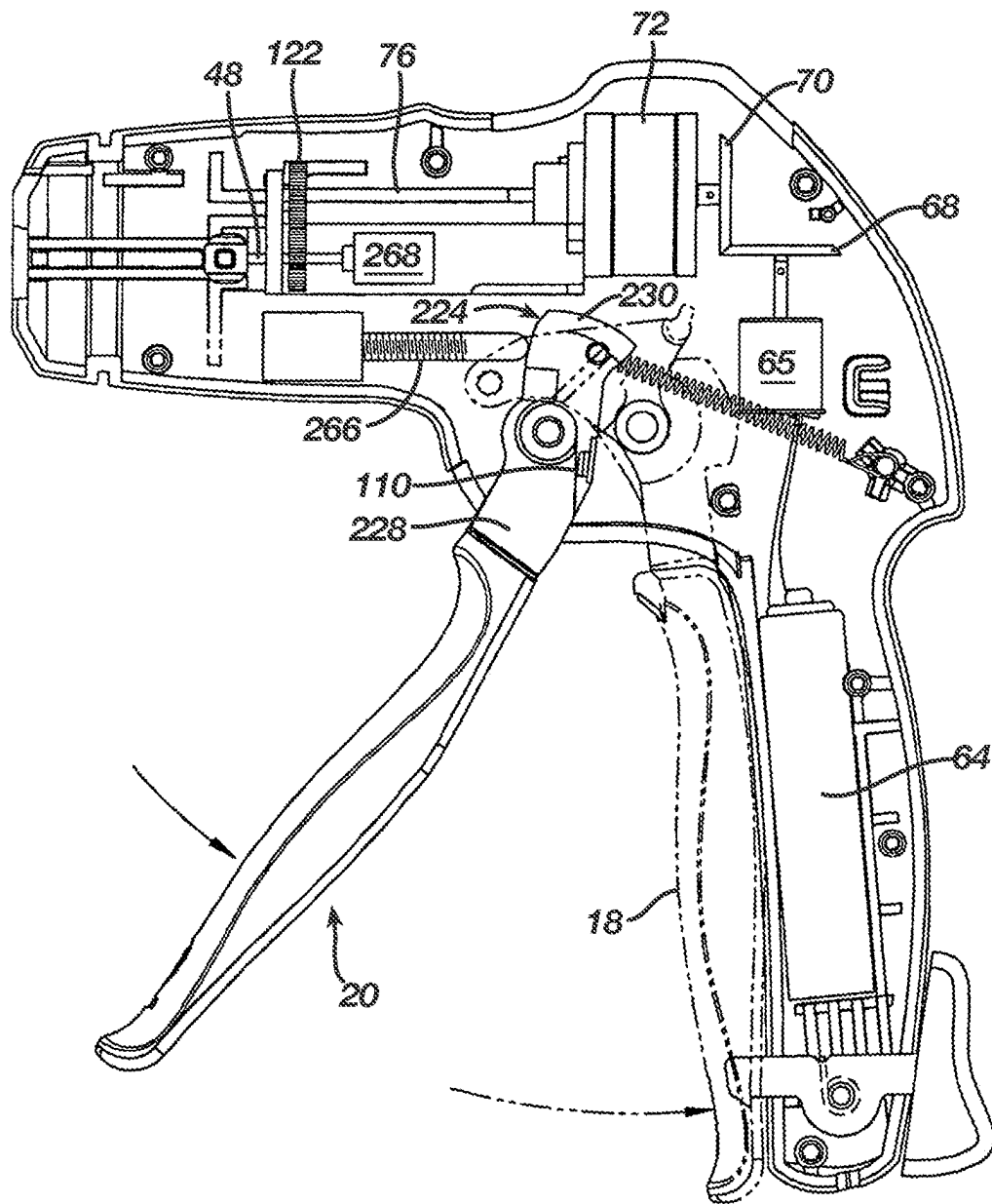


FIG. 35

**FIG. 36**

**FIG. 37**

**FIG. 38**

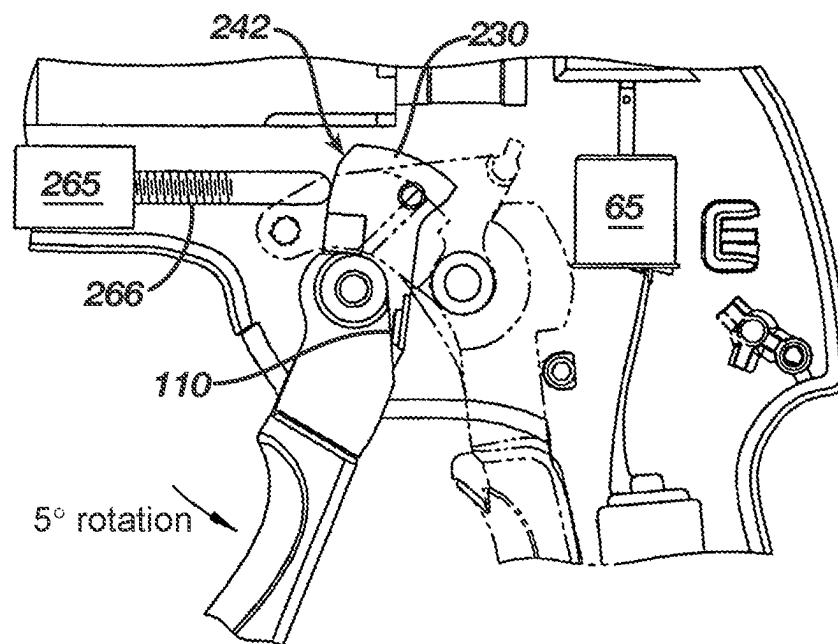


FIG. 39

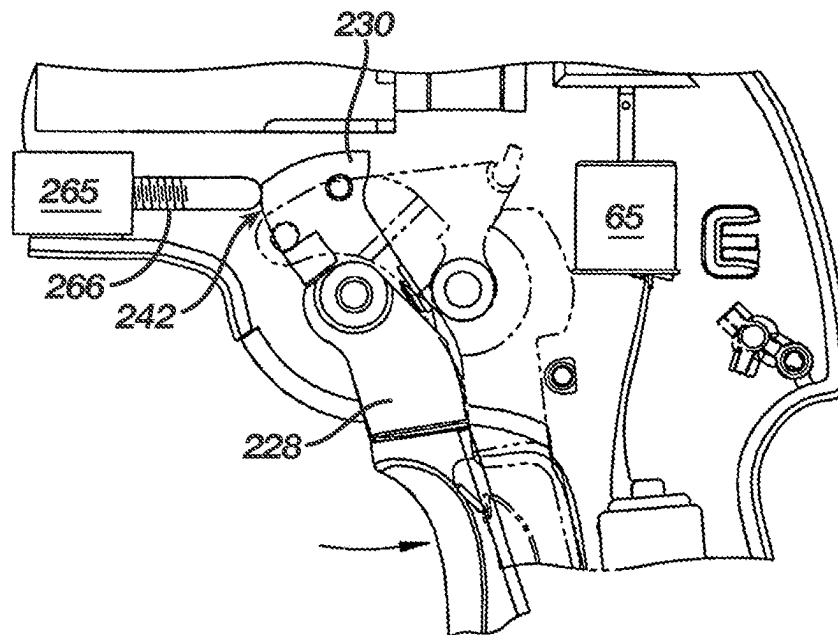
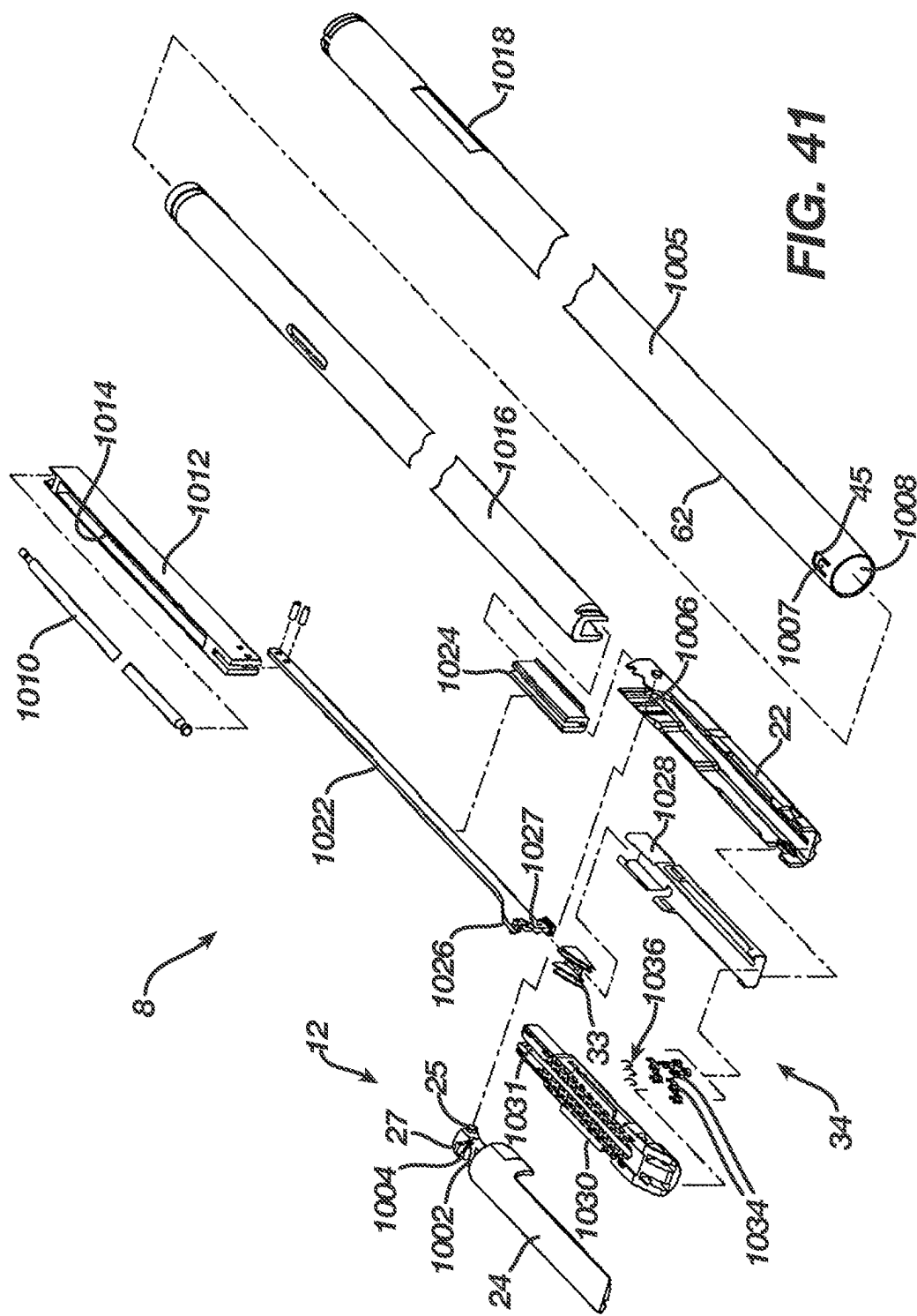


FIG. 40



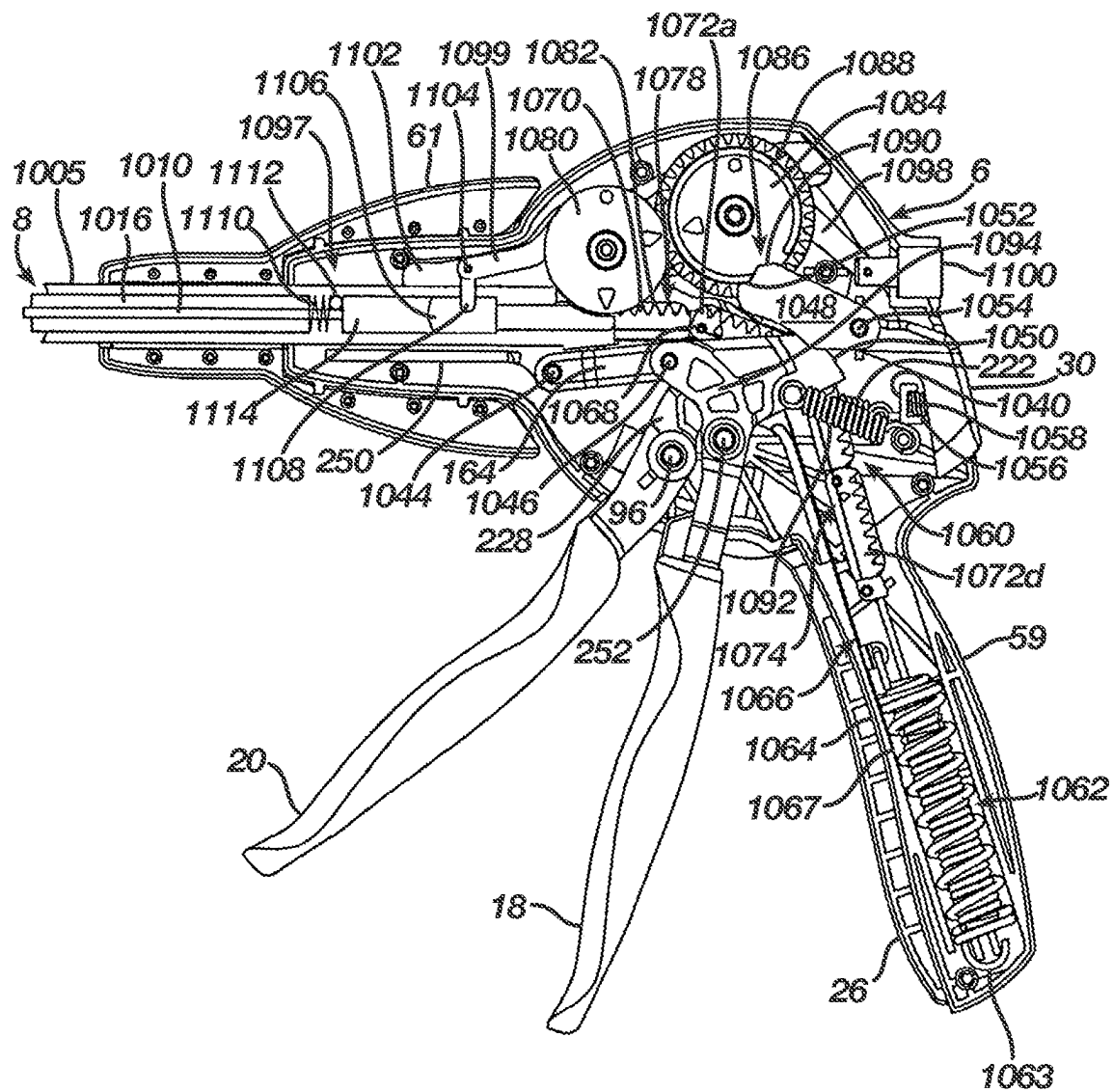
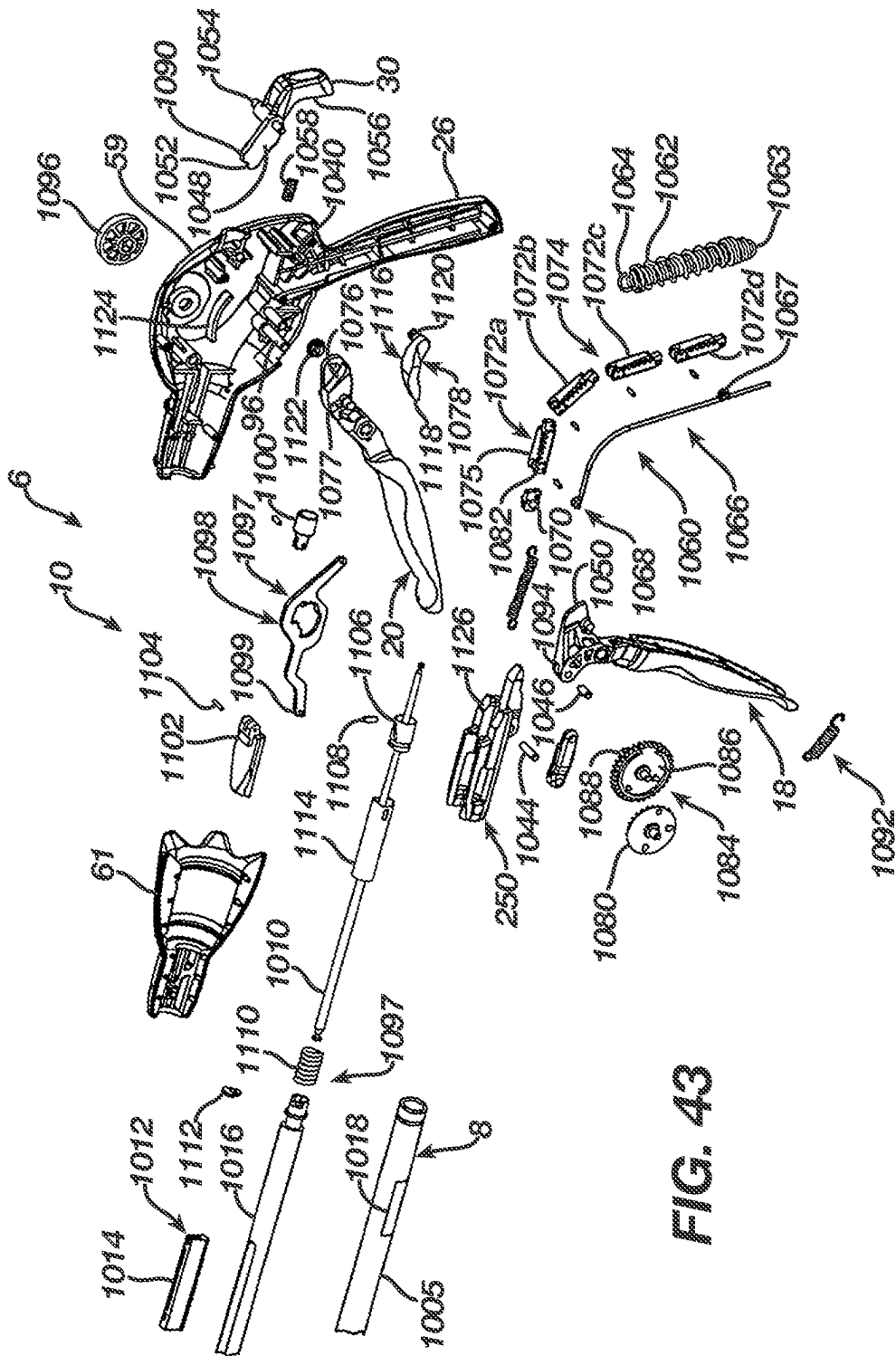


FIG. 42



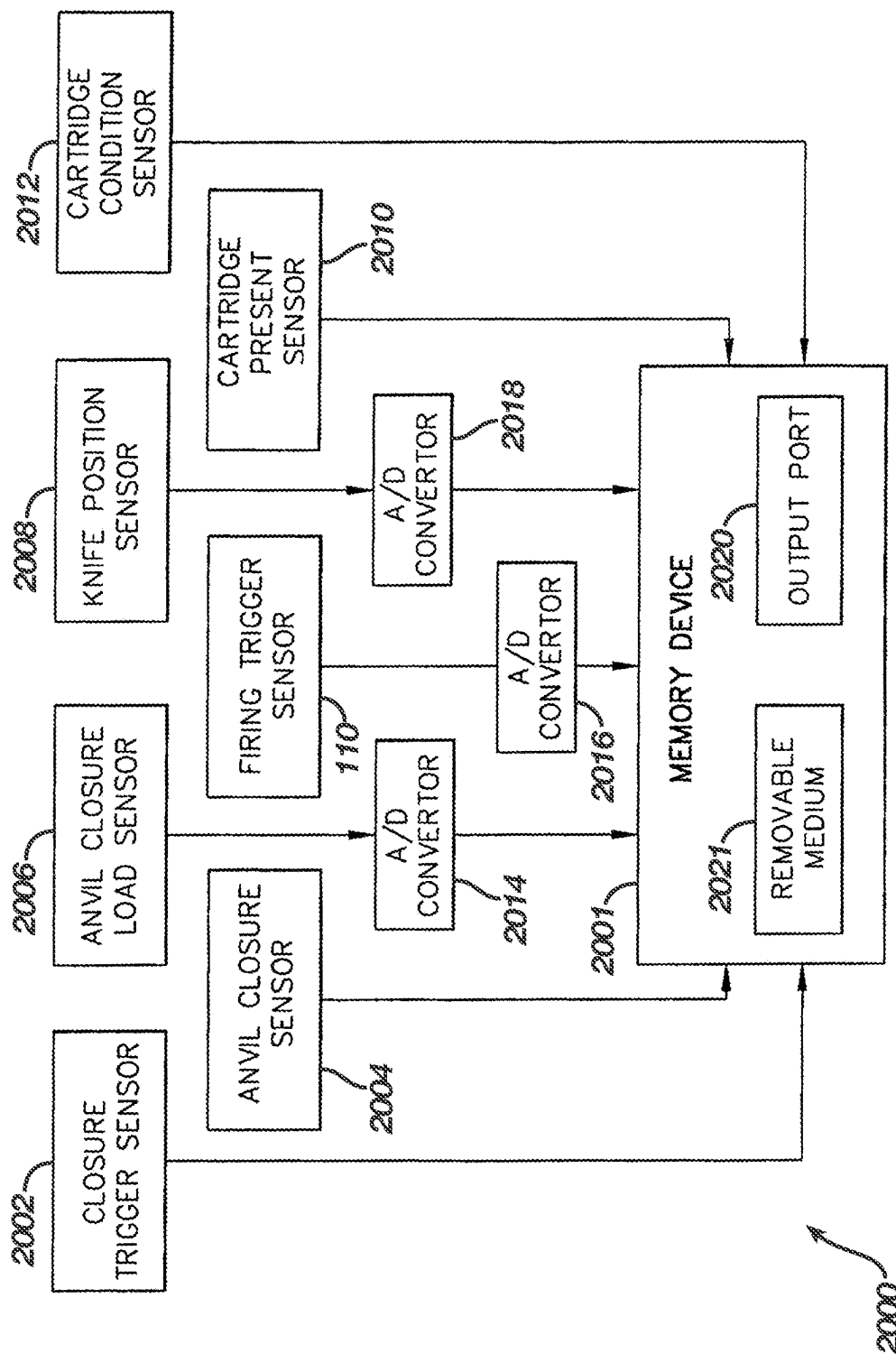


FIG. 44

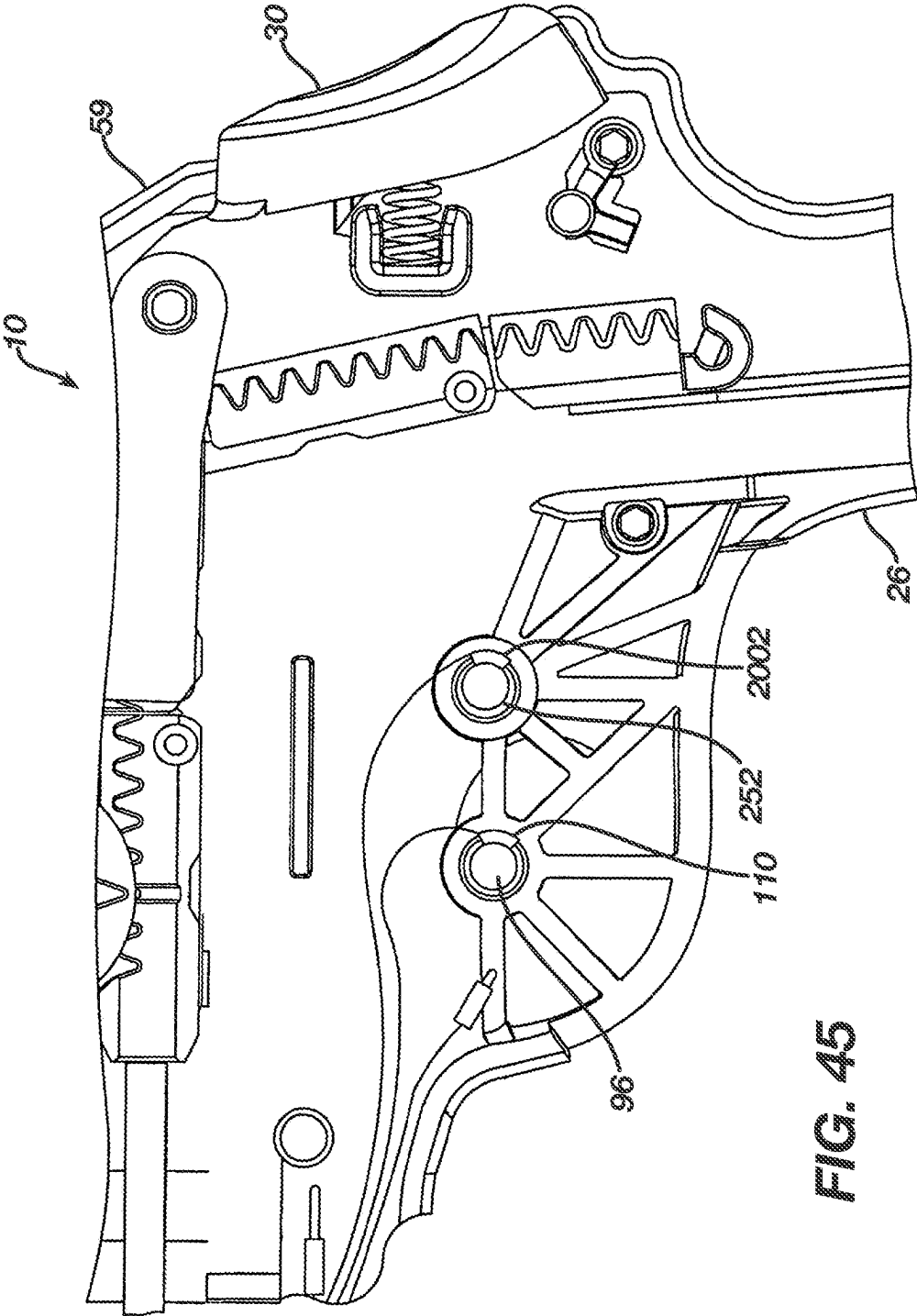


FIG. 45

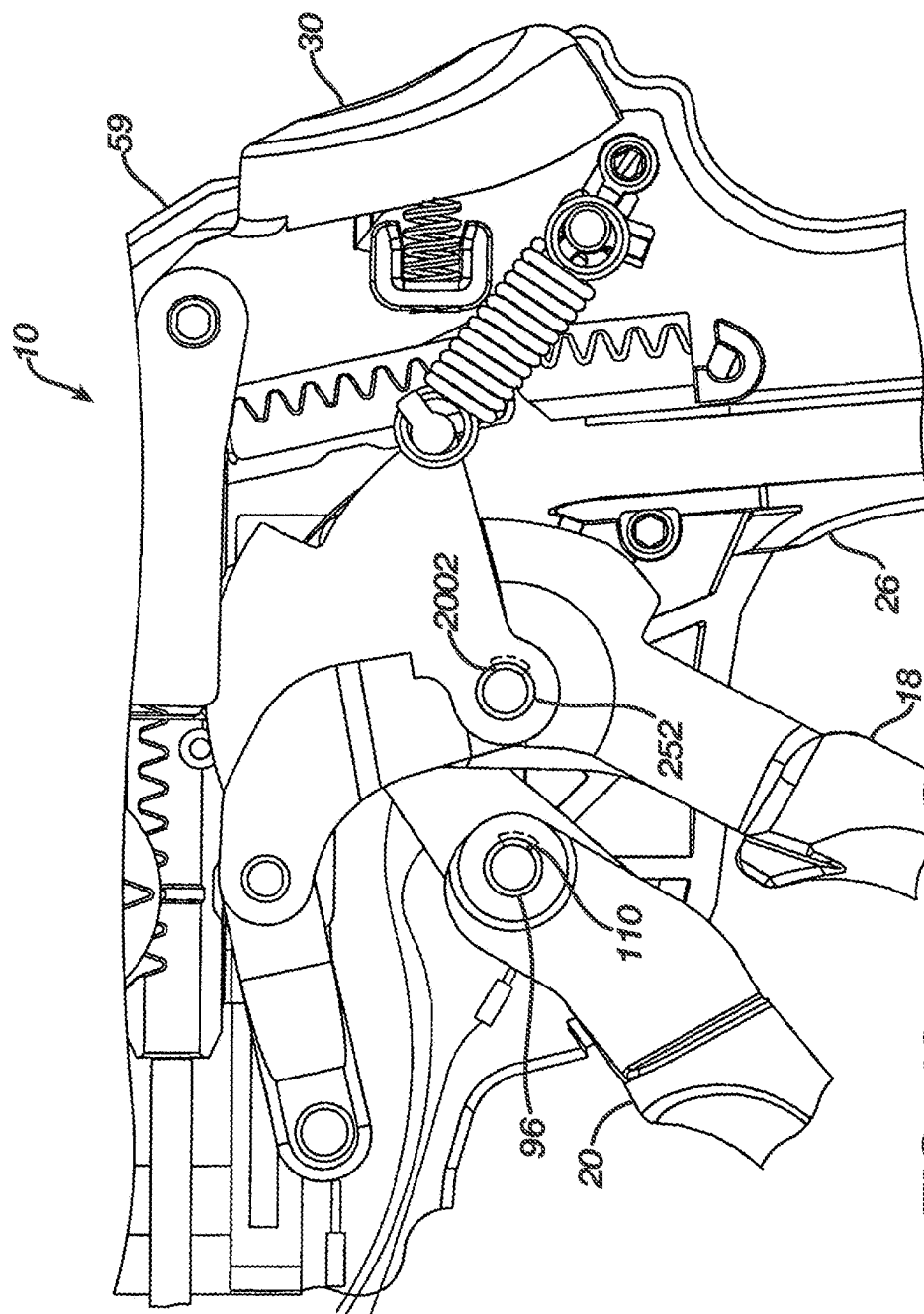
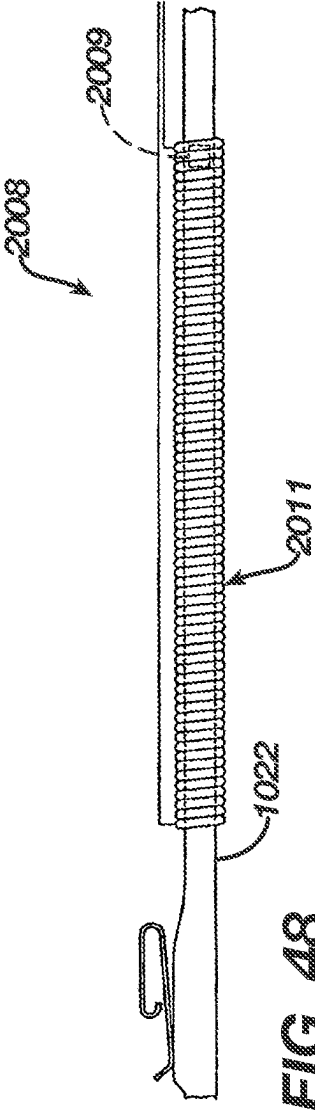
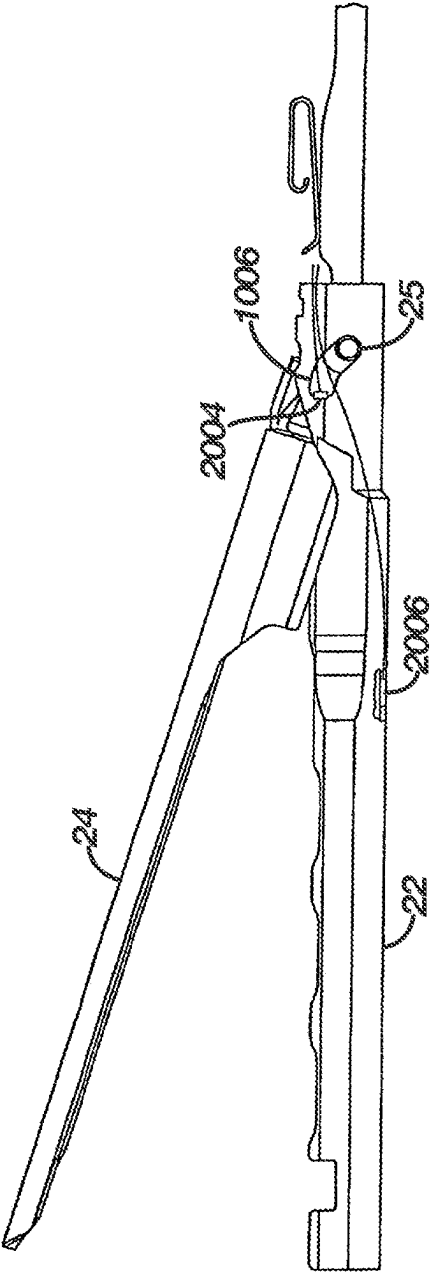


FIG. 46



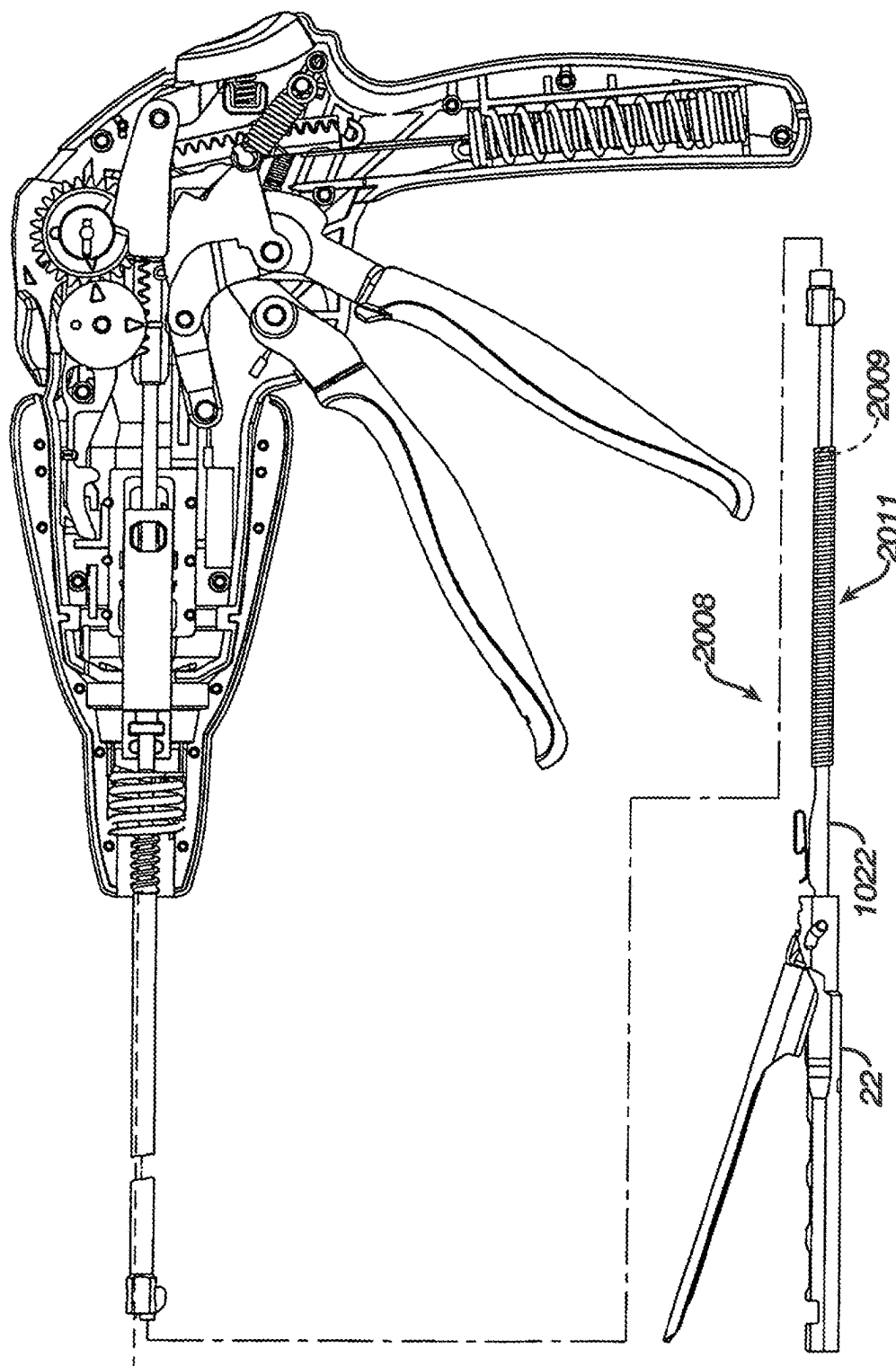


FIG. 49

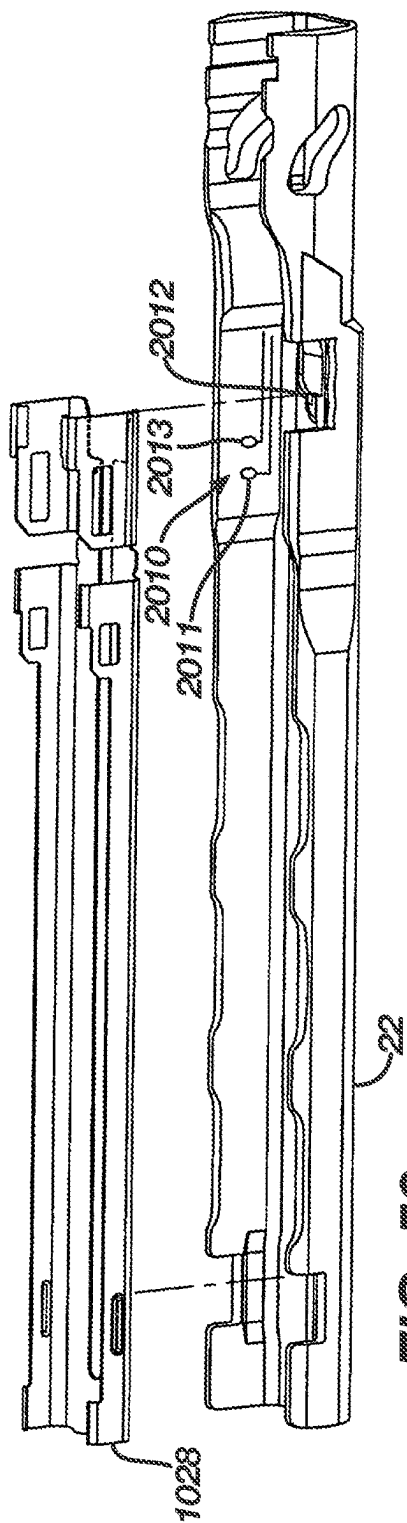


FIG. 50

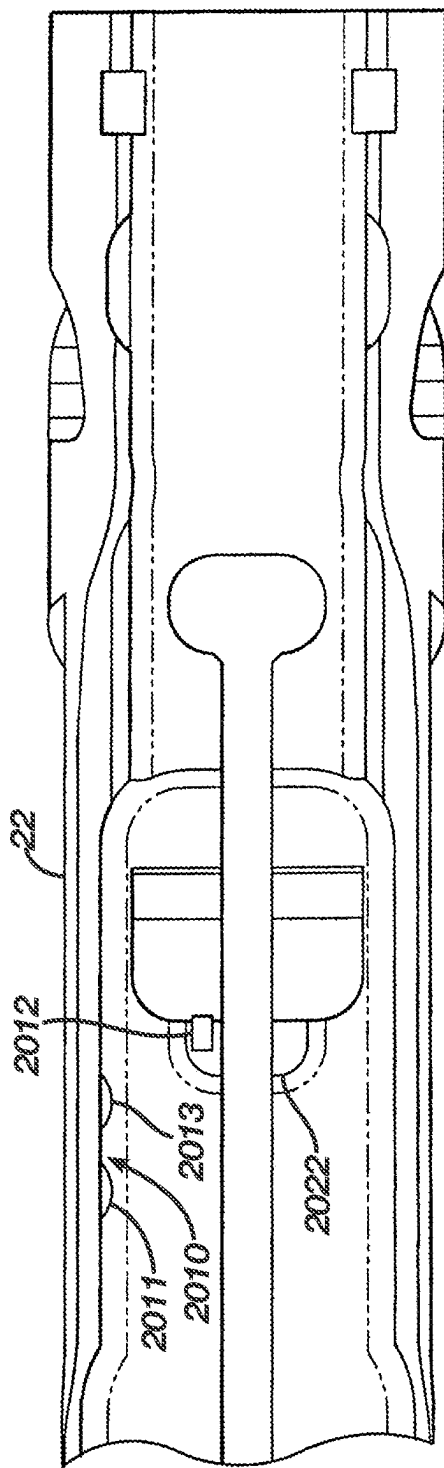
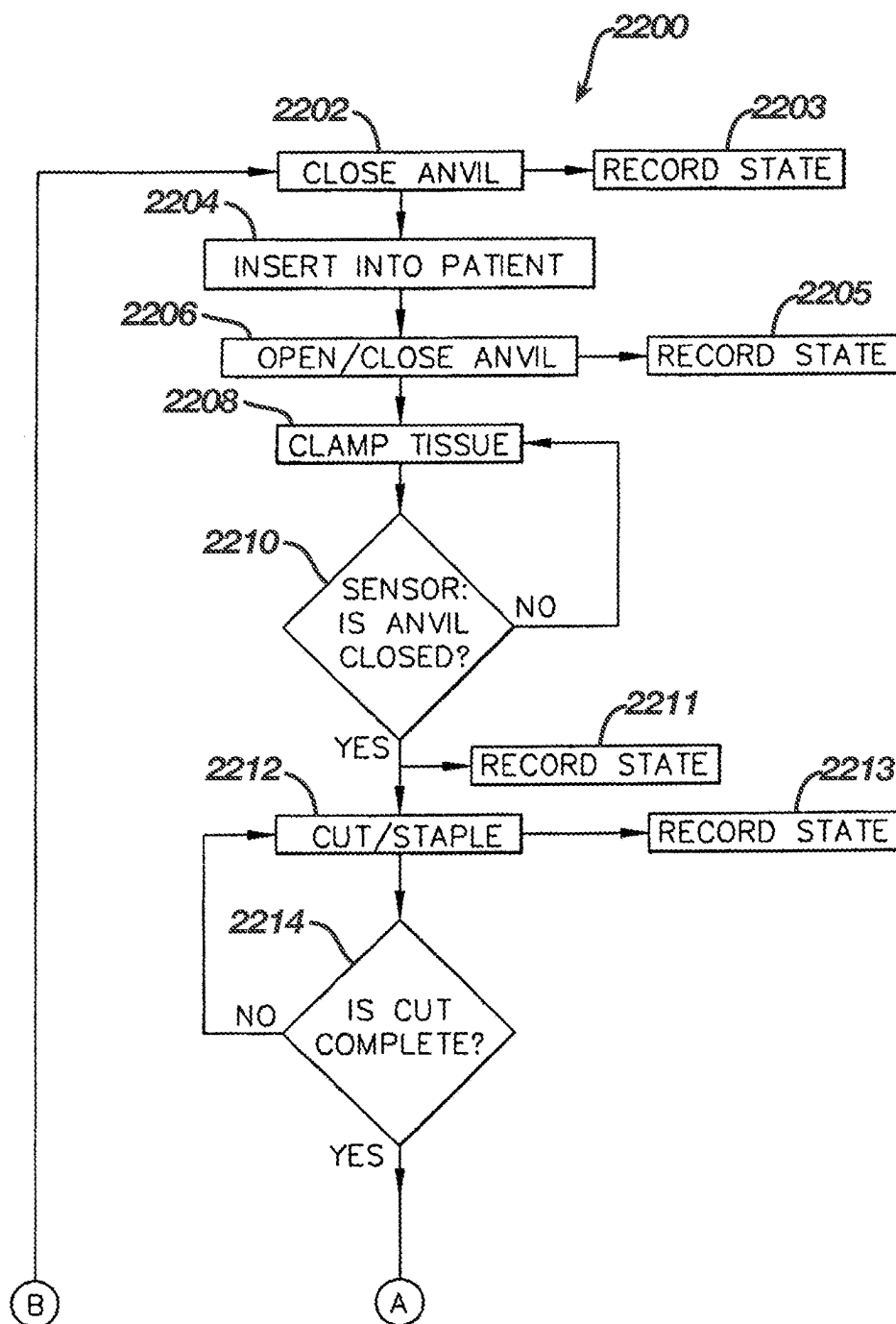
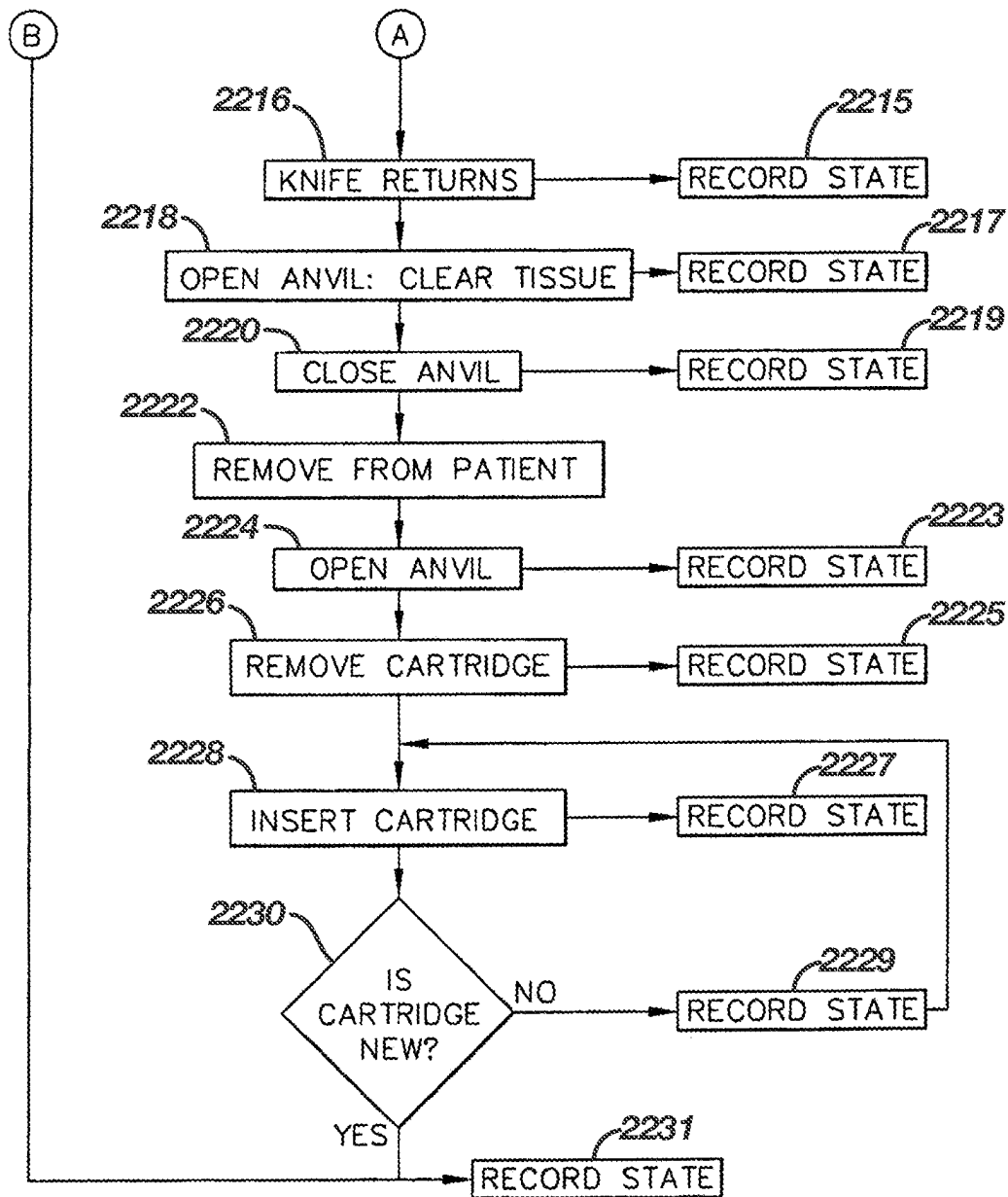


FIG. 51

**FIG. 52A**

**FIG. 52B**

2302				2304		2306		2308		2310		2312		2314		2316	
EVENT #				CLOSURE LOAD		FIRING STROKE		FIRING LOAD (MAX)		KNIFE POSITION %		LOCKOUT MEMORY		LOCKOUTS IS		CARTRIDGE PRESENT	
												ANVIL CLOSED/OPEN		SLED PRESENT YES/NO		YES - ON -1 NO - OFF -0	
1	10											0		1		1	
2	12											0		1		1	
3	15											0		1		1	
4	50											1		1		1	
EXAMPLE: 3 STEP FIRING																	
25	25	1		250	.33							1		0		1	
26	100	2		400	.66							1		0		1	
27	120	3		200	.75							1		0		1	
55	50											1		0		1	
56	50											1		0		1	

FIG. 53

FIG. 54

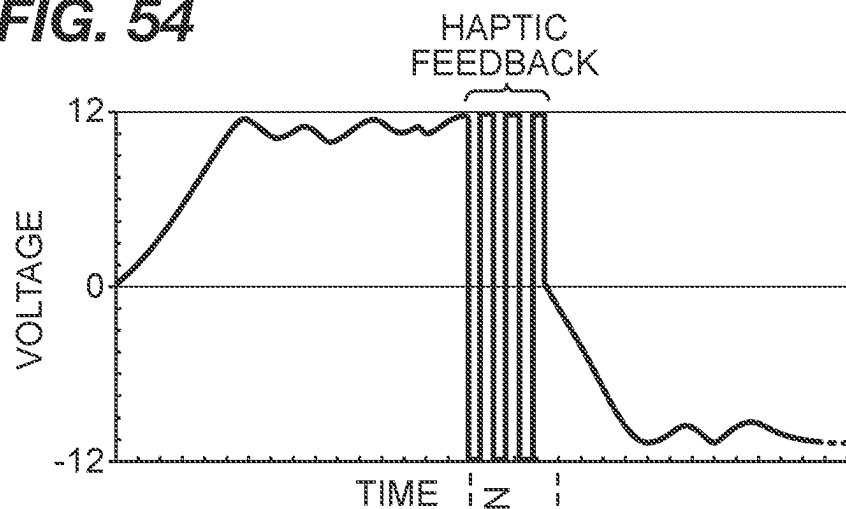
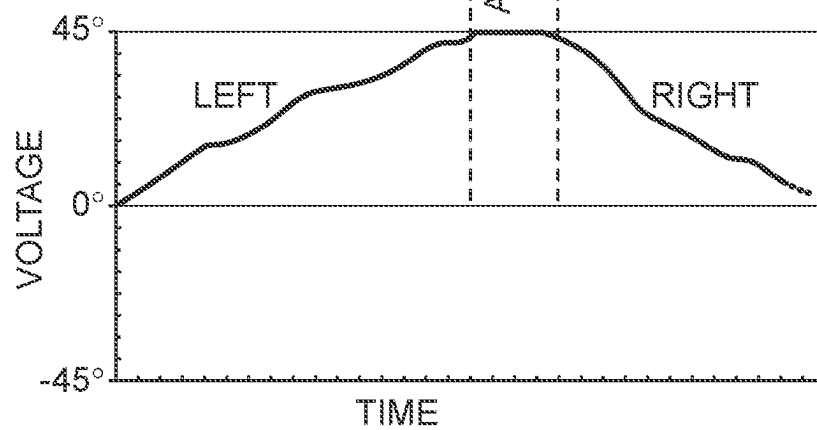
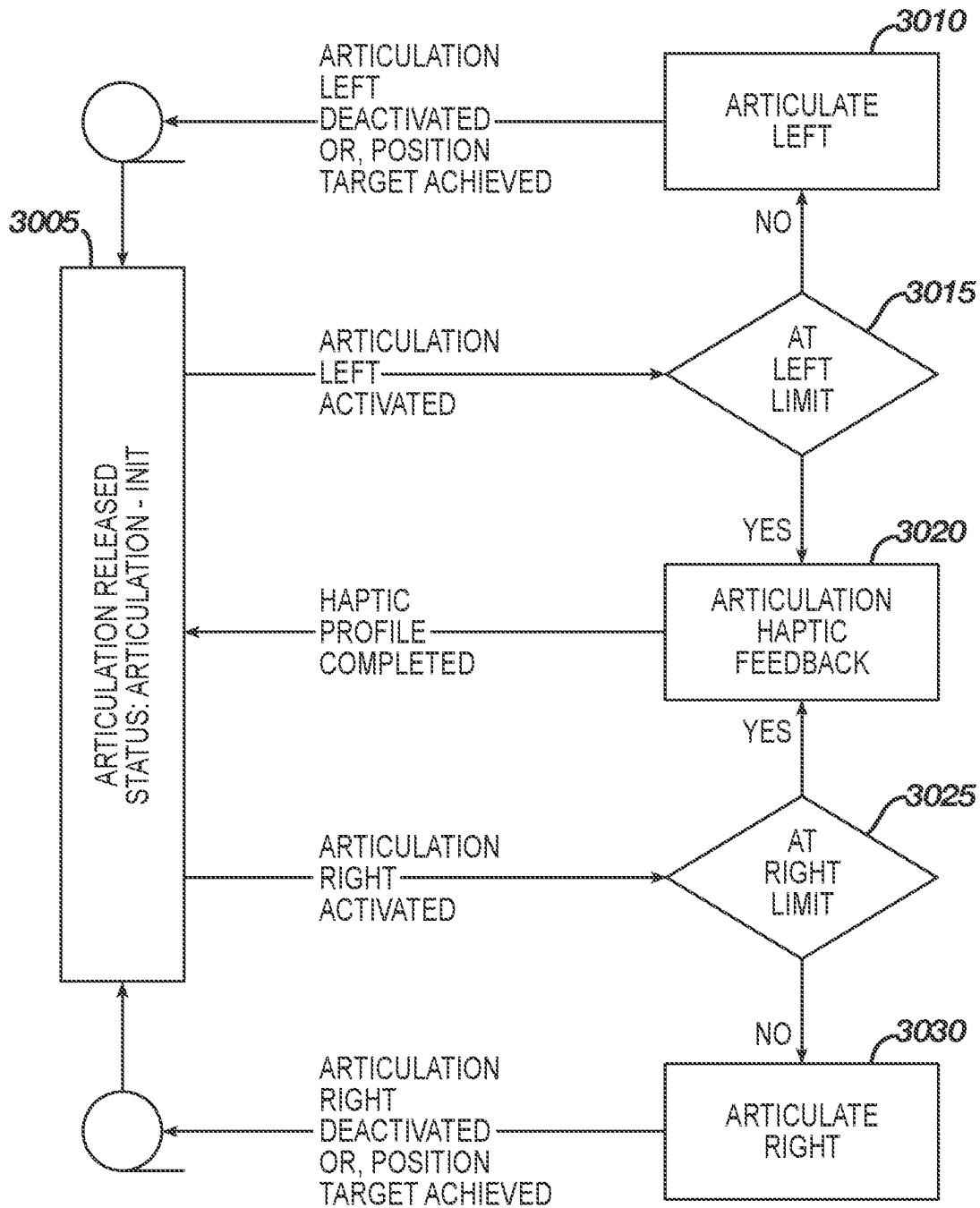


FIG. 55



**FIG. 56**

SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 18/117,130, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Mar. 3, 2023, which issued on Apr. 2, 2024 as U.S. Pat. No. 11,944,299, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/334,878, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed May 31, 2021, which issued on May 16, 2023 as U.S. Pat. No. 11,648,008, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/160,460, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Oct. 15, 2018, which issued on Jun. 1, 2021 as U.S. Pat. No. 11,020,113, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/223,791, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Jul. 29, 2016, which issued on Oct. 16, 2018 as U.S. Pat. No. 10,098,636, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/712,090, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Dec. 12, 2012, which issued on Sep. 13, 2016 as U.S. Pat. No. 9,439,649, the entire disclosures of which are hereby incorporated by reference herein. The entire disclosures of U.S. patent application Ser. No. 13/629,702, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Sep. 28, 2012, now U.S. Patent Application Publication No. 2013/0087597, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/424,648, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Mar. 20, 2012, which issued on Jun. 17, 2014 as U.S. Pat. No. 8,752,747, which is a divisional application claiming priority under 35 U.S.C. § 121 to U.S. patent application Ser. No. 12/949,099, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Nov. 18, 2010, which issued on May 1, 2012 as U.S. Pat. No. 8,167,185, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 11/343,803, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Jan. 31, 2006, which issued on Dec. 7, 2010 as U.S. Pat. No. 7,845,537, are hereby incorporated by reference herein.

The present application is related to the following U.S. patent applications filed on Jan. 31, 2006, which are incorporated herein by reference:

U.S. patent application Ser. No. 11/343,498, now U.S. Pat. No. 7,766,210, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH USER FEEDBACK SYSTEM;

U.S. patent application Ser. No. 11/343,573, now U.S. Pat. No. 7,416,101, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH LOADING FORCE FEEDBACK;

U.S. patent application Ser. No. 11/344,035, now U.S. Pat. No. 7,422,139, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH TACTILE POSITION FEEDBACK;

U.S. patent application Ser. No. 11/343,447, now U.S. Pat. No. 7,770,775, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH ADAPTIVE USER FEEDBACK;

U.S. patent application Ser. No. 11/343,562, now U.S. Pat. No. 7,568,603, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH ARTICULATABLE END EFFECTOR;

U.S. patent application Ser. No. 11/344,024, now U.S. Pat. No. 8,186,555, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH MECHANICAL CLOSURE SYSTEM;

U.S. patent application Ser. No. 11/343,321, now U.S. Patent Application Publication No. 2007/0175955, entitled SURGICAL CUTTING AND FASTENING INSTRUMENT WITH CLOSURE TRIGGER LOCKING MECHANISM;

U.S. patent application Ser. No. 11/343,563, now U.S. Patent Application Publication No. 2007/0175951, entitled GEARING SELECTOR FOR A POWERED SURGICAL CUTTING AND FASTENING STAPLING INSTRUMENT;

U.S. patent application Ser. No. 11/344,020, now U.S. Pat. No. 7,464,846, entitled SURGICAL INSTRUMENT HAVING A REMOVABLE BATTERY;

U.S. patent application Ser. No. 11/343,439, now U.S. Pat. No. 7,644,848, entitled ELECTRONIC LOCK-OUTS AND SURGICAL INSTRUMENT INCLUDING SAME;

U.S. patent application Ser. No. 11/343,547, now U.S. Pat. No. 7,753,904, entitled ENDOSCOPIC SURGICAL INSTRUMENT WITH A HANDLE THAT CAN ARTICULATE WITH RESPECT TO THE SHAFT;

U.S. patent application Ser. No. 11/344,021, now U.S. Pat. No. 7,464,849, entitled ELECTRO-MECHANICAL SURGICAL CUTTING AND FASTENING INSTRUMENT HAVING A ROTARY FIRING AND CLOSURE SYSTEM WITH PARALLEL CLOSURE AND ANVIL ALIGNMENT COMPONENTS;

U.S. patent application Ser. No. 11/343,546, now U.S. Patent Application Publication No. 2007/0175950, entitled DISPOSABLE STAPLE CARTRIDGE HAVING AN ANVIL WITH TISSUE LOCATOR FOR USE WITH A SURGICAL CUTTING AND FASTENING INSTRUMENT AND MODULAR END EFFECTOR SYSTEM THEREFOR; and

U.S. patent application Ser. No. 11/343,545, now U.S. Pat. No. 8,708,213, entitled SURGICAL INSTRUMENT HAVING A FEEDBACK SYSTEM.

BACKGROUND

The present invention relates in general to surgical instruments, and more particularly to minimally invasive surgical instruments capable of delivering a force feedback to the user.

Endoscopic surgical instruments are often preferred over traditional open surgical devices because a smaller incision tends to reduce the post-operative recovery time and complications. Consequently, significant development has gone into a range of endoscopic surgical instruments that are suitable for precise placement of a distal end effector at a desired surgical site through a cannula of a trocar. These distal end effectors engage the tissue in a number of ways to achieve a diagnostic or therapeutic effect (e.g., endocutter,

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grasper, cutter, staplers, clip applicator, access device, drug/gene therapy delivery device, and energy device using ultrasound, RF, laser, etc.).

Known surgical staplers include an end effector that simultaneously makes a longitudinal incision in tissue and applies lines of staples on opposing sides of the incision. The end effector includes a pair of cooperating jaw members that, if the instrument is intended for endoscopic or laparoscopic applications, are capable of passing through a cannula passageway. One of the jaw members receives a staple cartridge having at least two laterally spaced rows of staples. The other jaw member defines an anvil having staple-forming pockets aligned with the rows of staples in the cartridge. The instrument includes a plurality of reciprocating wedges which, when driven distally, pass through openings in the staple cartridge and engage drivers supporting the staples to effect the firing of the staples toward the anvil.

An example of a surgical stapler suitable for endoscopic applications is described in U.S. Pat. No. 5,465,895, entitled SURGICAL STAPLER INSTRUMENT, which discloses an endocutter with distinct closing and firing actions. A clinician using this device is able to close the jaw members upon tissue to position the tissue prior to firing. Once the clinician has determined that the jaw members are properly gripping tissue, the clinician can then fire the surgical stapler with a single firing stroke, or multiple firing strokes, depending on the device. Firing the surgical stapler causes severing and stapling of the tissue. The simultaneous severing and stapling avoids complications that may arise when performing such actions sequentially with different surgical tools that respectively only sever and staple.

One specific advantage of being able to close upon tissue before firing is that the clinician is able to verify via an endoscope that the desired location for the cut has been achieved, including a sufficient amount of tissue has been captured between opposing jaws. Otherwise, opposing jaws may be drawn too close together, especially pinching at their distal ends, and thus not effectively forming closed staples in the severed tissue. At the other extreme, an excessive amount of clamped tissue may cause binding and an incomplete firing.

When using endoscopic instruments, surgeons often need to be sure if the instrument has completed a function that the surgeon has initiated. For example, the surgeon may initiate articulation of an end-effector of an instrument and allow the articulation to continue to the most extreme articulated position, or an articulation limit. If the surgeon continues to try to articulate the end-effector past the articulation limit, the surgeon will waste valuable time in surgery attempting to complete a function that is past the instrument's limit. Similarly, if a surgeon is to try to fire an already fired instrument, valuable time will be wasted and instrument damage, in some cases, might occur. Therefore, it is advantageous to signal the user of a surgical instrument when the limit of a surgical function is reached.

SUMMARY

In one general aspect, the present invention is directed to a surgical instrument. The surgical instrument has an end effector and a trigger in communication with the end effector. The surgical instrument also has a sensor which communicates with a force delivery system when a limit point of a surgical function has been reached.

Also, in various embodiments, the sensor senses the position of an actuator which drives the surgical function.

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The actuator may translate or rotate, and may drive one or more of the various functions of the surgical instrument.

In various embodiments, the force deliver system may deliver a vibratory force. The force delivery system may utilize a motor which is driven by a sine wave or a square wave signal to cause the motor to move in a manner that causes vibratory force to signal the user and provide haptic feedback of the limit of a surgical function.

DRAWINGS

Various embodiments of the present invention are described herein by way of example in conjunction with the following figures, wherein

FIGS. 1 and 2 are perspective views of a surgical cutting and fastening instrument according to various embodiments of the present invention;

FIGS. 3-5 are exploded views of an end effector and shaft of the instrument according to various embodiments of the present invention;

FIG. 6 is a side view of the end effector according to various embodiments of the present invention;

FIG. 7 is an exploded view of the handle of the instrument according to various embodiments of the present invention;

FIGS. 8 and 9 are partial perspective views of the handle according to various embodiments of the present invention;

FIG. 10 is a side view of the handle according to various embodiments of the present invention;

FIGS. 10A and 10B illustrate a proportional sensor that may be used according to various embodiments of the present invention;

FIG. 11 is a schematic diagram of a circuit used in the instrument according to various embodiments of the present invention;

FIGS. 12-13 are side views of the handle according to other embodiments of the present invention;

FIGS. 14-22 illustrate different mechanisms for locking the closure trigger according to various embodiments of the present invention;

FIGS. 23A-B show a universal joint ("u-joint") that may be employed at the articulation point of the instrument according to various embodiments of the present invention;

FIGS. 24A-B shows a torsion cable that may be employed at the articulation point of the instrument according to various embodiments of the present invention;

FIGS. 25-31 illustrate a surgical cutting and fastening instrument with power assist according to another embodiment of the present invention;

FIGS. 32-36 illustrate a surgical cutting and fastening instrument with power assist according to yet another embodiment of the present invention;

FIGS. 37-40 illustrate a surgical cutting and fastening instrument with tactile feedback to embodiments of the present invention;

FIG. 41 illustrates an exploded view of an end effector and shaft of the instrument according to various embodiments of the present invention;

FIG. 42 illustrates a side view of the handle of a mechanically instrument according to various embodiments of the present invention;

FIG. 43 illustrates an exploded view of the handle of the mechanically actuated instrument of FIG. 42;

FIG. 44 illustrates a block diagram of a recording system for recording various conditions of the instrument according to various embodiments of the present invention;

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FIGS. 45-46 illustrate cut away side views of a handle of the instrument showing various sensors according to various embodiments of the present invention;

FIG. 47 illustrates the end effector of the instrument showing various sensors according to various embodiments of the present invention;

FIG. 48 illustrates a firing bar of the instrument including a sensor according to various embodiments of the present invention;

FIG. 49 illustrates a side view of the handle, end effector, and firing bar of the instrument showing a sensor according to various embodiments of the present invention;

FIG. 50 illustrates an exploded view of the staple channel and portions of a staple cartridge of the instrument showing various sensors according to various embodiments of the present invention;

FIG. 51 illustrates a top down view of the staple channel of the instrument showing various sensors according to various embodiments of the present invention;

FIGS. 52A and 52B illustrate a flow chart showing a method for operating the instrument according to various embodiments; and

FIG. 53 illustrates a memory chart showing exemplary recorded conditions of the instrument according to various embodiments of the present invention.

FIG. 54 illustrates a graph of voltage applied to a motor versus time showing a changing voltage versus time.

FIG. 55 illustrates a graph of motion of a member versus time

FIG. 56 illustrates a flow chart showing a method of tracking an extent of activation and providing haptic feedback determined by position

DETAILED DESCRIPTION

FIGS. 1 and 2 depict a surgical cutting and fastening instrument 10 according to various embodiments of the present invention. The illustrated embodiment is an endoscopic surgical instrument 10 and in general, the embodiments of the instrument 10 described herein are endoscopic surgical cutting and fastening instruments. It should be noted, however, that according to other embodiments of the present invention, the instrument 10 may be a non-endoscopic surgical cutting instrument, such as a laparoscopic instrument.

The surgical instrument 10 depicted in FIGS. 1 and 2 comprises a handle 6, a shaft 8, and an articulating end effector 12 pivotally connected to the shaft 8 at an articulation pivot 14. An articulation control 16 may be provided adjacent to the handle 6 to effect rotation of the end effector 12 about the articulation pivot 14. It will be appreciated that various embodiments may include a non-pivoting end effector, and therefore may not have an articulation pivot 14 or articulation control 16. Also, in the illustrated embodiment, the end effector 12 is configured to act as an endocutter for clamping, severing and stapling tissue, although, in other embodiments, different types of end effectors may be used, such as end effectors for other types of surgical devices, such as graspers, cutters, staplers, clip applicators, access devices, drug/gene therapy devices, ultrasound, RF or laser devices, etc.

The handle 6 of the instrument 10 may include a closure trigger 18 and a firing trigger 20 for actuating the end effector 12. It will be appreciated that instruments having end effectors directed to different surgical tasks may have different numbers or types of triggers or other suitable controls for operating the end effector 12. The end effector

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12 is shown separated from the handle 6 by a preferably elongate shaft 8. In one embodiment, a clinician or operator of the instrument 10 may articulate the end effector 12 relative to the shaft 8 by utilizing the articulation control 16, as described in more detail in U.S. patent application Ser. No. 11/329,020, entitled SURGICAL INSTRUMENT HAVING AN ARTICULATING END EFFECTOR, now U.S. Pat. No. 7,670,334, which is incorporated herein by reference.

The end effector 12 includes in this example, among other things, a staple channel 22 and a pivotally translatable clamping member, such as an anvil 24, which are maintained at a spacing that assures effective stapling and severing of tissue clamped in the end effector 12. The handle 6 includes a pistol grip 26 toward which a closure trigger 18 is pivotally drawn by the clinician to cause clamping or closing of the anvil 24 towards the staple channel 22 of the end effector 12 to thereby clamp tissue positioned between the anvil 24 and channel 22. The firing trigger 20 is farther outboard of the closure trigger 18. Once the closure trigger 18 is locked in the closure position as further described below, the firing trigger 20 may rotate slightly toward the pistol grip 26 so that it can be reached by the operator using one hand. Then the operator may pivotally draw the firing trigger 20 toward the pistol grip 26 to cause the stapling and severing of clamped tissue in the end effector 12. In other embodiments, different types of clamping members besides the anvil 24 could be used, such as, for example, an opposing jaw, etc.

It will be appreciated that the terms "proximal" and "distal" are used herein with reference to a clinician gripping the handle 6 of an instrument 10. Thus, the end effector 12 is distal with respect to the more proximal handle 6. It will be further appreciated that, for convenience and clarity, spatial terms such as "vertical" and "horizontal" are used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

The closure trigger 18 may be actuated first. Once the clinician is satisfied with the positioning of the end effector 12, the clinician may draw back the closure trigger 18 to its fully closed, locked position proximate to the pistol grip 26. The firing trigger 20 may then be actuated. The firing trigger 20 returns to the open position (shown in FIGS. 1 and 2) when the clinician removes pressure, as described more fully below. A release button on the handle 6, when depressed may release the locked closure trigger 18. The release button may be implemented in various forms such as, for example, release button 30 shown in FIGS. 42-43, slide release button 160 shown in FIG. 14, and/or button 172 shown in FIG. 16.

FIGS. 3-6 show embodiments of a rotary-driven end effector 12 and shaft 8 according to various embodiments. FIG. 3 is an exploded view of the end effector 12 according to various embodiments. As shown in the illustrated embodiment, the end effector 12 may include, in addition to the previously-mentioned channel 22 and anvil 24, a cutting instrument 32, a sled 33, a staple cartridge 34 that is removably seated in the channel 22, and a helical screw shaft 36. The cutting instrument 32 may be, for example, a knife. The anvil 24 may be pivotally opened and closed at pivot pins 25 connected to the proximate end of the channel 22. The anvil 24 may also include a tab 27 at its proximate end that is inserted into a component of the mechanical closure system (described further below) to open and close the anvil 24. When the closure trigger 18 is actuated, that is, drawn in by a user of the instrument 10, the anvil 24 may pivot about the pivot pins 25 into the clamped or closed position. If clamping of the end effector 12 is satisfactory, the operator

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may actuate the firing trigger 20, which, as explained in more detail below, causes the knife 32 and sled 33 to travel longitudinally along the channel 22, thereby cutting tissue clamped within the end effector 12. The movement of the sled 33 along the channel 22 causes the staples (not shown) of the staple cartridge 34 to be driven through the severed tissue and against the closed anvil 24, which turns the staples to fasten the severed tissue. In various embodiments, the sled 33 may be an integral component of the cartridge 34. U.S. Pat. No. 6,978,921, entitled SURGICAL STAPLING INSTRUMENT INCORPORATING AN E-BEAM FIRING MECHANISM, which is incorporated herein by reference, provides more details about such two-stroke cutting and fastening instruments. The sled 33 may be part of the cartridge 34, such that when the knife 32 retracts following the cutting operation, the sled 33 does not retract.

It should be noted that although the embodiments of the instrument 10 described herein employ an end effector 12 that staples the severed tissue, in other embodiments different techniques for fastening or sealing the severed tissue may be used. For example, end effectors that use RF energy or adhesives to fasten the severed tissue may also be used. U.S. Pat. No. 5,709,680, entitled ELECTROSURGICAL HEMOSTATIC DEVICE, and U.S. Pat. No. 5,688,270, entitled ELECTROSURGICAL HEMOSTATIC DEVICE WITH RECESSED AND/OR OFFSET ELECTRODES, which are incorporated herein by reference, disclose an endoscopic cutting instrument that uses RF energy to seal the severed tissue. U.S. patent application Ser. No. 11/267,811, entitled SURGICAL STAPLING INSTRUMENTS STRUCTURED FOR DELIVERY OF MEDICAL AGENTS, now U.S. Pat. No. 7,673,783, and U.S. patent application Ser. No. 11/267,383, entitled SURGICAL STAPLING INSTRUMENTS STRUCTURED FOR PUMP-ASSISTED DELIVERY OF MEDICAL AGENTS, now U.S. Pat. No. 7,607,557, which are also incorporated herein by reference, disclose cutting instruments that uses adhesives to fasten the severed tissue. Accordingly, although the description herein refers to cutting/stapling operations and the like below, it should be recognized that this is an exemplary embodiment and is not meant to be limiting. Other tissue fastening techniques may also be used.

FIGS. 4 and 5 are exploded views and FIG. 6 is a side view of the end effector 12 and shaft 8 according to various embodiments. As shown in the illustrated embodiment, the shaft 8 may include a proximate closure tube 40 and a distal closure tube 42 pivotably linked by a pivot link 44. The distal closure tube 42 includes an opening 45 into which the tab 27 on the anvil 24 is inserted in order to open and close the anvil 24, as further described below. Disposed inside the closure tubes 40, 42 may be a proximate spine tube 46. Disposed inside the proximate spine tube 46 may be a main rotational (or proximate) drive shaft 48 that communicates with a secondary (or distal) drive shaft 50 via a bevel gear assembly 52. The secondary drive shaft 50 is connected to a drive gear 54 that engages a proximate drive gear 56 of the helical screw shaft 36. The vertical bevel gear 52b may sit and pivot in an opening 57 in the distal end of the proximate spine tube 46. A distal spine tube 58 may be used to enclose the secondary drive shaft 50 and the drive gears 54, 56. Collectively, the main drive shaft 48, the secondary drive shaft 50, and the articulation assembly (e.g., the bevel gear assembly 52a-c) are sometimes referred to herein as the "main drive shaft assembly."

A bearing 38, positioned at a distal end of the staple channel 22, receives the helical drive screw 36, allowing the helical drive screw 36 to freely rotate with respect to the

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channel 22. The helical screw shaft 36 may interface a threaded opening (not shown) of the knife 32 such that rotation of the shaft 36 causes the knife 32 to translate distally or proximally (depending on the direction of the rotation) through the staple channel 22. Accordingly, when the main drive shaft 48 is caused to rotate by actuation of the firing trigger 20 (as explained in more detail below), the bevel gear assembly 52a-c causes the secondary drive shaft 50 to rotate, which in turn, because of the engagement of the drive gears 54, 56, causes the helical screw shaft 36 to rotate, which causes the knife driving member 32 to travel longitudinally along the channel 22 to cut any tissue clamped within the end effector 12. The sled 33 may be made of, for example, plastic, and may have a sloped distal surface. As the sled 33 traverses the channel 22, the sloped forward surface may push up or drive the staples in the staple cartridge through the clamped tissue and against the anvil 24. The anvil 24 turns the staples, thereby stapling the severed tissue. When the knife 32 is retracted, the knife 32 and sled 33 may become disengaged, thereby leaving the sled 33 at the distal end of the channel 22.

As described above, because of the lack of user feedback for the cutting/stapling operation, there is a general lack of acceptance among physicians of motor-driven endocutters where the cutting/stapling operation is actuated by merely pressing a button. In contrast, embodiments of the present invention provide a motor-driven endocutter with user-feedback of the deployment, force and/or position of the cutting instrument 32 in end effector 12.

FIGS. 7-10 illustrate an exemplary embodiment of a motor-driven endocutter, and in particular the handle thereof, that provides user-feedback regarding the deployment and loading force of the cutting instrument 32 in the end effector 12. In addition, the embodiment may use power provided by the user in retracting the firing trigger 20 to power the device (a so-called "power assist" mode). The embodiment may be used with the rotary driven end effector 12 and shaft 8 embodiments described above. As shown in the illustrated embodiment, the handle 6 includes exterior lower side pieces 59, 60 and exterior upper side pieces 61, 62 that fit together to form, in general, the exterior of the handle 6. A battery 64, such as a Li ion battery, may be provided in the pistol grip portion 26 of the handle 6. The battery 64 powers a motor 65 disposed in an upper portion of the pistol grip portion 26 of the handle 6. According to various embodiments, the motor 65 may be a DC brushed driving motor having a maximum rotation of, approximately, 5000 RPM. The motor 65 may drive a 90° bevel gear assembly 66 comprising a first bevel gear 68 and a second bevel gear 70. The bevel gear assembly 66 may drive a planetary gear assembly 72. The planetary gear assembly 72 may include a pinion gear 74 connected to a drive shaft 76. The pinion gear 74 may drive a mating ring gear 78 that drives a helical gear drum 80 via a drive shaft 82. A ring 84 may be threaded on the helical gear drum 80. Thus, when the motor 65 rotates, the ring 84 is caused to travel along the helical gear drum 80 by means of the interposed bevel gear assembly 66, planetary gear assembly 72 and ring gear 78.

The handle 6 may also include a run motor sensor 110 (see FIG. 10) in communication with the firing trigger 20 to detect when the firing trigger 20 has been drawn in (or "closed") toward the pistol grip portion 26 of the handle 6 by the operator to thereby actuate the cutting/stapling operation by the end effector 12. The sensor 110 may be a proportional sensor such as, for example, a rheostat or variable resistor. When the firing trigger 20 is drawn in, the sensor 110 detects the movement, and sends an electrical

signal indicative of the voltage (or power) to be supplied to the motor 65. When the sensor 110 is a variable resistor or the like, the rotation of the motor 65 may be generally proportional to the amount of movement of the firing trigger 20. That is, if the operator only draws or closes the firing trigger 20 in a little bit, the rotation of the motor 65 is relatively low. When the firing trigger 20 is fully drawn in (or in the fully closed position), the rotation of the motor 65 is at its maximum. In other words, the harder the user pulls on the firing trigger 20, the more voltage is applied to the motor 65, causing greater rates of rotation.

The handle 6 may include a middle handle piece 104 adjacent to the upper portion of the firing trigger 20. The handle 6 also may comprise a bias spring 112 connected between posts on the middle handle piece 104 and the firing trigger 20. The bias spring 112 may bias the firing trigger 20 to its fully open position. In that way, when the operator releases the firing trigger 20, the bias spring 112 will pull the firing trigger 20 to its open position, thereby removing actuation of the sensor 110, thereby stopping rotation of the motor 65. Moreover, by virtue of the bias spring 112, any time a user closes the firing trigger 20, the user will experience resistance to the closing operation, thereby providing the user with feedback as to the amount of rotation exerted by the motor 65. Further, the operator could stop retracting the firing trigger 20 to thereby remove force from the sensor 110, to thereby stop the motor 65. As such, the user may stop the deployment of the end effector 12, thereby providing a measure of control of the cutting/fastening operation to the operator.

The distal end of the helical gear drum 80 includes a distal drive shaft 120 that drives a ring gear 122, which mates with a pinion gear 124. The pinion gear 124 is connected to the main drive shaft 48 of the main drive shaft assembly. In that way, rotation of the motor 65 causes the main drive shaft assembly to rotate, which causes actuation of the end effector 12, as described above.

The ring 84 threaded on the helical gear drum 80 may include a post 86 that is disposed within a slot 88 of a slotted arm 90. The slotted arm 90 has an opening 92 its opposite end 94 that receives a pivot pin 96 that is connected between the handle exterior side pieces 59, 60. The pivot pin 96 is also disposed through an opening 100 in the firing trigger 20 and an opening 102 in the middle handle piece 104.

In addition, the handle 6 may include a reverse motor sensor (or end-of-stroke sensor) 130 and a stop motor (or beginning-of-stroke) sensor 142. In various embodiments, the reverse motor sensor 130 may be a limit switch located at the distal end of the helical gear drum 80 such that the ring 84 threaded on the helical gear drum 80 contacts and trips the reverse motor sensor 130 when the ring 84 reaches the distal end of the helical gear drum 80. The reverse motor sensor 130, when activated, sends a signal to the motor 65 to reverse its rotation direction, thereby withdrawing the knife 32 of the end effector 12 following the cutting operation.

The stop motor sensor 142 may be, for example, a normally-closed limit switch. In various embodiments, it may be located at the proximate end of the helical gear drum 80 so that the ring 84 trips the switch 142 when the ring 84 reaches the proximate end of the helical gear drum 80.

In operation, when an operator of the instrument 10 pulls back the firing trigger 20, the sensor 110 detects the deployment of the firing trigger 20 and sends a signal to the motor 65 to cause forward rotation of the motor 65, for example, at a rate proportional to how hard the operator pulls back the firing trigger 20. The forward rotation of the motor 65 in turn

causes the ring gear 78 at the distal end of the planetary gear assembly 72 to rotate, thereby causing the helical gear drum 80 to rotate, causing the ring 84 threaded on the helical gear drum 80 to travel distally along the helical gear drum 80. The rotation of the helical gear drum 80 also drives the main drive shaft assembly as described above, which in turn causes deployment of the knife 32 in the end effector 12. That is, the knife 32 and sled 33 are caused to traverse the channel 22 longitudinally, thereby cutting tissue clamped in the end effector 12. Also, the stapling operation of the end effector 12 is caused to happen in embodiments where a stapling-type end effector 12 is used.

By the time the cutting/stapling operation of the end effector 12 is complete, the ring 84 on the helical gear drum 80 will have reached the distal end of the helical gear drum 80, thereby causing the reverse motor sensor 130 to be tripped, which sends a signal to the motor 65 to cause the motor 65 to reverse its rotation. This in turn causes the knife 32 to retract, and also causes the ring 84 on the helical gear drum 80 to move back to the proximate end of the helical gear drum 80.

The middle handle piece 104 includes a backside shoulder 106 that engages the slotted arm 90 as best shown in FIGS. 8 and 9. The middle handle piece 104 also has a forward motion stop 107 that engages the firing trigger 20. The movement of the slotted arm 90 is controlled, as explained above, by rotation of the motor 65. When the slotted arm 90 rotates counter clockwise as the ring 84 travels from the proximate end of the helical gear drum 80 to the distal end, the middle handle piece 104 will be free to rotate counter clockwise. Thus, as the user draws in the firing trigger 20, the firing trigger 20 will engage the forward motion stop 107 of the middle handle piece 104, causing the middle handle piece 104 to rotate counter clockwise. Due to the backside shoulder 106 engaging the slotted arm 90, however, the middle handle piece 104 will only be able to rotate counter clockwise as far as the slotted arm 90 permits. In that way, if the motor 65 should stop rotating for some reason, the slotted arm 90 will stop rotating, and the user will not be able to further draw in the firing trigger 20 because the middle handle piece 104 will not be free to rotate counter clockwise due to the slotted arm 90.

FIGS. 10A and 10B illustrate two states of a variable sensor that may be used as the run motor sensor 110 according to various embodiments of the present invention. The sensor 110 may include a face portion 280, a first electrode (A) 282, a second electrode (B) 284, and a compressible dielectric material 286 between the electrodes 282, 284, such as, for example, an electroactive polymer (EAP). The sensor 110 may be positioned such that the face portion 280 contacts the firing trigger 20 when retracted. Accordingly, when the firing trigger 20 is retracted, the dielectric material 286 is compressed, as shown in FIG. 10B, such that the electrodes 282, 284 are closer together. Since the distance "b" between the electrodes 282, 284 is directly related to the impedance between the electrodes 282, 284, the greater the distance the more impedance, and the closer the distance the less impedance. In that way, the amount that the dielectric 286 is compressed due to retraction of the firing trigger 20 (denoted as force "F" in FIG. 42) is proportional to the impedance between the electrodes 282, 284, which can be used to proportionally control the motor 65.

Components of an exemplary closure system for closing (or clamping) the anvil 24 of the end effector 12 by retracting the closure trigger 18 are also shown in FIGS. 7-10. In the illustrated embodiment, the closure system includes a

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yoke 250 connected to the closure trigger 18 by a pivot pin 251 inserted through aligned openings in both the closure trigger 18 and the yoke 250. A pivot pin 252, about which the closure trigger 18 pivots, is inserted through another opening in the closure trigger 18 which is offset from where the pin 251 is inserted through the closure trigger 18. Thus, retraction of the closure trigger 18 causes the upper part of the closure trigger 18, to which the yoke 250 is attached via the pin 251, to rotate counterclockwise. The distal end of the yoke 250 is connected, via a pin 254, to a first closure bracket 256. The first closure bracket 256 connects to a second closure bracket 258. Collectively, the closure brackets 256, 258 define an opening in which the proximate end of the proximate closure tube 40 (see FIG. 4) is seated and held such that longitudinal movement of the closure brackets 256, 258 causes longitudinal motion by the proximate closure tube 40. The instrument 10 also includes a closure rod 260 disposed inside the proximate closure tube 40. The closure rod 260 may include a window 261 into which a post 263 on one of the handle exterior pieces, such as exterior lower side piece 59 in the illustrated embodiment, is disposed to fixedly connect the closure rod 260 to the handle 6. In that way, the proximate closure tube 40 is capable of moving longitudinally relative to the closure rod 260. The closure rod 260 may also include a distal collar 267 that fits into a cavity 269 in proximate spine tube 46 and is retained therein by a cap 271 (see FIG. 4).

In operation, when the yoke 250 rotates due to retraction of the closure trigger 18, the closure brackets 256, 258 cause the proximate closure tube 40 to move distally (i.e., away from the handle end of the instrument 10), which causes the distal closure tube 42 to move distally, which causes the anvil 24 to rotate about the pivot pins 25 into the clamped or closed position. When the closure trigger 18 is unlocked from the locked position, the proximate closure tube 40 is caused to slide proximally, which causes the distal closure tube 42 to slide proximally, which, by virtue of the tab 27 being inserted in the window 45 of the distal closure tube 42, causes the anvil 24 to pivot about the pivot pins 25 into the open or unclamped position. In that way, by retracting and locking the closure trigger 18, an operator may clamp tissue between the anvil 24 and channel 22, and may unclamp the tissue following the cutting/stapling operation by unlocking the closure trigger 20 from the locked position.

FIG. 11 is a schematic diagram of an electrical circuit of the instrument 10 according to various embodiments of the present invention. When an operator initially pulls in the firing trigger 20 after locking the closure trigger 18, the sensor 110 is activated, allowing current to flow there through. If the normally-open reverse motor sensor switch 130 is open (meaning the end of the end effector stroke has not been reached), current will flow to a single pole, double throw relay 132. Since the reverse motor sensor switch 130 is not closed, the inductor 134 of the relay 132 will not be energized, so the relay 132 will be in its non-energized state. The circuit also includes a cartridge lockout sensor 136. If the end effector 12 includes a staple cartridge 34, the sensor 136 will be in the closed state, allowing current to flow. Otherwise, if the end effector 12 does not include a staple cartridge 34, the sensor 136 will be open, thereby preventing the battery 64 from powering the motor 65.

When the staple cartridge 34 is present, the sensor 136 is closed, which energizes a single pole, single throw relay 138. When the relay 138 is energized, current flows through the relay 136, through the variable resistor sensor 110, and

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to the motor 65 via a double pole, double throw relay 140, thereby powering the motor 65 and allowing it to rotate in the forward direction.

When the end effector 12 reaches the end of its stroke, the reverse motor sensor 130 will be activated, thereby closing the switch 130 and energizing the relay 134. This causes the relay 134 to assume its energized state (not shown in FIG. 13), which causes current to bypass the cartridge lockout sensor 136 and variable resistor 110, and instead causes current to flow to both the normally-closed double pole, double throw relay 142 and back to the motor 65, but in a manner, via the relay 140, that causes the motor 65 to reverse its rotational direction.

Because the stop motor sensor switch 142 is normally-closed, current will flow back to the relay 134 to keep it closed until the switch 142 opens. When the knife 32 is fully retracted, the stop motor sensor switch 142 is activated, causing the switch 142 to open, thereby removing power from the motor 65.

In other embodiments, rather than a proportional-type sensor 110, an on-off type sensor could be used. In such embodiments, the rate of rotation of the motor 65 would not be proportional to the force applied by the operator. Rather, the motor 65 would generally rotate at a constant rate. But the operator would still experience force feedback because the firing trigger 20 is geared into the gear drive train.

FIG. 12 is a side-view of the handle 6 of a power-assist motorized endocutter according to another embodiment. The embodiment of FIG. 12 is similar to that of FIGS. 7-10 except that in the embodiment of FIG. 12, there is no slotted arm connected to the ring 84 threaded on the helical gear drum 80. Instead, in the embodiment of FIG. 12, the ring 84 includes a sensor portion 114 that moves with the ring 84 as the ring 84 advances down (and back) on the helical gear drum 80. The sensor portion 114 includes a notch 116. The reverse motor sensor 130 may be located at the distal end of the notch 116 and the stop motor sensor 142 may be located at the proximate end of the notch 116. As the ring 84 moves down the helical gear drum 80 (and back), the sensor portion 114 moves with it. Further, as shown in FIG. 12, the middle piece 104 may have an arm 118 that extends into the notch 12.

In operation, as an operator of the instrument 10 retracts in the firing trigger 20 toward the pistol grip 26, the run motor sensor 110 detects the motion and sends a signal to power the motor 65, which causes, among other things, the helical gear drum 80 to rotate. As the helical gear drum 80 rotates, the ring 84 threaded on the helical gear drum 80 advances (or retracts, depending on the rotation). Also, due to the pulling in of the firing trigger 20, the middle piece 104 is caused to rotate counter clockwise with the firing trigger 20 due to the forward motion stop 107 that engages the firing trigger 20. The counter clockwise rotation of the middle piece 104 cause the arm 118 to rotate counter clockwise with the sensor portion 114 of the ring 84 such that the arm 118 stays disposed in the notch 116. When the ring 84 reaches the distal end of the helical gear drum 80, the arm 118 will contact and thereby trip the reverse motor sensor 130. Similarly, when the ring 84 reaches the proximate end of the helical gear drum 80, the arm will contact and thereby trip the stop motor sensor 142. Such actions may reverse and stop the motor 65, respectively as described above.

FIG. 13 is a side-view of a power-assist motorized endocutter according to another embodiment. The embodiment of FIG. 13 is similar to that of FIGS. 7-10 except that in the embodiment of FIG. 13, there is no slot in the arm 90. Instead, the ring 84 threaded on the helical gear

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drum 80 includes a vertical channel 126. Instead of a slot, the arm 90 includes a post 128 that is disposed in the channel 126. As the helical gear drum 80 rotates, the ring 84 threaded on the helical gear drum 80 advances (or retracts, depending on the rotation). The arm 90 rotates counter clockwise as the ring 84 advances due to the post 128 being disposed in the channel 126, as shown in FIG. 13.

As mentioned above, in using a two-stroke motorized instrument, the operator first pulls back and locks the closure trigger 18. FIGS. 14 and 15 show one embodiment of a way to lock the closure trigger 18 to the pistol grip portion 26 of the handle 6. In the illustrated embodiment, the pistol grip portion 26 includes a hook 150 that is biased to rotate counter clockwise about a pivot point 151 by a torsion spring 152. Also, the closure trigger 18 includes a closure bar 154. As the operator draws in the closure trigger 18, the closure bar 154 engages a sloped portion 156 of the hook 150, thereby rotating the hook 150 upward (or clockwise in FIGS. 14-15) until the closure bar 154 completely passes the sloped portion 156 passes into a recessed notch 158 of the hook 150, which locks the closure trigger 18 in place. The operator may release the closure trigger 18 by pushing down on a slide button release 160 on the back or opposite side of the pistol grip portion 26. Pushing down the slide button release 160 rotates the hook 150 clockwise such that the closure bar 154 is released from the recessed notch 158.

FIG. 16 shows another closure trigger locking mechanism according to various embodiments. In the embodiment of FIG. 16, the closure trigger 18 includes a wedge 160 having an arrow-head portion 161. The arrow-head portion 161 is biased downward (or clockwise) by a leaf spring 162. The wedge 160 and leaf spring 162 may be made from, for example, molded plastic. When the closure trigger 18 is retracted, the arrow-head portion 161 is inserted through an opening 164 in the pistol grip portion 26 of the handle 6. A lower chamfered surface 166 of the arrow-head portion 161 engages a lower sidewall 168 of the opening 164, forcing the arrow-head portion 161 to rotate counter clockwise. Eventually the lower chamfered surface 166 fully passes the lower sidewall 168, removing the counter clockwise force on the arrow-head portion 161, causing the lower sidewall 168 to slip into a locked position in a notch 170 behind the arrow-head portion 161.

To unlock the closure trigger 18, a user presses down on a button 172 on the opposite side of the closure trigger 18, causing the arrow-head portion 161 to rotate counter clockwise and allowing the arrow-head portion 161 to slide out of the opening 164.

FIGS. 17-22 show a closure trigger locking mechanism according to another embodiment. As shown in this embodiment, the closure trigger 18 includes a flexible longitudinal arm 176 that includes a lateral pin 178 extending therefrom. The arm 176 and pin 178 may be made from molded plastic, for example. The pistol grip portion 26 of the handle 6 includes an opening 180 with a laterally extending wedge 182 disposed therein. When the closure trigger 18 is retracted, the pin 178 engages the wedge 182, and the pin 178 is forced downward (i.e., the arm 176 is rotated clockwise) by the lower surface 184 of the wedge 182, as shown in FIGS. 17 and 18. When the pin 178 fully passes the lower surface 184, the clockwise force on the arm 176 is removed, and the pin 178 is rotated counter clockwise such that the pin 178 comes to rest in a notch 186 behind the wedge 182, as shown in FIG. 19, thereby locking the closure trigger 18. The pin 178 is further held in place in the locked position by a flexible stop 188 extending from the wedge 184.

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To unlock the closure trigger 18, the operator may further squeeze the closure trigger 18, causing the pin 178 to engage a sloped backwall 190 of the opening 180, forcing the pin 178 upward past the flexible stop 188, as shown in FIGS. 20 and 21. The pin 178 is then free to travel out an upper channel 192 in the opening 180 such that the closure trigger 18 is no longer locked to the pistol grip portion 26, as shown in FIG. 22.

FIGS. 23A-B show a universal joint ("u-joint") 195. The second piece 195-2 of the u-joint 195 rotates in a horizontal plane in which the first piece 195-1 lies. FIG. 23A shows the u-joint 195 in a linear (180°) orientation and FIG. 23B shows the u-joint 195 at approximately a 150° orientation. The u-joint 195 may be used instead of the bevel gears 52a-c (see FIG. 4, for example) at the articulation point 14 of the main drive shaft assembly to articulate the end effector 12. FIGS. 24A-B show a torsion cable 197 that may be used in lieu of both the bevel gears 52a-c and the u-joint 195 to realize articulation of the end effector 12.

FIGS. 25-31 illustrate another embodiment of a motorized, two-stroke surgical cutting and fastening instrument 10 with power assist according to another embodiment of the present invention. The embodiment of FIGS. 25-31 is similar to that of FIGS. 6-10 except that instead of the helical gear drum 80, the embodiment of FIGS. 23-28 includes an alternative gear drive assembly. The embodiment of FIGS. 25-31 includes a gear box assembly 200 including a number of gears disposed in a frame 201, wherein the gears are connected between the planetary gear 72 and the pinion gear 124 at the proximate end of the drive shaft 48. As explained further below, the gear box assembly 200 provides feedback to the user via the firing trigger 20 regarding the deployment and loading force of the end effector 12. Also, the user may provide power to the system via the gear box assembly 200 to assist the deployment of the end effector 12. In that sense, like the embodiments described above, the embodiment of FIGS. 23-32 is another power assist motorized instrument 10 that provides feedback to the user regarding the loading force experienced by the instrument.

In the illustrated embodiment, the firing trigger 20 includes two pieces: a main body portion 202 and a stiffening portion 204. The main body portion 202 may be made of plastic, for example, and the stiffening portion 204 may be made out of a more rigid material, such as metal. In the illustrated embodiment, the stiffening portion 204 is adjacent to the main body portion 202, but according to other embodiments, the stiffening portion 204 could be disposed inside the main body portion 202. A pivot pin 207 may be inserted through openings in the firing trigger pieces 202, 204 and may be the point about which the firing trigger 20 rotates. In addition, a spring 222 may bias the firing trigger 20 to rotate in a counter clockwise direction. The spring 222 may have a distal end connected to a pin 224 that is connected to the pieces 202, 204 of the firing trigger 20. The proximate end of the spring 222 may be connected to one of the handle exterior lower side pieces 59, 60.

In the illustrated embodiment, both the main body portion 202 and the stiffening portion 204 includes gear portions 206, 208 (respectively) at their upper end portions. The gear portions 206, 208 engage a gear in the gear box assembly 200, as explained below, to drive the main drive shaft assembly and to provide feedback to the user regarding the deployment of the end effector 12.

The gear box assembly 200 may include as shown, in the illustrated embodiment, six (6) gears. A first gear 210 of the gear box assembly 200 engages the gear portions 206, 208 of the firing trigger 20. In addition, the first gear 210 engages

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a smaller second gear **212**, the smaller second gear **212** being coaxial with a large third gear **214**. The third gear **214** engages a smaller fourth gear **216**, the smaller fourth gear being coaxial with a fifth gear **218**. The fifth gear **218** is a 90° bevel gear that engages a mating 90° bevel gear **220** (best shown in FIG. **31**) that is connected to the pinion gear **124** that drives the main drive shaft **48**.

In operation, when the user retracts the firing trigger **20**, a run motor sensor (not shown) is activated, which may provide a signal to the motor **65** to rotate at a rate proportional to the extent or force with which the operator is retracting the firing trigger **20**. This causes the motor **65** to rotate at a speed proportional to the signal from the sensor. The sensor is not shown for this embodiment, but it could be similar to the run motor sensor **110** described above. The sensor could be located in the handle **6** such that it is depressed when the firing trigger **20** is retracted. Also, instead of a proportional-type sensor, an on/off type sensor may be used.

Rotation of the motor **65** causes the bevel gears **68**, **70** to rotate, which causes the planetary gear **72** to rotate, which causes, via the drive shaft **76**, the ring gear **122** to rotate. The ring gear **122** meshes with the pinion gear **124**, which is connected to the main drive shaft **48**. Thus, rotation of the pinion gear **124** drives the main drive shaft **48**, which causes actuation of the cutting/stapling operation of the end effector **12**.

Forward rotation of the pinion gear **124** in turn causes the bevel gear **220** to rotate, which causes, by way of the rest of the gears of the gear box assembly **200**, the first gear **210** to rotate. The first gear **210** engages the gear portions **206**, **208** of the firing trigger **20**, thereby causing the firing trigger **20** to rotate counter clockwise when the motor **65** provides forward drive for the end effector **12** (and to rotate counter clockwise when the motor **65** rotates in reverse to retract the end effector **12**). In that way, the user experiences feedback regarding loading force and deployment of the end effector **12** by way of the user's grip on the firing trigger **20**. Thus, when the user retracts the firing trigger **20**, the operator will experience a resistance related to the load force experienced by the end effector **12**. Similarly, when the operator releases the firing trigger **20** after the cutting/stapling operation so that it can return to its original position, the user will experience a clockwise rotation force from the firing trigger **20** that is generally proportional to the reverse speed of the motor **65**.

It should also be noted that in this embodiment the user can apply force (either in lieu of or in addition to the force from the motor **65**) to actuate the main drive shaft assembly (and hence the cutting/stapling operation of the end effector **12**) through retracting the firing trigger **20**. That is, retracting the firing trigger **20** causes the gear portions **206**, **208** to rotate counter clockwise, which causes the gears of the gear box assembly **200** to rotate, thereby causing the pinion gear **124** to rotate, which causes the main drive shaft **48** to rotate.

Although not shown in FIGS. **25-31**, the instrument **10** may further include reverse motor and stop motor sensors. As described above, the reverse motor and stop motor sensors may detect, respectively, the end of the cutting stroke (full deployment of the knife **32**) and the end of retraction operation (full retraction of the knife **32**). A similar circuit to that described above in connection with FIG. **11** may be used to appropriately power the motor **65**.

FIGS. **32-36** illustrate a two-stroke, motorized surgical cutting and fastening instrument **10** with power assist according to another embodiment. The embodiment of FIGS. **32-36** is similar to that of FIGS. **25-31** except that in

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the embodiment of FIGS. **32-36**, the firing trigger **20** includes a lower portion **228** and an upper portion **230**. Both portions **228**, **230** are connected to and pivot about a pivot pin **207** that is disposed through each portion **228**, **230**. The upper portion **230** includes a gear portion **232** that engages the first gear **210** of the gear box assembly **200**. The spring **222** is connected to the upper portion **230** such that the upper portion is biased to rotate in the clockwise direction. The upper portion **230** may also include a lower arm **234** that contacts an upper surface of the lower portion **228** of the firing trigger **20** such that when the upper portion **230** is caused to rotate clockwise the lower portion **228** also rotates clockwise, and when the lower portion **228** rotates counter clockwise the upper portion **230** also rotates counter clockwise. Similarly, the lower portion **228** includes a rotational stop **238** that engages a shoulder of the upper portion **230**. In that way, when the upper portion **230** is caused to rotate counter clockwise the lower portion **228** also rotates counter clockwise, and when the lower portion **228** rotates clockwise the upper portion **230** also rotates clockwise.

The illustrated embodiment also includes the run motor sensor **110** that communicates a signal to the motor **65** that, in various embodiments, may cause the motor **65** to rotate at a speed proportional to the force applied by the operator when retracting the firing trigger **20**. The sensor **110** may be, for example, a rheostat or some other variable resistance sensor, as explained herein. In addition, the instrument **10** may include reverse motor sensor **130** that is tripped or switched when contacted by a front face **242** of the upper portion **230** of the firing trigger **20**. When activated, the reverse motor sensor **130** sends a signal to the motor **65** to reverse direction. Also, the instrument **10** may include a stop motor sensor **142** that is tripped or actuated when contacted by the lower portion **228** of the firing trigger **20**. When activated, the stop motor sensor **142** sends a signal to stop the reverse rotation of the motor **65**.

In operation, when an operator retracts the closure trigger **18** into the locked position, the firing trigger **20** is retracted slightly (through mechanisms known in the art, including U.S. Pat. Nos. 6,978,921 and 6,905,057, which are incorporated herein by reference) so that the user can grasp the firing trigger **20** to initiate the cutting/stapling operation, as shown in FIGS. **32** and **33**. At that point, as shown in FIG. **33**, the gear portion **232** of the upper portion **230** of the firing trigger **20** moves into engagement with the first gear **210** of the gear box assembly **200**. When the operator retracts the firing trigger **20**, according to various embodiments, the firing trigger **20** may rotate a small amount, such as five degrees, before tripping the run motor sensor **110**, as shown in FIG. **34**. Activation of the sensor **110** causes the motor **65** to forward rotate at a rate proportional to the retraction force applied by the operator. The forward rotation of the motor **65** causes, as described above, the main drive shaft **48** to rotate, which causes the knife **32** in the end effector **12** to be deployed (i.e., begin traversing the channel **22**). Rotation of the pinion gear **124**, which is connected to the main drive shaft **48**, causes the gears **210-220** in the gear box assembly **200** to rotate. Since the first gear **210** is in engagement with the gear portion **232** of the upper portion **230** of the firing trigger **20**, the upper portion **232** is caused to rotate counter clockwise, which causes the lower portion **228** to also rotate counter clockwise.

When the knife **32** is fully deployed (i.e., at the end of the cutting stroke), the front face **242** of the upper portion **230** trips the reverse motor sensor **130**, which sends a signal to the motor **65** to reverse rotational direction. This causes the main drive shaft assembly to reverse rotational direction

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to retract the knife 32. Reverse rotation of the main drive shaft assembly also causes the gears 210-220 in the gear box assembly to reverse direction, which causes the upper portion 230 of the firing trigger 20 to rotate clockwise, which causes the lower portion 228 of the firing trigger 20 to rotate clockwise until the lower portion 228 trips or actuates the stop motor sensor 142 when the knife 32 is fully retracted, which causes the motor 65 to stop. In that way, the user experiences feedback regarding deployment of the end effector 12 by way of the user's grip on the firing trigger 20. Thus, when the user retracts the firing trigger 20, the operator will experience a resistance related to the deployment of the end effector 12 and, in particular, to the loading force experienced by the knife 32. Similarly, when the operator releases the firing trigger 20 after the cutting/stapling operation so that it can return to its original position, the user will experience a clockwise rotation force from the firing trigger 20 that is generally proportional to the reverse speed of the motor 65.

It should also be noted that in this embodiment the user can apply force (either in lieu of or in addition to the force from the motor 65) to actuate the main drive shaft assembly (and hence the cutting/stapling operation of the end effector 12) through retracting the firing trigger 20. That is, retracting the firing trigger 20 causes the gear portion 232 of the upper portion 230 to rotate counter clockwise, which causes the gears of the gear box assembly 200 to rotate, thereby causing the pinion gear 124 to rotate, which causes the main drive shaft assembly to rotate.

The above-described embodiments employed power-assist user feedback systems, with or without adaptive control (e.g., using a sensor 110, 130, and 142 outside of the closed loop system of the motor 65, gear drive train, and end effector 12) for a two-stroke, motorized surgical cutting and fastening instrument. That is, force applied by the user in retracting the firing trigger 20 may be added to the force applied by the motor 65 by virtue of the firing trigger 20 being geared into (either directly or indirectly) the gear drive train between the motor 65 and the main drive shaft 48. In other embodiments of the present invention, the user may be provided with tactile feedback regarding the position of the knife 32 in the end effector, but without having the firing trigger 20 geared into the gear drive train. FIGS. 37-40 illustrate a motorized surgical cutting and fastening instrument with such a tactile position feedback system.

In the illustrated embodiment of FIGS. 37-40, the firing trigger 20 may have a lower portion 228 and an upper portion 230, similar to the instrument 10 shown in FIGS. 32-36. Unlike the embodiment of FIG. 32-36, however, the upper portion 230 does not have a gear portion that mates with part of the gear drive train. Instead, the instrument includes a second motor 265 with a threaded rod 266 threaded therein. The threaded rod 266 reciprocates longitudinally in and out of the motor 265 as the motor 265 rotates, depending on the direction of rotation. The instrument 10 also includes an encoder 268 that is responsive to the rotations of the main drive shaft 48 for translating the incremental angular motion of the main drive shaft 48 (or other component of the main drive assembly) into a corresponding series of digital signals, for example. In the illustrated embodiment, the pinion gear 124 includes a proximate drive shaft 270 that connects to the encoder 268.

The instrument 10 also includes a control circuit (not shown), which may be implemented using a microcontroller or some other type of integrated circuit, that receives the digital signals from the encoder 268. Based on the signals from the encoder 268, the control circuit may calculate the

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stage of deployment of the knife 32 in the end effector 12. That is, the control circuit can calculate if the knife 32 is fully deployed, fully retracted, or at an intermittent stage. Based on the calculation of the stage of deployment of the end effector 12, the control circuit may send a signal to the second motor 265 to control its rotation to thereby control the reciprocating movement of the threaded rod 266.

In operation, as shown in FIG. 37, when the closure trigger 18 is not locked into the clamped position, the firing trigger 20 rotated away from the pistol grip portion 26 of the handle 6 such that the front face 242 of the upper portion 230 of the firing trigger 20 is not in contact with the proximate end of the threaded rod 266. When the operator retracts the closure trigger 18 and locks it in the clamped position, the firing trigger 20 rotates slightly towards the closure trigger 20 so that the operator can grasp the firing trigger 20, as shown in FIG. 38. In this position, the front face 242 of the upper portion 230 contacts the proximate end of the threaded rod 266.

As the user then retracts the firing trigger 20, after an initial rotational amount (e.g. 5 degrees of rotation) the run motor sensor 110 may be activated such that, as explained above, the sensor 110 sends a signal to the motor 65 to cause it to rotate at a forward speed proportional to the amount of retraction force applied by the operator to the firing trigger 20. Forward rotation of the motor 65 causes the main drive shaft 48 to rotate via the gear drive train, which causes the knife 32 and sled 33 to travel down the channel 22 and sever tissue clamped in the end effector 12. The control circuit receives the output signals from the encoder 268 regarding the incremental rotations of the main drive shaft assembly and sends a signal to the second motor 265 to cause the second motor 265 to rotate, which causes the threaded rod 266 to retract into the motor 265. This allows the upper portion 230 of the firing trigger 20 to rotate counter clockwise, which allows the lower portion 228 of the firing trigger to also rotate counter clockwise. In that way, because the reciprocating movement of the threaded rod 266 is related to the rotations of the main drive shaft assembly, the operator of the instrument 10, by way of his/her grip on the firing trigger 20, experiences tactile feedback as to the position of the end effector 12. The retraction force applied by the operator, however, does not directly affect the drive of the main drive shaft assembly because the firing trigger 20 is not geared into the gear drive train in this embodiment.

By virtue of tracking the incremental rotations of the main drive shaft assembly via the output signals from the encoder 268, the control circuit can calculate when the knife 32 is fully deployed (i.e., fully extended). At this point, the control circuit may send a signal to the motor 65 to reverse direction to cause retraction of the knife 32. The reverse direction of the motor 65 causes the rotation of the main drive shaft assembly to reverse direction, which is also detected by the encoder 268. Based on the reverse rotation detected by the encoder 268, the control circuit sends a signal to the second motor 265 to cause it to reverse rotational direction such that the threaded rod 266 starts to extend longitudinally from the motor 265. This motion forces the upper portion 230 of the firing trigger 20 to rotate clockwise, which causes the lower portion 228 to rotate clockwise. In that way, the operator may experience a clockwise force from the firing trigger 20, which provides feedback to the operator as to the retraction position of the knife 32 in the end effector 12. The control circuit can determine when the knife 32 is fully retracted. At this point, the control circuit may send a signal to the motor 65 to stop rotation.

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According to other embodiments, rather than having the control circuit determine the position of the knife 32, reverse motor and stop motor sensors may be used, as described above. In addition, rather than using a proportional sensor 110 to control the rotation of the motor 65, an on/off switch or sensor can be used. In such an embodiment, the operator would not be able to control the rate of rotation of the motor 65. Rather, it would rotate at a preprogrammed rate.

FIGS. 41-43 illustrate an exemplary embodiment of a mechanically actuated endocutter, and in particular the handle 6, shaft 8 and end effector 12 thereof. Further details of a mechanically actuated endocutter may be found in U.S. patent application Ser. No. 11/052,632 entitled, SURGICAL STAPLING INSTRUMENT INCORPORATING A MULTI-STROKE FIRING MECHANISM WITH AUTOMATIC END OF FIRING TRAVEL RETRACTION, now U.S. Pat. No. 7,083,075, which is incorporated herein by reference. With reference to FIG. 41, the end effector 12 responds to the closure motion from the handle 6 (not depicted in FIG. 41) first by including an anvil face 1002 connecting to an anvil proximal end 1004 that includes laterally projecting anvil pivot pins 25 that are proximal to a vertically projecting anvil tab 27. The anvil pivot pins 25 translate within kidney shaped openings 1006 in the staple channel 22 to open and close anvil 24 relative to channel 22. The tab 27 engages a bent tab 1007 extending inwardly in tab opening 45 on a distal end 1008 of the closure tube 1005, the latter distally terminating in a distal edge 1008 that pushes against the anvil face 1002. Thus, when the closure tube 1005 moves proximally from its open position, the bent tab 1007 of the closure tube 1005 draws the anvil tab 27 proximally, and the anvil pivot pins 25 follow the kidney shaped openings 1006 of the staple channel 22 causing the anvil 24 to simultaneously translate proximally and rotate upward to the open position. When the closure tube 1005 moves distally, the bent tab 1007 in the tab opening 45 releases from the anvil tab 27 and the distal edge 1008 pushes on the anvil face 1002, closing the anvil 24.

With continued reference to FIG. 41, the shaft 8 and end effector 12 also include components that respond to a firing motion of a firing rod 1010. In particular, the firing rod 1010 rotatably engages a firing trough member 1012 having a longitudinal recess 1014. Firing trough member 1012 moves longitudinally within frame 1016 in direct response to longitudinal motion of firing rod 1010. A longitudinal slot 1018 in the closure tube 1005 operably couples with the right and left exterior side handle pieces 61, 62 of the handle 6 (not shown in FIG. 41). The length of the longitudinal slot 1018 in the closure tube 1005 is sufficiently long to allow relative longitudinal motion with the handle pieces 61, 62 to accomplish firing and closure motions respectively with the coupling of the handle pieces 61, 62 passing on through a longitudinal slot 1020 in the frame 1016 to slidably engage the longitudinal recess 1014 in the frame trough member 1012.

The distal end of the frame trough member 1012 is attached to a proximal end of a firing bar 1022 that moves within the frame 1016, specifically within a guide 1024 therein, to distally project the knife 32 into the end effector 12. The end effector 12 includes a staple cartridge 34 that is actuated by the knife 32. The staple cartridge 34 has a tray 1028 that holds a staple cartridge body 1030, a wedge sled driver 33, staple drivers 1034 and staples 1036. It will be appreciated that the wedge sled driver 33 longitudinally moves within a firing recess (not shown) located between the cartridge tray 1028 and the cartridge body 1030. The wedge sled driver 33 presents camming surfaces that contact

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and lift the staple drivers 1034 upward, driving the staples 1036. The staple cartridge body 1030 further includes a proximally open, vertical slot 1031 for passage of the knife 32. Specifically, a cutting surface 1027 is provided along a distal end of knife 32 to cut tissue after it is stapled.

It should be appreciated that the shaft 8 is shown in FIG. 4 as a non-articulating shaft. Nonetheless, applications of the present invention may include instruments capable of articulation, for example, as such shown above with reference to FIGS. 1-4 and described in the following U.S. patents and patent applications, the disclosure of each being hereby incorporated by reference in their entirety: (1) SURGICAL INSTRUMENT INCORPORATING AN ARTICULATION MECHANISM HAVING ROTATION ABOUT THE LONGITUDINAL AXIS, U.S. Pat. No. 7,111,769; (2) SURGICAL STAPLING INSTRUMENT INCORPORATING AN ARTICULATION JOINT FOR A FIRING BAR TRACK, U.S. Pat. No. 6,786,382; (3) A SURGICAL INSTRUMENT WITH A LATERAL-MOVING ARTICULATION CONTROL, U.S. Pat. No. 6,981,628; (4) SURGICAL STAPLING INSTRUMENT INCORPORATING A TAPERED FIRING BAR FOR INCREASED FLEXIBILITY AROUND THE ARTICULATION JOINT, U.S. Pat. No. 6,964,363; and (5) SURGICAL STAPLING INSTRUMENT HAVING ARTICULATION JOINT SUPPORT PLATES FOR SUPPORTING A FIRING BAR, U.S. Pat. No. 7,055,731.

FIGS. 42-43 show an embodiment of the handle 6 that is configured for use in a mechanically actuated endocutter along with the embodiment of the shaft 8 and end effector 12 as shown above in FIG. 41. It will be appreciated that any suitable handle design may be used to mechanically close and fire the end effector 12. In FIGS. 42-43, the handle 6 of the surgical stapling and severing instrument 10 includes a linked transmission firing mechanism 1060 that provides features such as increased strength, reduced handle size, minimized binding, etc.

Closure of the end effector 12 (not shown in FIGS. 42-43) is caused by depressing the closure trigger 18 toward the pistol grip 26 of handle 6. The closure trigger 18 pivots about a closure pivot pin 252 that is coupled to right and left exterior lower side pieces 59, 60 the handle 6, causing an upper portion 1094 of the closure trigger 18 to move forward. The closure tube 1005 receives this closure movement via the closure yoke 250 that is pinned to a closure link 1042 and to the upper portion 1094 of the closure trigger 18 respectively by a closure yoke pin 1044 and a closure link pin 1046.

In the fully open position of FIG. 42, the upper portion 1094 of the closure trigger 18 contacts and holds a locking arm 1048 of the pivoting closure release button 30 in the position shown. When the closure trigger 18 reaches its fully depressed position, the closure trigger 18 releases the locking arm 1048 and an abutting surface 1050 rotates into engagement with a distal rightward notch 1052 of the pivoting locking arm 1048, holding the closure trigger 18 in this clamped or closed position. A proximal end of the locking arm 1048 pivots about a lateral pivotal connection 1054 with the pieces 59, 60 to expose the closure release button 30. An intermediate, distal side 1056 of the closure release button 30 is urged proximally by a compression spring 1058, which is compressed between a housing structure 1040 and closure release button 30. The result is that the closure release button 30 urges the locking arm 1048 counterclockwise (when viewed from the left) into locking contact with the abutting surface 1050 of closure trigger 18,

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which prevents unclamping of closure trigger 18 when the linked transmission firing system 1040 is in an un-retracted condition.

With the closure trigger 18 retracted and fully depressed, the firing trigger 20 is unlocked and may be depressed toward the pistol grip 26, multiple times in this embodiment, to effect firing of the end effector 12. As depicted, the linked transmission firing mechanism 1060 is initially retracted, urged to remain in this position by a combination tension/compression spring 1062 that is constrained within the pistol grip 26 of the handle 6, with its nonmoving end 1063 connected to the pieces 59, 60 and a moving end 1064 connected to a downwardly flexed and proximal, retracted end 1067 of a steel band 1066.

A distally-disposed end 1068 of the steel band 1066 is attached to a link coupling 1070 for structural loading, which in turn is attached to a front link 1072a of a plurality of links 1072a-1072d that form a linked rack 1074. Linked rack 1074 is flexible yet has distal links that form a straight rigid rack assembly that may transfer a significant firing force through the firing rod 1010 in the shaft 6, yet readily retract into the pistol grip 26 to minimize the longitudinal length of the handle 6. It should be appreciated that the combination tension/compression spring 1062 increases the amount of firing travel available while essentially reducing the minimum length by half over a single spring.

The firing trigger 20 pivots about a firing trigger pin 96 that is connected to the handle pieces 59, 60. An upper portion 228 of the firing trigger 20 moves distally about the firing trigger pin 96 as the firing trigger 20 is depressed towards pistol grip 26, stretching a proximally placed firing trigger tension spring 222 proximally connected between the upper portion 228 of the firing trigger 20 and the pieces 59, 60. The upper portion 228 of the firing trigger 20 engages the linked rack 1074 during each firing trigger depression by a traction biasing mechanism 1078 that also disengages when the firing trigger 20 is released. Firing trigger tension spring 222 urges the firing trigger 20 distally when released and disengages the traction biasing mechanism 1078.

As the linked transmission firing mechanism 1040 actuates, an idler gear 1080 is rotated clockwise (as viewed from the left side) by engagement with a toothed upper surface 1082 of the linked rack 1074. This rotation is coupled to an indicator gear 1084, which thus rotates counterclockwise in response to the idler gear 1080. Both the idler gear 1080 and indicator gear 1084 are rotatably connected to the pieces 59, 60 of the handle 6. The gear relationship between the linked rack 1074, idler gear 1080 and indicator gear 1084 may be advantageously selected so that the toothed upper surface 1082 has tooth dimensions that are suitably strong and that the indicator gear 1084 makes no more than one revolution during the full firing travel of the linked transmission firing mechanism 1060.

As described in greater detail below, the indicator gear 1084 performs at least four functions. First, when the linked rack 1074 is fully retracted and both triggers 18, 20 are open as shown in FIG. 42, an opening 1086 in a circular ridge 1088 on the left side of the indicator gear 1084 is presented to an upper surface 1090 of the locking arm 1048. Locking arm 1048 is biased into the opening 1086 by contact with the closure trigger 18, which in turn is urged to the open position by a closure tension spring 1092. Closure trigger tension spring 1092 is connected proximally to the upper portion 1094 of the closure trigger 18 and the handle pieces 59, 60, and thus has energy stored during closing of the closure trigger 18 that urges the closure trigger 18 distally to its unclosed position.

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A second function of the indicator gear 1084 is that it is connected to the indicating retraction knob 1096 externally disposed on the handle 6. Thus, the indicator gear 1084 communicates the relative position of the firing mechanism 1060 to the indicating retraction knob 1096 so that the surgeon has a visual indication of how many strokes of the firing trigger 20 are required to complete firing.

A third function of the indicator gear 1084 is to longitudinally and angularly move an anti-backup release lever 1098 of an anti-backup mechanism (one-way clutch mechanism) 1097 as the surgical stapling and severing instrument 10 is operated. During the firing strokes, proximal movement of anti-backup release lever 1098 by indicator gear 1084 activates the anti-backup mechanism 1097 that allows distal movement of firing bar 1010 and prevents proximal motion of firing bar 1010. This movement also extends the anti-backup release button 1100 from the proximal end of the handle pieces 59, 60 for the operator to actuate should the need arise for the linked transmission firing mechanism 1060 to be retracted during the firing strokes. After completion of the firing strokes, the indicator gear 1084 reverses direction of rotation as the firing mechanism 1060 retracts. The reversed rotation deactivates the anti-backup mechanism 1097, withdraws the anti-backup release button 1100 into the handle 6, and rotates the anti-backup release lever 1098 laterally to the right to allow continued reverse rotation of the indicator gear 1084.

A fourth function of the indicator gear 1084 is to receive a manual rotation from the indicating retraction knob 1096 (clockwise in the depiction of FIG. 42) to retract the firing mechanism 1060 with anti-backup mechanism 1097 unlocked, thereby overcoming any binding in the firing mechanism 1060 that is not readily overcome by the combination tension/compression spring 1062. This manual retraction assistance may be employed after a partial firing of the firing mechanism 1060 that would otherwise be prevented by the anti-backup mechanism 1097 that withdraws the anti-backup release button 1100 so that the latter may not laterally move the anti-backup release lever 1098.

Continuing with FIGS. 42-43, anti-backup mechanism 1097 consists of the operator accessible anti-backup release lever 1098 operably coupled at the proximal end to the anti-backup release button 1100 and at the distal end to an anti-backup yoke 1102. In particular, a distal end 1099 of the anti-backup release lever 1098 is engaged to the anti-backup yoke 1102 by an anti-backup yoke pin 1104. The anti-backup yoke 1102 moves longitudinally to impart a rotation to an anti-backup cam slot tube 1106 that is longitudinally constrained by the handle pieces 59, 90 and that encompasses the firing rod 1010 distally to the connection of the firing rod 1010 to the link coupling 1070 of the linked rack 1074. The anti-backup yoke 1102 communicates the longitudinal movement from the anti-backup release lever 1098 via a cam slot tube pin 1108 to the anti-backup cam slot tube 1106. That is, longitudinal movement of cam slot tube pin 1108 in an angled slot in the anti-backup cam slot tube 1106 rotates the anti-backup cam slot tube 1106.

Trapped between a proximal end of the frame 1016 and the anti-backup cam slot tube 1106 respectively are an anti-backup compression spring 1110, an anti-backup plate 1112, and an anti-backup cam tube 1114. As depicted, proximal movement of the firing rod 1010 causes the anti-backup plate 1112 to pivot top to the rear, presenting an increased frictional contact to the firing rod 1010 that resists further proximal movement of the firing rod 1010.

This anti-backup plate 1112 pivots in a manner similar to that of a screen door lock that holds open a screen door when

the anti-backup cam slot tube **1106** is closely spaced to the anti-backup cam tube **1114**. Specifically, the anti-backup compression spring **1110** is able to act upon a top surface of the plate **1112** to tip the anti-backup plate **1112** to its locked position. Rotation of the anti-backup cam slot tube **1106** causes a distal camming movement of the anti-backup cam tube **1114** thereby forcing the top of the anti-backup plate **1112** distally, overcoming the force from the anti-backup compression spring **1110**, thus positioning the anti-backup plate **1112** in an untipped (perpendicular), unlocked position that allows proximal retraction of the firing rod **1010**.

With particular reference to FIG. **43**, the traction biasing mechanism **1078** is depicted as being composed of a pawl **1116** that has a distally projecting narrow tip **1118** and a rightwardly projecting lateral pin **1120** at its proximal end that is rotatably inserted through a hole **1076** in the upper portion **230** of the firing trigger **20**. On the right side of the firing trigger **20** the lateral pin **1120** receives a biasing member, depicted as biasing wheel **1122**. As the firing trigger **20** translates fore and aft, the biasing wheel **1122** traverses an arc proximate to the right half piece **59** of the handle **6**, overrunning at its distal portion of travel a biasing ramp **1124** integrally formed in the right half piece **59**. The biasing wheel **1122** may advantageously be formed from a resilient, frictional material that induces a counterclockwise rotation (when viewed from the left) into the lateral pin **1120** of the pawl **1116**, thus traction biasing the distally projecting narrow tip **1118** downward into a ramped central track **1075** of the nearest link **1072a-d** to engage the linked rack **1074**.

As the firing trigger **20** is released, the biasing wheel **1122** thus tractionally biases the pawl **1116** in the opposite direction, raising the narrow tip **1118** from the ramped central track **1075** of the linked rack **1074**. To ensure disengagement of the tip **1118** under high load conditions and at nearly full distal travel of the pawl **1116**, the right side of the pawl **1116** ramps up onto a proximally and upwardly facing beveled surface **1126** on the right side of the closure yoke **250** to disengage the narrow tip **1118** from the ramped central track **1075**. If the firing trigger **20** is released at any point other than full travel, the biasing wheel **1122** is used to lift the narrow tip **1118** from the ramped central track **1075**. Whereas a biasing wheel **1122** is depicted, it should be appreciated that the shape of the biasing member or wheel **1122** is illustrative and may be varied to accommodate a variety of shapes that use friction or traction to engage or disengage the firing of the end effector **12**.

Various embodiments of the surgical instrument **10** have the capability to record instrument conditions at one or more times during use. FIG. **44** shows a block diagram of a system **2000** for recording conditions of the instrument **10**. It will be appreciated that the system **2000** may be implemented in embodiments of the instrument **10** having motorized or motor-assisted firing, for example, as described above with reference to FIGS. **1-40**, as well as embodiments of the instrument **10** having mechanically actuated firing, for example, as described above with reference to FIGS. **41-43**.

The system **2000** may include various sensors **2002**, **2004**, **2006**, **2008**, **2010**, **2012** for sensing instrument conditions. The sensors may be positioned, for example, on or within the instrument **10**. In various embodiments, the sensors may be dedicated sensors that provide output only for the system **2000**, or may be dual-use sensors that perform other functions within the instrument **10**. For example, sensors **110**, **130**, **142** described above may be configured to also provide output to the system **2000**.

Directly or indirectly, each sensor provides a signal to the memory device **2001**, which records the signals as described

in more detail below. The memory device **2001** may be any kind of device capable of storing or recording sensor signals. For example, the memory device **2001** may include a microprocessor, an Electrically Erasable Programmable Read Only Memory (EEPROM), or any other suitable storage device. The memory device **2001** may record the signals provided by the sensors in any suitable way. For example, in one embodiment, the memory device **2001** may record the signal from a particular sensor when that signal changes states. In another embodiment, the memory device **2001** may record a state of the system **2000**, e.g., the signals from all of the sensors included in the system **2000**, when the signal from any sensor changes states. This may provide a snap-shot of the state of the instrument **10**. In various embodiments, the memory device **2001** and/or sensors may be implemented to include 1-WIRE bus products available from DALLAS SEMICONDUCTOR such as, for example, a 1-WIRE EEPROM.

In various embodiments, the memory device **2001** is externally accessible, allowing an outside device, such as a computer, to access the instrument conditions recorded by the memory device **2001**. For example, the memory device **2001** may include a data port **2020**. The data port **2020** may provide the stored instrument conditions according to any wired or wireless communication protocol in, for example, serial or parallel format. The memory device **2001** may also include a removable medium **2021** in addition to or instead of the output port **2020**. The removable medium **2021** may be any kind of suitable data storage device that can be removed from the instrument **10**. For example, the removable medium **2021** may include any suitable kind of flash memory, such as a Personal Computer Memory Card International Association (PCMCIA) card, a COMPACTFLASH card, a MULTIMEDIA card, a FLASHMEDIA card, etc. The removable medium **2021** may also include any suitable kind of disk-based storage including, for example, a portable hard drive, a compact disk (CD), a digital video disk (DVD), etc.

The closure trigger sensor **2002** senses a condition of the closure trigger **18**. FIGS. **45** and **46** show an exemplary embodiment of the closure trigger sensor **2002**. In FIGS. **45** and **46**, the closure trigger sensor **2002** is positioned between the closure trigger **18** and closure pivot pin **252**. It will be appreciated that pulling the closure trigger **18** toward the pistol grip **26** causes the closure trigger **18** to exert a force on the closure pivot pin **252**. The sensor **2002** may be sensitive to this force, and generate a signal in response thereto, for example, as described above with respect to sensor **110** and FIGS. **10A** and **10B**. In various embodiments, the closure trigger sensor **2002** may be a digital sensor that indicates only whether the closure trigger **18** is actuated or not actuated. In other various embodiments, the closure trigger sensor **2002** may be an analog sensor that indicates the force exerted on the closure trigger **18** and/or the position of the closure trigger **18**. If the closure trigger sensor **2002** is an analog sensor, an analog-to-digital converter may be logically positioned between the sensor **2002** and the memory device **2001**. Also, it will be appreciated that the closure trigger sensor **2002** may take any suitable form and be placed at any suitable location that allows sensing of the condition of the closure trigger.

The anvil closure sensor **2004** may sense whether the anvil **24** is closed. FIG. **47** shows an exemplary anvil closure sensor **2004**. The sensor **2004** is positioned next to, or within the kidney shaped openings **1006** of the staple channel **22** as shown. As the anvil **24** is closed, anvil pivot pins **25** slides through the kidney shaped openings **1006** and into contact

with the sensor **2004**, causing the sensor **2004** to generate a signal indicating that the anvil **24** is closed. The sensor **2004** may be any suitable kind of digital or analog sensor including a proximity sensor, etc. It will be appreciated that when the anvil closure sensor **2004** is an analog sensor, an analog-to-digital converter may be included logically between the sensor **2004** and the memory device **2001**.

Anvil closure load sensor **2006** is shown placed on an inside bottom surface of the staple channel **22**. In use, the sensor **2006** may be in contact with a bottom side of the staple cartridge **34** (not shown in FIG. **46**). As the anvil **24** is closed, it exerts a force on the staple cartridge **34** which is transferred to the sensor **2006**. In response, the sensor **2006** generates a signal. The signal may be an analog signal proportional to the force exerted on the sensor **2006** by the staple cartridge **34** and due to the closing of the anvil **24**. Referring the FIG. **44**, the analog signal may be provided to an analog-to-digital converter **2014**, which converts the analog signal to a digital signal before providing it to the memory device **2001**. It will be appreciated that embodiments where the sensor **2006** is a digital or binary sensor may not include analog-to-digital converter **2014**.

The firing trigger sensor **110** senses the position and/or state of the firing trigger **20**. In motorized or motor-assisted embodiments of the instrument, the firing trigger sensor may double as the run motor sensor **110** described above. In addition, the firing trigger sensor **110** may take any of the forms described above, and may be analog or digital. FIGS. **45** and **46** show an additional embodiment of the firing trigger sensor **110**. In FIGS. **45** and **46**, the firing trigger sensor is mounted between firing trigger **20** and firing trigger pivot pin **96**. When firing trigger **20** is pulled, it will exert a force on firing trigger pivot pin **96** that is sensed by the sensor **110**. Referring to FIG. **44**, in embodiments where the output of the firing trigger sensor **110** is analog, analog-to-digital converter **2016** is included logically between the firing trigger sensor **110** and the memory device **2001**.

The knife position sensor **2008** senses the position of the knife **32** or cutting surface **1027** within the staple channel **22**. FIGS. **47** and **48** show embodiments of a knife position sensor **2008** that are suitable for use with the mechanically actuated shaft **8** and end effector **12** shown in FIG. **41**. The sensor **2008** includes a magnet **2009** coupled to the firing bar **1022** of the instrument **10**. A coil **2011** is positioned around the firing bar **1022**, and may be installed; for example, along the longitudinal recess **1014** of the firing trough member **1012** (see FIG. **41**). As the knife **32** and cutting surface **1027** are reciprocated through the staple channel **22**, the firing bar **1022** and magnet **2009** may move back and forth through the coil **2011**. This motion relative to the coil induces a voltage in the coil proportional to the position of the firing rod within the coil and the cutting edge **1027** within the staple channel **22**. This voltage may be provided to the memory device **2001**, for example, via analog-to-digital converter **2018**.

In various embodiments, the knife position sensor **2008** may instead be implemented as a series of digital sensors (not shown) placed at various positions on or within the shaft **8**. The digital sensors may sense a feature of the firing bar **1022** such as, for example, magnet **2009**, as the feature reciprocates through the shaft **8**. The position of the firing bar **1022** within the shaft **8**, and by extension, the position of the knife **32** within the staple channel **22**, may be approximated as the position of the last digital sensor tripped.

It will be appreciated that the knife position may also be sensed in embodiments of the instrument **10** having a rotary driven end effector **12** and shaft **8**, for example, as described

above, with reference to FIGS. **3-6**. An encoder, such as encoder **268**, may be configured to generate a signal proportional to the rotation of the helical screw shaft **36**, or any other drive shaft or gear. Because the rotation of the shaft **36** and other drive shafts and gears is proportional to the movement of the knife **32** through the channel **22**, the signal generated by the encoder **268** is also proportional to the movement of the knife **32**. Thus, the output of the encoder **268** may be provided to the memory device **2001**.

It will be appreciated that the position of an actuator that drives a surgical function of the end effector **12** may also be sensed on powered embodiments of the instrument **10**, as described above, for example, with reference to FIGS. **3-10B**. Actuators, or moving elements, may drive surgical functions of the end effector **12** such as, for example, cutting, stapling, closure, and articulation. When a position sensor senses that an actuator, such as, for example, firing bar **1022** or helical screw shaft **36**, is at a position indicative that the surgical function is at an end-of-stroke or at a limit position, the position sensor signals a microprocessor in the force delivery system that the limit has been reached. The microprocessor, such as one that may be included in memory device **2001**, can cause a change in voltage driving the motor. The change in voltage may increase, decrease, or reverse the motor's speed of rotation in a manner to cause a change in force, and thus a force feedback, to a user holding the handle. The user then senses that the surgical function is at the limit position. Alternately, the signal from the position sensor, the voltage change, and the force feedback may occur when the surgical function is approaching the limit position without having yet achieved the limit position.

The force delivered to the user may be a vibratory force. FIGS. **54** and **55** demonstrate a changing voltage, and a changing force, that is delivered to the user in the case of a vibratory feedback showing end-of-stroke for an actuator causing articulation. If an end effector such as end effector **12** is articulated to the right to its articulation limit, a position sensor can sense actuator movement corresponding to the limit position and cause a microprocessor to send a voltage to the motor driving articulation. The signal can comprise a rapidly varying square wave or sine wave to drive the motor in a vibratory manner sensible by the user holding the handle. Likewise, an end-effector articulated to the left to its articulation limit can also be signaled.

FIG. **56** shows a flow diagram illustrating processor logic that can be repeated multiple times per second to determine if an actuator causing articulation can be advanced further or if vibratory feedback is warranted. A processor following this logic would begin at **3005** at the initiation of articulation. If a user depresses a switch such as switch **110** to initiate articulation to the right, the processor logic, at decision point **3025**, uses a signal from a position sensor to check to see if end effector **12** is at the right limit. If the end effector **12** is at the right limit, the processor logic proceeds to state **3020**, where the processor sends the rapidly varying square or sign wave. If the end effector **12** is not at the right limit, the processor logic proceeds to **3030**, allowing further articulation until the user ends articulation or until the right limit is reached. Processor logic proceeds similarly to articulate the end effector **12** to the extreme left limit. Similar logic may also be used with motors and actuators that drive the surgical functions of cutting, stapling, closure, and opening.

The cartridge present sensor **2010** may sense the presence of the staple cartridge **34** within the staple channel **22**. In motorized or motor-assisted instruments, the cartridge present sensor **2010** may double as the cartridge lock-out sensor **136** described above with reference to FIG. **11**. FIGS. **50** and

51 show an embodiment of the cartridge present sensor 2010. In the embodiment shown, the cartridge present sensor 2010 includes two contacts, 2011 and 2013. When no cartridge 34 is present, the contacts 2011, 2013 form an open circuit. When a cartridge 34 is present, the cartridge tray 1028 of the staple cartridge 34 contacts the contacts 2011, 2013, a closed circuit is formed. When the circuit is open, the sensor 2010 may output a logic zero. When the circuit is closed, the sensor 2010 may output a logic one. The output of the sensor 2010 is provided to memory device 2001, as shown in FIG. 44.

The cartridge condition sensor 2012 may indicate whether a cartridge 34 installed within the staple channel 22 has been fired or spent. As the knife 32 is translated through the end effector 12, it pushes the sled 33, which fires the staple cartridge. Then the knife 32 is translated back to its original position, leaving the sled 33 at the distal end of the cartridge. Without the sled 33 to guide it, the knife 32 may fall into lock-out pocket 2022. Sensor 2012 may sense whether the knife 32 is present in the lock-out pocket 2022, which indirectly indicates whether the cartridge 34 has been spent. It will be appreciated that in various embodiments, sensor 2012 may directly sense the present of the sled at the proximate end of the cartridge 34, thus eliminating the need for the knife 32 to fall into the lock-out pocket 2022.

FIGS. 52A and 52B depict a process flow 2200 for operating embodiments of the surgical instrument 10 configured as an endocutter and having the capability to record instrument conditions according to various embodiments. At box 2202, the anvil 24 of the instrument 10 may be closed. This causes the closure trigger sensor 2002 and/or the anvil closure sensor 2006 to change state. In response, the memory device 2001 may record the state of all of the sensors in the system 2000 at box 2203. At box 2204, the instrument 10 may be inserted into a patient. When the instrument is inserted, the anvil 24 may be opened and closed at box 2206, for example, to manipulate tissue at the surgical site. Each opening and closing of the anvil 24 causes the closure trigger sensor 2002 and/or the anvil closure sensor 2004 to change state. In response, the memory device 2001 records the state of the system 2000 at box 2205.

At box 2208, tissue is clamped for cutting and stapling. If the anvil 24 is not closed at decision block 2210, continued clamping is required. If the anvil 24 is closed, then the sensors 2002, 2004 and/or 2006 may change state, prompting the memory device 2001 to record the state of the system at box 2213. This recording may include a closure pressure received from sensor 2006. At box 2212, cutting and stapling may occur. Firing trigger sensor 110 may change state as the firing trigger 20 is pulled toward the pistol grip 26. Also, as the knife 32 moves through the staple channel 22, knife position sensor 2008 will change state. In response, the memory device 2001 may record the state of the system 2000 at box 2013.

When the cutting and stapling operations are complete, the knife 32 may return to a pre-firing position. Because the cartridge 34 has now been fired, the knife 32 may fall into lock-out pocket 2022, changing the state of cartridge condition sensor 2012 and triggering the memory device 2001 to record the state of the system 2000 at box 2015. The anvil 24 may then be opened to clear the tissue. This may cause one or more of the closure trigger sensor 2002, anvil closure sensor 2004 and anvil closure load sensor 2006 to change state, resulting in a recordation of the state of the system 2000 at box 2017. After the tissue is cleared, the anvil 24 may be again closed at box 2220. This causes another state change for at least sensors 2002 and 2004, which in turn

causes the memory device 2001 to record the state of the system at box 2019. Then the instrument 10 may be removed from the patient at box 2222.

If the instrument 10 is to be used again during the same procedure, the anvil may be opened at box 2224, triggering another recordation of the system state at box 2223. The spent cartridge 34 may be removed from the end effector 12 at box 2226. This causes cartridge present sensor 2010 to change state and cause a recordation of the system state at box 2225. Another cartridge 34 may be inserted at box 2228. This causes a state change in the cartridge present sensor 2010 and a recordation of the system state at box 2227. If the other cartridge 34 is a new cartridge, indicated at decision block 2230, its insertion may also cause a state change to cartridge condition sensor 2012. In that case, the system state may be recorded at box 2231.

FIG. 53 shows an exemplary memory map 2300 from the memory device 2001 according to various embodiments. The memory map 2300 includes a series of columns 2302, 2304, 2306, 2308, 2310, 2312, 2314, 2316 and rows (not labeled). Column 2302 shows an event number for each of the rows. The other columns represent the output of one sensor of the system 2000. All of the sensor readings recorded at a given time may be recorded in the same row under the same event number. Hence, each row represents an instance where one or more of the signals from the sensors of the system 2000 are recorded.

Column 2304 lists the closure load recorded at each event. This may reflect the output of anvil closure load sensor 2006. Column 2306 lists the firing stroke position. This may be derived from the knife position sensor 2008. For example, the total travel of the knife 32 may be divided into partitions. The number listed in column 2306 may represent the partition where the knife 32 is currently present. The firing load is listed in column 2308. This may be derived from the firing trigger sensor 110. The knife position is listed at column 2310. The knife position may be derived from the knife position sensor 2008 similar to the firing stroke. Whether the anvil 24 is open or closed may be listed at column 2312. This value may be derived from the output of the anvil closure sensor 2004 and/or the anvil closure load sensor 2006. Whether the sled 33 is present, or whether the cartridge 34 is spent, may be indicated at column 2314. This value may be derived from the cartridge condition sensor 2012. Finally, whether the cartridge 34 is present may be indicated a column 2316. This value may be derived from cartridge present sensor 2010. It will be appreciated that various other values may be stored at memory device 2001 including, for example, the end and beginning of firing strokes, for example, as measured by sensors 130, 142.

While the present invention has been illustrated by description of several embodiments and while the illustrative embodiments have been described in considerable detail, it is not the intention of the applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications may readily appear to those skilled in the art.

For example, although the embodiments described above have advantages for an endoscopically employed surgical severing and stapling instrument 100, a similar embodiments may be used in other clinical procedures. It is generally accepted that endoscopic procedures are more common than laparoscopic procedures. Accordingly, the present invention has been discussed in terms of endoscopic procedures and apparatus. However, use herein of terms such as "endoscopic", should not be construed to limit the present invention to a surgical instrument for use only in conjunction

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with an endoscopic tube (i.e., trocar). On the contrary, it is believed that the present invention may find use in any procedure where access is limited to a small incision, including but not limited to laparoscopic procedures, as well as open procedures.

Any patent, publication, or information, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated material does not conflict with existing definitions, statements, or other disclosure material set forth in this document. As such the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference.

What is claimed is:

1. A surgical assembly, comprising:

a shaft;

an end effector;

a driver actuatable by a motor, wherein the driver is movable through a drive motion, and wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; and

a control circuit communicably coupled to the motor, the control circuit to:

receive a user input corresponding to an initiation of the drive motion;

initiate the drive motion based on the user input; continuously monitor the position of the driver throughout the drive motion; and

deliver a feedback to a user over a period of time based on the position of the driver, wherein the feedback comprises an indication that the driver is approaching the conclusion of the drive motion.

2. The surgical assembly of claim 1, wherein the feedback comprises a tactile indication that the driver is approaching the conclusion of the drive motion.

3. The surgical assembly of claim 2, further comprising a motor to generate the tactile indication that the driver is approaching the conclusion of the drive motion.

4. The surgical assembly of claim 1, wherein the control circuit is further to transmit a modulating signal profile based on the position of the driver.

5. The surgical assembly of claim 4, wherein the modulating signal profile corresponds to a haptic feedback to the user.

6. The surgical assembly of claim 1, wherein the feedback comprises an indication that the driver is approaching a limit position of the drive motion.

7. The surgical assembly of claim 6, further comprising an articulation joint, wherein the driver comprises an articulation cable, and wherein the drive motion comprises an articulation motion.

8. The surgical assembly of claim 1, wherein the driver comprises an I-beam.

9. The surgical assembly of claim 8, wherein the drive motion comprises a closure stroke.

10. The surgical assembly of claim 8, wherein the drive motion comprises a firing stroke.

11. A surgical assembly, comprising:

a shaft;

an end effector;

a driver actuatable by a motor, wherein the driver is movable through a drive motion, and wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; and

a control circuit communicably coupled to the motor, the control circuit to:

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receive a user input corresponding to an initiation of the drive motion;

initiate the drive motion based on the user input;

continuously monitor the status of the driver throughout the drive motion; and

deliver a feedback to a user over a period of time based on the status of the driver, wherein the feedback comprises an indication that the driver is approaching the conclusion of the drive motion.

12. The surgical assembly of claim 11, wherein the feedback comprises a tactile indication that the driver is approaching the conclusion of the drive motion.

13. The surgical assembly of claim 12, further comprising a motor to generate the tactile indication that the driver is approaching the conclusion of the drive motion.

14. The surgical assembly of claim 11, wherein the control circuit is further to transmit a modulating signal profile based on the status of the driver.

15. The surgical assembly of claim 14, wherein the modulating signal profile corresponds to a haptic feedback to the user.

16. The surgical assembly of claim 11, wherein the feedback comprises an indication that the driver is approaching a limit position of the drive motion.

17. The surgical assembly of claim 16, wherein the driver comprises an I-beam.

18. The surgical assembly of claim 17, wherein the drive motion comprises a closure stroke.

19. The surgical assembly of claim 17, wherein the drive motion comprises a firing stroke.

20. The surgical assembly of claim 11, further comprising an articulation joint, wherein the driver comprises an articulation cable, wherein the drive motion comprises an articulation motion, and wherein the status of the driver comprises the position of the articulation cable throughout the articulation motion.

21. A surgical assembly, comprising:

a shaft;

an end effector, wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; a motorized driver movable through a motorized drive motion; and

a control circuit to:

initiate the motorized drive motion;

continuously track a position of the motorized driver throughout the motorized drive motion; and

deliver a feedback force to a user during the motorized drive motion based on the position of the motorized driver.

22. The surgical assembly of claim 21, wherein the motorized driver comprises an I-beam.

23. The surgical assembly of claim 22, wherein the motorized drive motion comprises a closure stroke.

24. The surgical assembly of claim 23, further comprising a sensor to detect the position of the motorized driver.

25. The surgical assembly of claim 22, wherein the motorized drive motion comprises a firing stroke.

26. The surgical assembly of claim 25, further comprising a sensor to detect the position of the motorized driver.

27. The surgical assembly of claim 26, wherein the feedback force comprises a vibration.

28. The surgical assembly of claim 27, further comprising a motor to generate the vibration.

29. The surgical assembly of claim 21, wherein the feedback force comprises an indication that the driver is approaching a limit position of the motorized drive motion.

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30. The surgical assembly of claim **21**, further comprising an articulation joint, wherein the driver comprises an articulation cable, and wherein the motorized drive motion comprises an articulation motion.

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